
Robotics & State Estimation (RiSE) Tutorials

Lecture Notes

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Robotics & State Estimation Tutorials

<https://github.com/yangyulin/rise-tutorial>



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1 Lecture 1: Foundations — Probability, Linear Algebra, and Linear Systems

1.1 Probability

1.1.1 Definition

For an event A , the probability $P(A)$ is

$$P(A) = \frac{N_A}{N}, \quad (1)$$

where N_A is the number of occurrences of A and N is the number of all possible outcomes.

1.1.2 Conditional probability

The probability of event A given that event B is true is

$$P(A | B) = \frac{P(A, B)}{P(B)}, \quad P(B) \neq 0, \quad (2)$$

which yields the product rule

$$P(A, B) = P(A | B) P(B) \quad (3)$$

$$= P(B | A) P(A). \quad (4)$$

1.1.3 Total probability theorem

If $\{A_1, \dots, A_n\}$ is a partition (mutually exclusive, $P(A_i, A_j) = 0$ for $i \neq j$) and B is an arbitrary event, then

$$P(B) = P(B | A_1)P(A_1) + \dots + P(B | A_n)P(A_n) \quad (5)$$

$$= \sum_i P(B | A_i) P(A_i). \quad (6)$$

1.1.4 Bayes' theorem

$$P(A_i | B) = \frac{P(B | A_i) P(A_i)}{P(B)} \quad (7)$$

$$= \frac{P(B | A_i) P(A_i)}{\sum_i P(B | A_i) P(A_i)}. \quad (8)$$

1.1.5 Independence

With the inclusion–exclusion relation $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, two events A and B are independent when

$$P(A, B) = P(A | B) P(B) \quad (9)$$

$$= P(B | A) P(A) \quad (10)$$

$$= P(A) P(B) \quad (11)$$

$$= P(B) P(A). \quad (12)$$

1.2 Random Variables

A random variable defines a function that quantifies an experiment. For a sample space $S = \{s_1, \dots, s_n\}$, a random variable is a map $x : S \rightarrow \mathbb{R}$ with $x(s_i) = x_i$.

1.2.1 Distribution

The (cumulative) distribution of a random variable x is

$$F_x(x) = P(x \leq x), \quad (13)$$

i.e. the probability that x takes a value lower than or equal to x .

1.2.2 Probability density function (pdf)

$$p(x) = \lim_{\varepsilon \rightarrow 0} \frac{F_x(x) - F_x(x - \varepsilon)}{\varepsilon} \quad (14)$$

$$= \frac{dF_x(x)}{dx} \quad (15)$$

$$= \frac{dP(x \leq x)}{dx}, \quad (16)$$

hence

$$P(x \leq x) = \int_{-\infty}^x p(\xi) d\xi, \quad (17)$$

$$P(x \leq \infty) = \int_{-\infty}^{+\infty} p(\xi) d\xi = 1, \quad (18)$$

and for an interval

$$P(x_1 \leq x < x_2) = \int_{x_1}^{x_2} p(\xi) d\xi. \quad (19)$$

1.2.3 Example distributions

- Gaussian: $x \sim \mathcal{N}(m, \sigma) \iff p(x) = \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{(x-m)^2}{2\sigma^2}}.$

- Uniform: $p(x) = \begin{cases} 0 & x < x_1 \\ \frac{1}{x_2 - x_1} & x_1 \leq x \leq x_2 \\ 0 & x > x_2 \end{cases}$

1.2.4 Conditional distribution and density

$$P(x \leq x \mid A) = \frac{P(x \leq x, A)}{P(A)}, \quad (20)$$

$$p(x \mid A) = \frac{dP(x \leq x \mid A)}{dx}. \quad (21)$$

1.2.5 Total probability theorem for pdfs

For a mutually exclusive partition $\{A_1, \dots, A_n\}$,

$$p(x) = p(x \mid A_1)P(A_1) + \dots + p(x \mid A_n)P(A_n). \quad (22)$$

1.2.6 Continuous Bayes' theorem

Using the infinitesimal relations $P(x = x) = p(x) \delta x$ and $P(x = x \mid A) = p(x \mid A) \delta x$,

$$P(A \mid x = x) = \frac{P(A, x = x)}{P(x = x)} \quad (23)$$

$$= \frac{P(x = x \mid A) P(A)}{P(x = x)} \quad (24)$$

$$= \frac{p(x \mid A)}{p(x)} P(A), \quad (25)$$

so that $P(A \mid x = x) p(x) = p(x \mid A) P(A)$, which gives the continuous total-probability and Bayes' relations

$$P(A) = \int_{-\infty}^{+\infty} P(A \mid x = x) p(x) dx, \quad (26)$$

$$p(x \mid A) = \frac{P(A \mid x = x)}{P(A)} p(x) \quad (27)$$

$$= \frac{P(A \mid x = x) p(x)}{\int_{-\infty}^{+\infty} P(A \mid x = x) p(x) dx}. \quad (28)$$

1.3 Expectation, Variance, and Random Vectors

1.3.1 Mean

The expected value (mean) of a continuous random variable, and its conditional version, are

$$m_x = \mathbb{E}(x) \quad (29)$$

$$= \int_{-\infty}^{+\infty} x p(x) dx, \quad (30)$$

$$m_{x|A} = \mathbb{E}(x \mid A) \quad (31)$$

$$= \int_{-\infty}^{+\infty} x p(x \mid A) dx. \quad (32)$$

1.3.2 Variance

$$\sigma_x^2 = \mathbb{E}((x - m_x)^2) \quad (33)$$

$$= \int_{-\infty}^{+\infty} (x - m_x)^2 p(x) dx \quad (34)$$

$$= \mathbb{E}(x^2) - (\mathbb{E}(x))^2, \quad (35)$$

with the conditional variance

$$\sigma_{x|A}^2 = \mathbb{E}\{(x - m_{x|A})^2 \mid A\} \quad (36)$$

$$= \int_{-\infty}^{+\infty} (x - m_{x|A})^2 p(x \mid A) dx. \quad (37)$$

1.3.3 A set of random variables

The joint pdf of $x_1, \dots, x_n$ satisfies

$$\int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} p(x_1, \dots, x_n) dx_1 \cdots dx_n = 1. \quad (38)$$

If x_i, x_j are independent, $p(x_i, x_j) = p(x_i)p(x_j)$. Expectation is linear: for $y = \alpha x_i + \beta x_j$,

$$\mathbb{E}(y) = \alpha \mathbb{E}(x_i) + \beta \mathbb{E}(x_j). \quad (39)$$

1.3.4 Covariance

$$\sigma_{ij}^2 = \mathbb{E}((x_i - m_i)(x_j - m_j)) \quad (40)$$

$$= \mathbb{E}(x_i x_j) - \mathbb{E}(x_i) \mathbb{E}(x_j). \quad (41)$$

When $\sigma_{ij} = 0$ we have $\mathbb{E}(x_i x_j) = \mathbb{E}(x_i) \mathbb{E}(x_j)$ and x_i, x_j are *uncorrelated*.

Discussion questions.

1. If x_i, x_j are independent, then $p(x_i, x_j) = p(x_i)p(x_j) \Rightarrow \mathbb{E}(x_i x_j) = \mathbb{E}(x_i)\mathbb{E}(x_j)$ — so independence implies uncorrelated.
2. If x_i, x_j are uncorrelated, are they necessarily independent?
3. If x_i, x_j are jointly Gaussian, is independence equivalent to being uncorrelated?

1.3.5 Random vector

Let $\mathbf{x} = [x_1, \dots, x_n]^\top$, where all elements are random variables, with mean $\mathbb{E}(\mathbf{x}) \triangleq \mathbf{m}_x$ and covariance

$$\mathbf{P} = \mathbb{E}((\mathbf{x} - \mathbf{m}_x)(\mathbf{x} - \mathbf{m}_x)^\top) \quad (42)$$

$$= \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbb{E}(\mathbf{x}) \mathbb{E}(\mathbf{x}^\top), \quad (43)$$

with entries

$$P_{ii} = \mathbb{E}((x_i - m_i)^2) \quad (44)$$

$$= \int_{-\infty}^{+\infty} (x_i - m_i)^2 p(x_i) dx_i, \quad (45)$$

$$P_{ij} = \mathbb{E}((x_i - m_i)(x_j - m_j)). \quad (46)$$

Since $P_{ij} = P_{ji}$, the covariance $\mathbf{P}$ is a symmetric matrix; it is also positive (semi-)definite.

1.3.6 Gaussian vector pdf

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{n/2} \sqrt{|\mathbf{P}|}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mathbf{m}_x)^\top \mathbf{P}^{-1}(\mathbf{x} - \mathbf{m}_x)\right), \quad (47)$$

where $|\mathbf{P}|$ is the determinant of $\mathbf{P}$, $\mathbf{m}_x = \mathbb{E}(\mathbf{x})$, and $\mathbf{P} = \mathbb{E}((\mathbf{x} - \mathbf{m}_x)(\mathbf{x} - \mathbf{m}_x)^\top)$.

Homework. Let $\mathbf{x} = [x_1, \dots, x_n]^\top$ with $p(\mathbf{x}) = \mathcal{N}(\mathbf{m}_x, \mathbf{P}_{xx})$. Define

$$\begin{aligned} q &= (\mathbf{x} - \mathbf{m}_x)^\top \mathbf{P}_{xx}^{-1}(\mathbf{x} - \mathbf{m}_x), \\ \mathbf{y} &= \mathbf{P}_{xx}^{-1/2}(\mathbf{x} - \mathbf{m}_x), \\ q &= \mathbf{y}^\top \mathbf{y} = \sum_i y_i^2. \end{aligned}$$

What are the mean and covariance of q ?

1.4 Maximum Likelihood and Maximum a Posteriori Estimation

For a measurement model with additive noise $z_i = h_i(\mathbf{x}) + n_i$, stacked as $\mathbf{z} = \mathbf{h}(\mathbf{x}) + \mathbf{n}$, two estimators are

- **Maximum likelihood estimation (MLE):** $\max_{\mathbf{x}} p(\mathbf{z} | \mathbf{x})$.
- **Maximum a posteriori (MAP):** $\max_{\mathbf{x}} p(\mathbf{x} | \mathbf{z})$, with $p(\mathbf{x} | \mathbf{z}) = \frac{p(\mathbf{z} | \mathbf{x}) p(\mathbf{x})}{p(\mathbf{z})}$.

For zero-mean Gaussian noise $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \mathbf{P})$,

$$\max_{\mathbf{x}} p(\mathbf{z} | \mathbf{x}) = \max_{\mathbf{x}} \frac{1}{(2\pi)^{n/2} \sqrt{|\mathbf{P}|}} \exp\left(-(\mathbf{z} - \mathbf{h}(\mathbf{x}))^\top \mathbf{P}^{-1}(\mathbf{z} - \mathbf{h}(\mathbf{x}))\right), \quad (48)$$

which is equivalent to the weighted least-squares problem

$$\min_{\mathbf{x}} C = (\mathbf{z} - \mathbf{h}(\mathbf{x}))^\top \mathbf{P}^{-1}(\mathbf{z} - \mathbf{h}(\mathbf{x})). \quad (49)$$

Linearizing about $\hat{\mathbf{x}}$ with $\mathbf{z} = \mathbf{h}(\mathbf{x}) + \mathbf{n} = \mathbf{h}(\hat{\mathbf{x}}) + \mathbf{H}(\mathbf{x} - \hat{\mathbf{x}}) + \mathbf{n}$, and defining the residual $\tilde{\mathbf{z}} = \mathbf{z} - \mathbf{h}(\hat{\mathbf{x}})$ and error $\tilde{\mathbf{x}} = \mathbf{x} - \hat{\mathbf{x}}$,

$$\min_{\tilde{\mathbf{x}}} (\tilde{\mathbf{z}} - \mathbf{H}\tilde{\mathbf{x}})^\top \mathbf{P}^{-1}(\tilde{\mathbf{z}} - \mathbf{H}\tilde{\mathbf{x}}) = \min_{\tilde{\mathbf{x}}} (\tilde{\mathbf{z}}^\top \mathbf{P}^{-1} \tilde{\mathbf{z}} - 2\tilde{\mathbf{z}}^\top \mathbf{P}^{-1} \mathbf{H}\tilde{\mathbf{x}} + \tilde{\mathbf{x}}^\top \mathbf{H}^\top \mathbf{P}^{-1} \mathbf{H}\tilde{\mathbf{x}}). \quad (50)$$

Setting the gradient to zero gives the normal equations $\mathbf{H}^\top \mathbf{P}^{-1} \mathbf{H} \tilde{\mathbf{x}} = \mathbf{H}^\top \mathbf{P}^{-1} \tilde{\mathbf{z}}$, hence

$$\boxed{\begin{aligned} \mathbf{x}^\oplus &= \hat{\mathbf{x}} + \tilde{\mathbf{x}} \\ &= \hat{\mathbf{x}} + (\mathbf{H}^\top \mathbf{P}^{-1} \mathbf{H})^{-1} \mathbf{H}^\top \mathbf{P}^{-1} \tilde{\mathbf{z}}. \end{aligned}} \quad (51)$$

$$(52)$$

This is the classical weighted least-squares solution.

Questions.

1. What is the covariance of the new estimate $\mathbf{x}^\oplus$?
2. For a prior $\mathbf{x}_p = \mathbf{x} + \mathbf{n}_p$ together with $z_i = h_i(\mathbf{x}) + n_i$, $i = 1, \dots, n$, what is the MAP solution $\max_{\mathbf{x}} p(\mathbf{x} | \mathbf{z})$?

1.4.1 Optimization

Nonlinear least-squares problems are solved iteratively, e.g. Gauss–Newton (GN) and Levenberg–Marquardt (LM).

1.5 Linear Systems

A linear (continuous-time) state-space system is

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}, \quad (53)$$

$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}, \quad (54)$$

where $\mathbf{x}$ is the state, $\mathbf{u}$ the input, and $\mathbf{y}$ the output. Its discrete-time counterpart is

$$\mathbf{x}_{k+1} = \mathbf{A}'\mathbf{x}_k + \mathbf{B}'\mathbf{u}_k, \quad (55)$$

$$\mathbf{y}_{k+1} = \mathbf{C}'\mathbf{x}_{k+1} + \mathbf{D}'\mathbf{u}_{k+1}. \quad (56)$$

If $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$ are constant, the system is *linear time-invariant* (LTI); if they depend on time, it is *linear time-variant* (LTV).

1.5.1 Homogeneous solution

Consider $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ with $\mathbf{x}(t_0) = \mathbf{x}_0$ and $\mathbf{y} = \mathbf{C}\mathbf{x}$. From $\dot{\mathbf{x}} - \mathbf{A}\mathbf{x} = \mathbf{0}$,

$$\frac{d}{dt}(e^{-\mathbf{A}t}\mathbf{x}) = e^{-\mathbf{A}t}\dot{\mathbf{x}} + e^{-\mathbf{A}t}(-\mathbf{A})\mathbf{x} \quad (57)$$

$$= e^{-\mathbf{A}t}(\dot{\mathbf{x}} - \mathbf{A}\mathbf{x}) \quad (58)$$

$$= \mathbf{0}. \quad (59)$$

Integrating from 0 to t gives $e^{-\mathbf{A}t}\mathbf{x}(t) - \mathbf{x}(t_0) = \mathbf{0}$, so

$$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}_0 \quad (60)$$

$$= \Phi(t, 0)\mathbf{x}_0, \quad (61)$$

$$\Phi(t, 0) = e^{\mathbf{A}t}, \quad (62)$$

where $\Phi(t, 0)$ is the *state transition matrix*, and $\mathbf{y} = \mathbf{C}\Phi(t, 0)\mathbf{x}_0$.

1.5.2 Properties of the state transition matrix

$$\Phi(t_2, t_0) = \Phi(t_2, t_1)\Phi(t_1, t_0), \quad (63)$$

$$\frac{d}{dt}\Phi(t, 0) = \mathbf{A}\Phi(t, 0). \quad (64)$$

The second follows from $\dot{\mathbf{x}} = \dot{\Phi}(t, 0)\mathbf{x}_0 = \mathbf{A}\mathbf{x} = \mathbf{A}\Phi(t, 0)\mathbf{x}_0$.

1.5.3 Output sequence and observability

Sampling the homogeneous response at $t_0, t_1, \dots, t_n$ with $\mathbf{x}_k = \Phi(k, 0)\mathbf{x}_0$ and $\mathbf{y}_k = \mathbf{C}\mathbf{x}_k$ stacks into

$$\begin{bmatrix} \mathbf{y}_1 \\ \vdots \\ \mathbf{y}_n \end{bmatrix} = \begin{bmatrix} \mathbf{C}\Phi(1, 0) \\ \vdots \\ \mathbf{C}\Phi(n, 0) \end{bmatrix} \mathbf{x}_0, \quad (65)$$

which relates the measured outputs to the initial state (an observability-type relation).

1.5.4 Full LTV solution

For the complete system $\dot{\mathbf{x}} = \mathbf{A}(t)\mathbf{x} + \mathbf{B}(t)\mathbf{u}(t)$, write $\dot{\mathbf{x}} - \mathbf{A}(t)\mathbf{x} = \mathbf{B}(t)\mathbf{u}(t)$ and use the integrating factor $e^{-\int_0^t \mathbf{A}(\tau) d\tau}$:

$$\frac{d}{dt} \left(e^{-\int_0^t \mathbf{A}(\tau) d\tau} \mathbf{x} \right) = e^{-\int_0^t \mathbf{A}(\tau) d\tau} (\dot{\mathbf{x}} - \mathbf{A}(t)\mathbf{x}) \quad (66)$$

$$= e^{-\int_0^t \mathbf{A}(\tau) d\tau} \mathbf{B}(t)\mathbf{u}(t). \quad (67)$$

Integrating,

$$e^{-\int_0^t \mathbf{A}(\tau) d\tau} \mathbf{x}(t) - \mathbf{x}_0 = \int_0^t e^{-\int_0^\tau \mathbf{A}(s) ds} \mathbf{B}(\tau)\mathbf{u}(\tau) d\tau, \quad (68)$$

so that, with $\Phi(t, 0) = e^{\int_0^t \mathbf{A}(\tau) d\tau}$ and $\Phi(t, \tau) = e^{\int_\tau^t \mathbf{A}(s) ds}$,

$$\mathbf{x}(t) = \Phi(t, 0) \mathbf{x}_0 + \int_0^t \Phi(t, \tau) \mathbf{B}(\tau) \mathbf{u}(\tau) d\tau, \quad (69)$$

$$\mathbf{y}(t) = \mathbf{C}(t) \left(\Phi(t, 0) \mathbf{x}_0 + \int_0^t \Phi(t, \tau) \mathbf{B}(\tau) \mathbf{u}(\tau) d\tau \right) + \mathbf{D}(t)\mathbf{u}(t). \quad (70)$$

This is one of the most important properties of linear systems.

1.6 Homework Problems and Solutions

1.6.1 Variance identity

Problem. Show that the variance of a scalar random variable can be written as $\sigma_x^2 = \mathbb{E}(\mathbf{x}^2) - (\mathbb{E}(\mathbf{x}))^2$, and discuss when it vanishes.

Solution. Expanding the definition with $m_x = \mathbb{E}(\mathbf{x})$,

$$\sigma_x^2 = \mathbb{E}((\mathbf{x} - m_x)^2) \quad (71)$$

$$= \mathbb{E}(\mathbf{x}^2 - 2m_x\mathbf{x} + m_x^2) \quad (72)$$

$$= \mathbb{E}(\mathbf{x}^2) - 2m_x\mathbb{E}(\mathbf{x}) + m_x^2 \quad (73)$$

$$= \mathbb{E}(\mathbf{x}^2) - m_x^2. \quad (74)$$

Since $\sigma_x^2 \geq 0$, we have $\mathbb{E}(\mathbf{x}^2) \geq (\mathbb{E}(\mathbf{x}))^2$, with equality if and only if $\sigma_x^2 = 0$, i.e. $\mathbf{x} = m_x$ almost surely (a deterministic variable) — a degenerate special case.

1.6.2 Covariance matrix identity

Problem. For a random vector $\mathbf{x}$ with mean $\mathbf{m}_x$, show that $\mathbf{P} = \mathbb{E}((\mathbf{x} - \mathbf{m}_x)(\mathbf{x} - \mathbf{m}_x)^\top) = \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbb{E}(\mathbf{x})\mathbb{E}(\mathbf{x}^\top)$.

Solution. Expanding the outer product and using linearity of expectation, together with $\mathbb{E}(\mathbf{x}) = \mathbf{m}_x$,

$$\mathbf{P} = \mathbb{E}(\mathbf{x}\mathbf{x}^\top - \mathbf{x}\mathbf{m}_x^\top - \mathbf{m}_x\mathbf{x}^\top + \mathbf{m}_x\mathbf{m}_x^\top) \quad (75)$$

$$= \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbb{E}(\mathbf{x})\mathbf{m}_x^\top - \mathbf{m}_x\mathbb{E}(\mathbf{x}^\top) + \mathbf{m}_x\mathbf{m}_x^\top \quad (76)$$

$$= \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbf{m}_x\mathbf{m}_x^\top - \mathbf{m}_x\mathbf{m}_x^\top + \mathbf{m}_x\mathbf{m}_x^\top \quad (77)$$

$$= \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbf{m}_x\mathbf{m}_x^\top \quad (78)$$

$$= \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbb{E}(\mathbf{x})\mathbb{E}(\mathbf{x}^\top). \quad (79)$$

1.6.3 Scalar covariance identity

Problem. For two random variables x_i, x_j , show that $\sigma_{ij} = \mathbb{E}(x_i x_j) - \mathbb{E}(x_i)\mathbb{E}(x_j)$.

Solution. With $m_i = \mathbb{E}(x_i)$, $m_j = \mathbb{E}(x_j)$, expand

$$\sigma_{ij} = \mathbb{E}((x_i - m_i)(x_j - m_j)) \quad (80)$$

$$= \iint (x_i - m_i)(x_j - m_j) p(x_i, x_j) dx_i dx_j \quad (81)$$

$$\begin{aligned} &= \iint x_i x_j p(x_i, x_j) dx_i dx_j - m_j \iint x_i p(x_i, x_j) dx_i dx_j \\ &\quad - m_i \iint x_j p(x_i, x_j) dx_i dx_j + m_i m_j \iint p(x_i, x_j) dx_i dx_j. \end{aligned} \quad (82)$$

Using the marginalization identities (and symmetrically for x_j)

$$\int p(x_i, x_j) dx_i = p(x_j), \quad \iint p(x_i, x_j) dx_i dx_j = 1, \quad (83)$$

we get

$$m_i \iint x_j p(x_i, x_j) dx_i dx_j = m_i \int x_j p(x_j) dx_j \quad (84)$$

$$= m_i m_j, \quad (85)$$

and likewise for the other term, so

$$\sigma_{ij} = \mathbb{E}(x_i x_j) - m_i m_j - m_i m_j + m_i m_j \quad (86)$$

$$= \mathbb{E}(x_i x_j) - \mathbb{E}(x_i)\mathbb{E}(x_j). \quad (87)$$

1.6.4 Independence versus uncorrelatedness

Problem. Resolve the three discussion questions: (1) does independence imply uncorrelatedness? (2) does uncorrelatedness imply independence? (3) for jointly Gaussian variables, are the two equivalent?

Solution. (1) *Independent $\Rightarrow$ uncorrelated.* If $p(x_i, x_j) = p(x_i)p(x_j)$, then

$$\mathbb{E}(x_i x_j) = \iint x_i x_j p(x_i, x_j) dx_i dx_j \quad (88)$$

$$= \int x_i p(x_i) dx_i \cdot \int x_j p(x_j) dx_j \quad (89)$$

$$= \mathbb{E}(x_i)\mathbb{E}(x_j), \quad (90)$$

so $\sigma_{ij} = 0$.

(2) *Uncorrelated $\not\Rightarrow$ independent.* Uncorrelatedness gives only

$$\mathbb{E}(x_i x_j) - \mathbb{E}(x_i)\mathbb{E}(x_j) = \iint x_i x_j (p(x_i, x_j) - p(x_i)p(x_j)) dx_i dx_j \quad (91)$$

$$= 0, \quad (92)$$

which does *not* force $p(x_i, x_j) = p(x_i)p(x_j)$ in general. Hence uncorrelated does not imply independence.

(3) *Jointly Gaussian.* For $x_i \sim \mathcal{N}(m_i, \sigma_i^2)$ and $x_j \sim \mathcal{N}(m_j, \sigma_j^2)$, the product of marginals is

$$p(x_i)p(x_j) = \frac{1}{2\pi\sigma_i\sigma_j} \exp\left\{-\frac{1}{2} \begin{bmatrix} x_i - m_i \\ x_j - m_j \end{bmatrix}^\top \begin{bmatrix} \sigma_i^2 & 0 \\ 0 & \sigma_j^2 \end{bmatrix}^{-1} \begin{bmatrix} x_i - m_i \\ x_j - m_j \end{bmatrix}\right\}. \quad (93)$$

The joint Gaussian $[x_i, x_j]^\top \sim \mathcal{N}([m_i, m_j]^\top, \mathbf{P}_{xx})$ with $\mathbf{P}_{xx} = \begin{bmatrix} \sigma_i^2 & \sigma_{ij} \\ \sigma_{ji} & \sigma_j^2 \end{bmatrix}$ is

$$p(x_i, x_j) = \frac{1}{2\pi\sqrt{|\mathbf{P}_{xx}|}} \exp\left\{-\frac{1}{2} \begin{bmatrix} x_i - m_i \\ x_j - m_j \end{bmatrix}^\top \mathbf{P}_{xx}^{-1} \begin{bmatrix} x_i - m_i \\ x_j - m_j \end{bmatrix}\right\}. \quad (94)$$

If $\sigma_{ij} = \sigma_{ji} = 0$, then $\mathbf{P}_{xx}$ is diagonal and $p(x_i, x_j) = p(x_i)p(x_j)$. Hence, for (jointly) Gaussian variables, *independent* $\Leftrightarrow$ *uncorrelated*.

1.6.5 The χ^2 distribution

Problem. Let $\mathbf{x} = [x_1, \dots, x_n]^\top$ with $p(\mathbf{x}) = \mathcal{N}(\mathbf{m}_x, \mathbf{P}_{xx})$ and define $q = (\mathbf{x} - \mathbf{m}_x)^\top \mathbf{P}_{xx}^{-1} (\mathbf{x} - \mathbf{m}_x)$. Find the mean and variance of q .

Solution. Factor $\mathbf{P}_{xx} = \mathbf{L}\mathbf{L}^\top$ (Cholesky), so $\mathbf{P}_{xx}^{-1} = \mathbf{L}^{-\top}\mathbf{L}^{-1}$, and define the whitened variable $\mathbf{y} = \mathbf{L}^{-1}(\mathbf{x} - \mathbf{m}_x)$. Then

$$q = (\mathbf{x} - \mathbf{m}_x)^\top \mathbf{L}^{-\top} \mathbf{L}^{-1} (\mathbf{x} - \mathbf{m}_x) \quad (95)$$

$$= \mathbf{y}^\top \mathbf{y} \quad (96)$$

$$= \sum_{i=1}^n y_i^2. \quad (97)$$

The whitened variable has zero mean and identity covariance,

$$\mathbf{m}_y = \mathbb{E}(\mathbf{y}) \quad (98)$$

$$= \mathbf{L}^{-1}(\mathbb{E}(\mathbf{x}) - \mathbf{m}_x) \quad (99)$$

$$= \mathbf{0}, \quad (100)$$

$$\mathbf{P}_{yy} = \mathbb{E}(\mathbf{y}\mathbf{y}^\top) \quad (101)$$

$$= \mathbf{L}^{-1} \mathbb{E}((\mathbf{x} - \mathbf{m}_x)(\mathbf{x} - \mathbf{m}_x)^\top) \mathbf{L}^{-\top} \quad (102)$$

$$= \mathbf{L}^{-1} \mathbf{P}_{xx} \mathbf{L}^{-\top} \quad (103)$$

$$= \mathbf{L}^{-1} \mathbf{L}\mathbf{L}^\top \mathbf{L}^{-\top} \quad (104)$$

$$= \mathbf{I}_n, \quad (105)$$

so $y_i \sim \mathcal{N}(0, 1)$ are independent. Therefore the mean of q is

$$\mathbb{E}(q) = \mathbb{E}(\mathbf{y}^\top \mathbf{y}) \quad (106)$$

$$= \sum_{i=1}^n \mathbb{E}(y_i^2) \quad (107)$$

$$= n. \quad (108)$$

For the variance, with $\mathbb{E}(y_i^2) = 1$ the cross terms vanish by independence, giving

$$\sigma_{qq}^2 = \mathbb{E}((q - n)^2) \quad (109)$$

$$= \mathbb{E}\left(\left(\sum_i (y_i^2 - 1)\right)^2\right) \quad (110)$$

$$= \sum_i \mathbb{E}((y_i^2 - 1)^2) \quad (111)$$

$$= \sum_i (\mathbb{E}(y_i^4) - 2\mathbb{E}(y_i^2) + 1). \quad (112)$$

The fourth moment of a standard normal follows from integration by parts (with $u = y_i^3$, $dv = y_i p(y_i) dy_i$, $v = -p(y_i)$):

$$\mathbb{E}(y_i^4) = \int_{-\infty}^{+\infty} y_i^4 p(y_i) dy_i \quad (113)$$

$$= \underbrace{[-y_i^3 p(y_i)]_{-\infty}^{+\infty}}_{=0} + \int_{-\infty}^{+\infty} 3y_i^2 p(y_i) dy_i \quad (114)$$

$$= 3\mathbb{E}(y_i^2) \quad (115)$$

$$= 3. \quad (116)$$

Hence each term equals $3 - 2 + 1 = 2$, and

$$\sigma_{qq}^2 = \sum_{i=1}^n 2 = 2n. \quad (117)$$

So $q \sim \chi^2(n)$ has mean n and variance $2n$.

2 Lecture 2: Rotation, Lie Groups, SE(3), SE₂(3), and Quaternions

Further reading. This lecture draws on several standard references: matrix Lie groups in Barfoot [1, Ch. 7]; the Lie-group treatment of $SO(3)/SE(3)$ in the GTSAM notes [2], Atanasov's ECE276A slides [3], and the micro Lie theory of Solà, Deray & Atchuthan [4]; and quaternion algebra/kinematics in Trawny & Roumeliotis [5, Sec. 1] and Solà [6, Sec. 1–3].

2.1 Preliminaries: useful series

The matrix exponential and the sine/cosine series underpin everything that follows:

$$e^x = \exp(x) = \sum_{k=0}^{\infty} \frac{1}{k!} x^k \quad (118)$$

$$= I + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \frac{1}{4!}x^4 + \dots, \quad (119)$$

$$\sin(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1} \quad (120)$$

$$= x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \dots, \quad (121)$$

$$\cos(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} x^{2k} \quad (122)$$

$$= 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \dots. \quad (123)$$

For small x (so that $x^2, x^3 \rightarrow 0$), the important first-order approximations are $e^x \approx I + x$, $\sin(x) \approx x$, and $\cos(x) \approx 1$.

2.2 Frames, notation, and transformations

We work with coordinate frames $\{A\}$ and $\{B\}$. A point/vector $\mathbf{L}$ expressed in $\{A\}$ is ${}^A\mathbf{L}$, and in $\{B\}$ is ${}^B\mathbf{L}$. The angle $\theta_{A \rightarrow B}$ is the rotation about the z -axis taking $\{A\}$ to $\{B\}$ (Fig. 1).

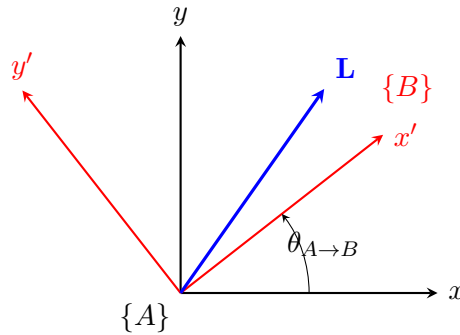


Figure 1: Frame $\{B\}$ is frame $\{A\}$ rotated by $\theta_{A \rightarrow B}$ about the z -axis; the vector $\mathbf{L}$ can be expressed in either frame.

The rotation matrix ${}^A_B\mathbf{R}$ maps a vector from $\{B\}$ to $\{A\}$:

$${}^A\mathbf{L} = {}^A_B\mathbf{R} {}^B\mathbf{L} \quad (124)$$

$$= {}^A_B\mathbf{R}(\theta_{A \rightarrow B}) {}^B\mathbf{L}. \quad (125)$$

Adding the translation ${}^A\mathbf{p}_B$ (the origin of $\{B\}$ expressed in $\{A\}$, Fig. 2) gives the full rigid-body transformation

$$\{A\}, \{B\} \Rightarrow \{{}_B^A\mathbf{R}, {}^A\mathbf{p}_B\} \Rightarrow {}_B^A\mathbf{T} = \begin{bmatrix} {}_B^A\mathbf{R} & {}^A\mathbf{p}_B \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix}. \quad (126)$$

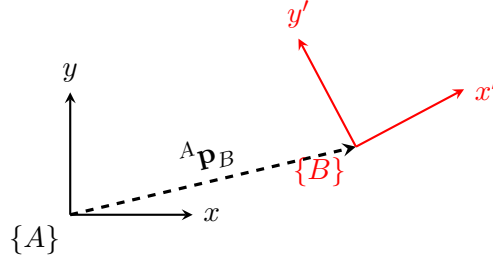


Figure 2: A rigid-body transformation combines the rotation ${}_B^A\mathbf{R}$ with the translation ${}^A\mathbf{p}_B$.

2.3 Defining rotation

2.3.1 Euler angles

With roll θ (about x), pitch r (about y), yaw ψ (about z),

$$\mathbf{R}(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}, \quad \mathbf{R}(r) = \begin{bmatrix} \cos r & 0 & \sin r \\ 0 & 1 & 0 \\ -\sin r & 0 & \cos r \end{bmatrix}, \quad \mathbf{R}(\psi) = \begin{bmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad (127)$$

and the yaw–pitch–roll composition is

$$\mathbf{R} = \mathbf{R}(\psi) \mathbf{R}(r) \mathbf{R}(\theta) \quad (128)$$

$$= \begin{bmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos r & 0 & \sin r \\ 0 & 1 & 0 \\ -\sin r & 0 & \cos r \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}. \quad (129)$$

2.3.2 Infinitesimal rotation and the skew-symmetric matrix

For small angles $\delta\theta, \delta r, \delta\psi \rightarrow 0$ each elementary rotation linearizes ($\cos \rightarrow 1$, $\sin \rightarrow \text{angle}$):

$$\mathbf{R}(\delta\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -\delta\theta \\ 0 & \delta\theta & 1 \end{bmatrix}, \quad \mathbf{R}(\delta r) = \begin{bmatrix} 1 & 0 & \delta r \\ 0 & 1 & 0 \\ -\delta r & 0 & 1 \end{bmatrix}, \quad \mathbf{R}(\delta\psi) = \begin{bmatrix} 1 & -\delta\psi & 0 \\ \delta\psi & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (130)$$

Multiplying the last two and dropping all second-order products ($\delta r \delta\theta \approx 0$ etc.),

$$\mathbf{R}(\delta r) \mathbf{R}(\delta\theta) = \begin{bmatrix} 1 & 0 & \delta r \\ 0 & 1 & 0 \\ -\delta r & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -\delta\theta \\ 0 & \delta\theta & 1 \end{bmatrix} \quad (131)$$

$$= \begin{bmatrix} 1 & 0 & \delta r \\ 0 & 1 & -\delta\theta \\ -\delta r & \delta\theta & 1 \end{bmatrix}, \quad (132)$$

and then

$$\begin{aligned}\mathbf{R} &= \mathbf{R}(\delta\psi)\mathbf{R}(\delta r)\mathbf{R}(\delta\theta) = \begin{bmatrix} 1 & -\delta\psi & 0 \\ \delta\psi & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & \delta r \\ 0 & 1 & -\delta\theta \\ -\delta r & \delta\theta & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & -\delta\psi & \delta r \\ \delta\psi & 1 & -\delta\theta \\ -\delta r & \delta\theta & 1 \end{bmatrix} \end{aligned} \quad (133)$$

$$= \mathbf{I} + \begin{bmatrix} 0 & -\delta\psi & \delta r \\ \delta\psi & 0 & -\delta\theta \\ -\delta r & \delta\theta & 0 \end{bmatrix} \quad (134)$$

$$= \mathbf{I} + [\boldsymbol{\theta}]_{\times}, \quad (135)$$

where the small-angle vector is $\boldsymbol{\theta} = [\delta\theta, \delta r, \delta\psi]^{\top}$. The matrix $[\boldsymbol{\theta}]_{\times}$ is skew-symmetric, $[\boldsymbol{\theta}]_{\times} = -[\boldsymbol{\theta}]_{\times}^{\top}$. This motivates writing a finite rotation as $\mathbf{R}(\boldsymbol{\theta}) \approx \mathbf{I} + [\boldsymbol{\theta}]_{\times} \approx \exp([\boldsymbol{\theta}]_{\times})$, and defining $\text{Exp}(\boldsymbol{\theta}) \triangleq \exp([\boldsymbol{\theta}]_{\times})$. The question is then whether $\mathbf{R}(\boldsymbol{\theta}) = \text{Exp}(\boldsymbol{\theta})$.

2.4 Axis-angle and Rodrigues' formula

Let $\mathbf{k}$ be the unit rotation axis and θ the rotation angle, with $\boldsymbol{\theta} = \theta\mathbf{k} = [\theta_1, \theta_2, \theta_3]^{\top}$, so

$$\mathbf{k} = \begin{bmatrix} k_1 \\ k_2 \\ k_3 \end{bmatrix}, \quad [\mathbf{k}]_{\times} = \begin{bmatrix} 0 & -k_3 & k_2 \\ k_3 & 0 & -k_1 \\ -k_2 & k_1 & 0 \end{bmatrix}, \quad [\boldsymbol{\theta}]_{\times} = \begin{bmatrix} 0 & -\theta_3 & \theta_2 \\ \theta_3 & 0 & -\theta_1 \\ -\theta_2 & \theta_1 & 0 \end{bmatrix}. \quad (136)$$

The skew operator on a unit vector satisfies the cyclic power identities

$$[\mathbf{k}]_{\times}^2 = \mathbf{k}\mathbf{k}^{\top} - \mathbf{I}_3, \quad [\mathbf{k}]_{\times}^3 = -[\mathbf{k}]_{\times}, \quad [\mathbf{k}]_{\times}^4 = -[\mathbf{k}]_{\times}^2, \quad (137)$$

$$[\mathbf{k}]_{\times}^5 = [\mathbf{k}]_{\times}, \quad [\mathbf{k}]_{\times}^6 = [\mathbf{k}]_{\times}^2, \quad [\mathbf{k}]_{\times}^7 = -[\mathbf{k}]_{\times}. \quad (138)$$

Substituting $[\boldsymbol{\theta}]_{\times} = \theta [\mathbf{k}]_{\times}$ into the exponential series term by term,

$$\begin{aligned}\text{Exp}(\boldsymbol{\theta}) &= \exp(\theta [\mathbf{k}]_{\times}) = \mathbf{I} + \theta [\mathbf{k}]_{\times} + \frac{\theta^2}{2!} [\mathbf{k}]_{\times}^2 + \frac{\theta^3}{3!} [\mathbf{k}]_{\times}^3 + \frac{\theta^4}{4!} [\mathbf{k}]_{\times}^4 \\ &\quad + \frac{\theta^5}{5!} [\mathbf{k}]_{\times}^5 + \frac{\theta^6}{6!} [\mathbf{k}]_{\times}^6 + \frac{\theta^7}{7!} [\mathbf{k}]_{\times}^7 + \cdots \\ &= \mathbf{I} + \theta [\mathbf{k}]_{\times} + \frac{\theta^2}{2!} [\mathbf{k}]_{\times}^2 - \frac{\theta^3}{3!} [\mathbf{k}]_{\times} - \frac{\theta^4}{4!} [\mathbf{k}]_{\times}^2 \\ &\quad + \frac{\theta^5}{5!} [\mathbf{k}]_{\times} + \frac{\theta^6}{6!} [\mathbf{k}]_{\times}^2 - \frac{\theta^7}{7!} [\mathbf{k}]_{\times} + \cdots \\ &= \mathbf{I} + \underbrace{\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \cdots \right)}_{\sin \theta} [\mathbf{k}]_{\times} \\ &\quad + \underbrace{\left(\frac{\theta^2}{2!} - \frac{\theta^4}{4!} + \frac{\theta^6}{6!} - \cdots \right)}_{1 - \cos \theta} [\mathbf{k}]_{\times}^2, \end{aligned} \quad (139)$$

which gives **Rodrigues' rotation formula**

$$\boxed{\mathbf{R}(\boldsymbol{\theta}) = \text{Exp}(\boldsymbol{\theta}) = \exp(\theta [\mathbf{k}]_{\times}) = \mathbf{I} + \sin \theta [\mathbf{k}]_{\times} + (1 - \cos \theta) [\mathbf{k}]_{\times}^2.} \quad (140)$$

For a small angle $\delta\theta \rightarrow 0$, $\mathbf{R}(\delta\boldsymbol{\theta}) \approx \mathbf{I} + [\delta\boldsymbol{\theta}]_{\times}$ (small-angle approximation).

2.4.1 Logarithmic map

The log map is the inverse of the exponential map:

$$\log(\mathbf{R}) = [\boldsymbol{\theta}]_{\times}, \quad \text{Log}(\mathbf{R}) = \boldsymbol{\theta}_{3 \times 1}, \quad \theta = \arccos\left(\frac{\text{tr} \mathbf{R} - 1}{2}\right), \quad [\mathbf{k}]_{\times} = \frac{\mathbf{R} - \mathbf{R}^{\top}}{2 \sin \theta}. \quad (141)$$

2.5 SO(3) kinematics

The two directions of the maps are: from the vector space $\mathbb{R}^3$ via the Lie algebra to the Lie group, $\boldsymbol{\theta} \rightarrow [\boldsymbol{\theta}]_{\times} \rightarrow \mathbf{R} = \text{Exp}(\boldsymbol{\theta})$; and back, $\mathbf{R} \rightarrow [\boldsymbol{\theta}]_{\times} \rightarrow \boldsymbol{\theta} = \text{Log}(\mathbf{R})$.

(1) Angular-velocity kinematics. Differentiating the orthogonality constraint $\mathbf{R}\mathbf{R}^{\top} = \mathbf{I}_3$,

$$\frac{d}{dt}(\mathbf{R}\mathbf{R}^{\top}) = \mathbf{0} \Rightarrow \dot{\mathbf{R}}\mathbf{R}^{\top} + \mathbf{R}\dot{\mathbf{R}}^{\top} = \mathbf{0} \Rightarrow \dot{\mathbf{R}}\mathbf{R}^{\top} = -\mathbf{R}\dot{\mathbf{R}}^{\top} = -(\dot{\mathbf{R}}\mathbf{R}^{\top})^{\top}, \quad (142)$$

so $\dot{\mathbf{R}}\mathbf{R}^{\top}$ is skew-symmetric. Writing it as $[\boldsymbol{\omega}]_{\times}$ with $[\boldsymbol{\omega}]_{\times} = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix}$,

$$\dot{\mathbf{R}}\mathbf{R}^{\top} = [\boldsymbol{\omega}]_{\times} \Rightarrow \dot{\mathbf{R}} = [\boldsymbol{\omega}]_{\times} \mathbf{R}. \quad (143)$$

(2) Equivalent limit form. With $\mathbf{R}(t + \Delta t) = \mathbf{R}(t) \exp([\boldsymbol{\omega}]_{\times} \Delta t)$,

$$\dot{\mathbf{R}}(t) = \lim_{\Delta t \rightarrow 0} \frac{\mathbf{R}(t + \Delta t) - \mathbf{R}(t)}{\Delta t} \quad (144)$$

$$= \lim_{\Delta t \rightarrow 0} \frac{\mathbf{R}(t) \exp([\boldsymbol{\omega}]_{\times} \Delta t) - \mathbf{R}(t)}{\Delta t} \quad (145)$$

$$= \mathbf{R}(t) \lim_{\Delta t \rightarrow 0} \frac{\exp([\boldsymbol{\omega}]_{\times} \Delta t) - \mathbf{I}}{\Delta t} \quad (146)$$

$$= \mathbf{R}(t) [\boldsymbol{\omega}]_{\times}. \quad (147)$$

(3) Integration. With $\mathbf{R}_i = \exp([\boldsymbol{\omega}_i]_{\times} \Delta t_i)$, composing increments gives

$$\mathbf{R}_{1 \rightarrow k} = \mathbf{R}_k \mathbf{R}_{k-1} \cdots \mathbf{R}_2 \mathbf{R}_1 \quad (148)$$

$$= \exp([\boldsymbol{\omega}_k]_{\times} \Delta t_k) \cdots \exp([\boldsymbol{\omega}_1]_{\times} \Delta t_1). \quad (149)$$

(4) Interpolation. For ${}^G_{t_1}\mathbf{R}, {}^G_{t_2}\mathbf{R}$ and $t \in [t_1, t_2]$, define the relative rotation and its angle, and the interpolation ratio:

$${}^{t_1}_{t_2}\mathbf{R} = {}^G_{t_1}\mathbf{R}^{\top} {}^G_{t_2}\mathbf{R}, \quad (150)$$

$${}^{t_1}_{t_2}\boldsymbol{\theta} = \text{Log}({}^{t_1}_{t_2}\mathbf{R}) = \text{Log}({}^G_{t_1}\mathbf{R}^{\top} {}^G_{t_2}\mathbf{R}), \quad (151)$$

$$\lambda = \frac{t - t_1}{t_2 - t_1}. \quad (152)$$

Then

$${}^G_t\mathbf{R} = {}^G_{t_1}\mathbf{R} {}^{t_1}_t\mathbf{R} \quad (153)$$

$$= {}^G_{t_1}\mathbf{R} \text{Exp}(\lambda {}^{t_1}_{t_2}\boldsymbol{\theta}). \quad (154)$$

(5) Right Jacobian. For a rotation $\mathbf{R}(\boldsymbol{\theta}) = \text{Exp}(\boldsymbol{\theta})$, perturb the angle by $\delta\boldsymbol{\theta} \in \mathbb{R}^3$ and let $\delta\boldsymbol{\varphi}$ be the corresponding small rotation in $\mathfrak{so}(3)$:

$$\mathbf{R}(\boldsymbol{\theta}) \text{Exp}(\delta\boldsymbol{\varphi}) = \mathbf{R}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}) \Rightarrow \text{Exp}(\boldsymbol{\theta}) \text{Exp}(\delta\boldsymbol{\varphi}) = \text{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}). \quad (155)$$

Solving for $\delta\boldsymbol{\varphi}$,

$$\text{Exp}(\delta\boldsymbol{\varphi}) = \text{Exp}(\boldsymbol{\theta})^{-1} \text{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}) \Rightarrow \delta\boldsymbol{\varphi} = \text{Log}(\text{Exp}(\boldsymbol{\theta})^{-1} \text{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta})), \quad (156)$$

and the *right Jacobian* is the derivative of $\delta\boldsymbol{\varphi}$ with respect to $\delta\boldsymbol{\theta}$:

$$\mathbf{J}_r(\boldsymbol{\theta}) \triangleq \frac{\partial \delta\boldsymbol{\varphi}}{\partial \delta\boldsymbol{\theta}} = \lim_{\delta\boldsymbol{\theta} \rightarrow 0} \frac{\text{Log}(\text{Exp}(\boldsymbol{\theta})^{-1} \text{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}))}{\delta\boldsymbol{\theta}}. \quad (157)$$

Equivalently, $\text{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}) = \text{Exp}(\boldsymbol{\theta}) \text{Exp}(\mathbf{J}_r(\boldsymbol{\theta}) \delta\boldsymbol{\theta})$, and inverting,

$$\text{Exp}(\boldsymbol{\theta}) \text{Exp}(\delta\boldsymbol{\varphi}) = \text{Exp}(\boldsymbol{\theta} + \mathbf{J}_r^{-1}(\boldsymbol{\theta}) \delta\boldsymbol{\varphi}) \Rightarrow \text{Log}(\text{Exp}(\boldsymbol{\theta}) \text{Exp}(\delta\boldsymbol{\varphi})) = \boldsymbol{\theta} + \mathbf{J}_r^{-1}(\boldsymbol{\theta}) \delta\boldsymbol{\varphi}, \quad (158)$$

with the closed form

$$\mathbf{J}_r(\boldsymbol{\theta}) = \mathbf{I} - \frac{1 - \cos\|\boldsymbol{\theta}\|}{\|\boldsymbol{\theta}\|^2} [\boldsymbol{\theta}]_{\times} + \frac{\|\boldsymbol{\theta}\| - \sin\|\boldsymbol{\theta}\|}{\|\boldsymbol{\theta}\|^3} [\boldsymbol{\theta}]_{\times}^2. \quad (159)$$

(6) Left Jacobian. Applying the perturbation on the left instead defines the left Jacobian:

$$\text{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}) = \text{Exp}(\boldsymbol{\theta}) \text{Exp}(\mathbf{J}_r(\boldsymbol{\theta}) \delta\boldsymbol{\theta}) = \text{Exp}(\mathbf{J}_l(\boldsymbol{\theta}) \delta\boldsymbol{\theta}) \text{Exp}(\boldsymbol{\theta}). \quad (160)$$

Using the adjoint identity $\text{Exp}(\boldsymbol{\theta}) \text{Exp}(\mathbf{x}) \text{Exp}(\boldsymbol{\theta})^{-1} = \text{Exp}(\mathbf{R}(\boldsymbol{\theta})\mathbf{x})$ on the middle expression,

$$\text{Exp}(\boldsymbol{\theta}) \text{Exp}(\mathbf{J}_r \delta\boldsymbol{\theta}) = \text{Exp}(\mathbf{R}(\boldsymbol{\theta})\mathbf{J}_r \delta\boldsymbol{\theta}) \text{Exp}(\boldsymbol{\theta}) \Rightarrow \mathbf{J}_l(\boldsymbol{\theta}) = \mathbf{R}(\boldsymbol{\theta}) \mathbf{J}_r(\boldsymbol{\theta}). \quad (161)$$

(7) Sign symmetry. From $\text{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}) = \text{Exp}(\boldsymbol{\theta}) \text{Exp}(\mathbf{J}_r(\boldsymbol{\theta})\delta\boldsymbol{\theta})$, replacing $\boldsymbol{\theta} \rightarrow -\boldsymbol{\theta}$, $\delta\boldsymbol{\theta} \rightarrow -\delta\boldsymbol{\theta}$,

$$\text{Exp}(-\boldsymbol{\theta} - \delta\boldsymbol{\theta}) = \text{Exp}(-\mathbf{J}_l(-\boldsymbol{\theta})\delta\boldsymbol{\theta}) \text{Exp}(-\boldsymbol{\theta}) = \text{Exp}(-\mathbf{J}_r(\boldsymbol{\theta})\delta\boldsymbol{\theta}) \text{Exp}(-\boldsymbol{\theta}) \Rightarrow \mathbf{J}_l(-\boldsymbol{\theta}) = \mathbf{J}_r(\boldsymbol{\theta}). \quad (162)$$

(8) Summary. Collecting (5)–(7),

$$\text{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}) = \text{Exp}(\boldsymbol{\theta}) \text{Exp}(\mathbf{J}_r(\boldsymbol{\theta})\delta\boldsymbol{\theta}), \quad \mathbf{J}_l(\boldsymbol{\theta}) = \mathbf{R}(\boldsymbol{\theta})\mathbf{J}_r(\boldsymbol{\theta}), \quad \mathbf{J}_l(-\boldsymbol{\theta}) = \mathbf{J}_r(\boldsymbol{\theta}). \quad (163)$$

(9) Identity $[\mathbf{R}\boldsymbol{\omega}]_{\times} = \mathbf{R} [\boldsymbol{\omega}]_{\times} \mathbf{R}^{\top}$. For any vector $\mathbf{v}$, using $\mathbf{R}(\mathbf{u} \times \mathbf{v}) = (\mathbf{R}\mathbf{u}) \times (\mathbf{R}\mathbf{v})$ and $\mathbf{R}^{\top}\mathbf{R} = \mathbf{I}$,

$$[\mathbf{R}\mathbf{u}]_{\times} (\mathbf{R}\mathbf{v}) = \mathbf{R} ([\mathbf{u}]_{\times} \mathbf{v}) \quad (164)$$

$$= \mathbf{R} ([\mathbf{u}]_{\times} \mathbf{R}^{\top} \mathbf{R}\mathbf{v}) \quad (165)$$

$$= \mathbf{R} [\mathbf{u}]_{\times} \mathbf{R}^{\top} (\mathbf{R}\mathbf{v}). \quad (166)$$

Since this holds for all $\mathbf{R}\mathbf{v}$, we conclude $[\mathbf{R}\mathbf{u}]_{\times} = \mathbf{R} [\mathbf{u}]_{\times} \mathbf{R}^{\top}$.

Homework. Prove $[\mathbf{R}\boldsymbol{\omega}]_{\times} = \mathbf{R} [\boldsymbol{\omega}]_{\times} \mathbf{R}^{\top}$.

(10) Identity $\text{Exp}(\mathbf{R}\boldsymbol{\varphi}) = \mathbf{R} \text{Exp}(\boldsymbol{\varphi}) \mathbf{R}^\top$. Using $[\mathbf{R}\boldsymbol{\varphi}]_\times^k = \mathbf{R} [\boldsymbol{\varphi}]_\times^k \mathbf{R}^\top$ (which follows from (9), since $[\mathbf{R}\boldsymbol{\varphi}]_\times^2 = \mathbf{R} [\boldsymbol{\varphi}]_\times \mathbf{R}^\top \mathbf{R} [\boldsymbol{\varphi}]_\times \mathbf{R}^\top = \mathbf{R} [\boldsymbol{\varphi}]_\times^2 \mathbf{R}^\top$, etc.),

$$\begin{aligned} \text{Exp}(\mathbf{R}\boldsymbol{\varphi}) &= \mathbf{I} + [\mathbf{R}\boldsymbol{\varphi}]_\times + \frac{1}{2!} [\mathbf{R}\boldsymbol{\varphi}]_\times^2 + \frac{1}{3!} [\mathbf{R}\boldsymbol{\varphi}]_\times^3 + \cdots \\ &= \mathbf{R}\mathbf{R}^\top + \mathbf{R} [\boldsymbol{\varphi}]_\times \mathbf{R}^\top + \frac{1}{2!} \mathbf{R} [\boldsymbol{\varphi}]_\times^2 \mathbf{R}^\top + \cdots \\ &= \mathbf{R}(\mathbf{I} + [\boldsymbol{\varphi}]_\times + \frac{1}{2!} [\boldsymbol{\varphi}]_\times^2 + \frac{1}{3!} [\boldsymbol{\varphi}]_\times^3 + \cdots) \mathbf{R}^\top \end{aligned} \quad (167)$$

$$= \mathbf{R} \text{Exp}(\boldsymbol{\varphi}) \mathbf{R}^\top. \quad (168)$$

Homework. Prove $\text{Exp}(\mathbf{R}\boldsymbol{\varphi}) = \mathbf{R} \text{Exp}(\boldsymbol{\varphi}) \mathbf{R}^\top$.

2.5.1 Worked applications

(11) Recovering ${}^C_I\mathbf{R}$ from vector pairs. Given camera frame $\{C\}$ and IMU frame $\{I\}$ with ${}^C\mathbf{v}_1 = {}^C_I\mathbf{R} {}^I\mathbf{v}_1$ and ${}^C\mathbf{v}_2 = {}^C_I\mathbf{R} {}^I\mathbf{v}_2$, the trick is to stack three independent directions (the third from the cross product) into an invertible matrix:

$$[{}^C\mathbf{v}_1 \ {}^C\mathbf{v}_2 \ {}^C\mathbf{v}_1 \times {}^C\mathbf{v}_2] = {}^C_I\mathbf{R} [{}^I\mathbf{v}_1 \ {}^I\mathbf{v}_2 \ {}^I\mathbf{v}_1 \times {}^I\mathbf{v}_2]. \quad (169)$$

(12) Given instead relative rotations ${}^{C_1}_{C_2}\mathbf{R} = {}^C_I\mathbf{R} {}^{I_1}_{I_2}\mathbf{R} {}^C_I\mathbf{R}^\top$, recover ${}^C_I\mathbf{R}$ from two such pairs.

(13) Discussion. For (11)/(12): What is the minimum number of data pairs needed? What requirements must the pairs satisfy? What if more than the minimum is available?

(14) Averaging noisy rotations. Given n measurements $\mathbf{R}_{m_i} = \mathbf{R} \exp([\mathbf{n}_i]_\times)$ with $\mathbf{n}_i \sim \mathcal{N}(\mathbf{0}_{3 \times 1}, \mathbf{P}_i)$, estimate $\mathbf{R}$ and its covariance.

2.6 Groups: SO(3) and SE(3)

SO(3) — special orthogonal group.

$$SO(3) = \{\mathbf{R} \in \mathbb{R}^{3 \times 3} \mid \mathbf{R}^\top \mathbf{R} = \mathbf{I}_3, \det(\mathbf{R}) = 1\}. \quad (170)$$

It satisfies the four group axioms: *closure* ($\mathbf{R}_1\mathbf{R}_2 \in SO(3)$), *identity* ($\mathbf{I}_3 \in SO(3)$), *inverse* ($\mathbf{R}^{-1} = \mathbf{R}^\top \in SO(3)$), and *associativity* ($(\mathbf{R}_1\mathbf{R}_2)\mathbf{R}_3 = \mathbf{R}_1(\mathbf{R}_2\mathbf{R}_3)$).

SE(3) — special Euclidean group.

$$SE(3) = \left\{ \mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix} \mid \mathbf{R} \in SO(3), \mathbf{p} \in \mathbb{R}^3 \right\}. \quad (171)$$

The four axioms: *closure*

$$\mathbf{T}_1\mathbf{T}_2 = \begin{bmatrix} \mathbf{R}_1 & \mathbf{p}_1 \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{R}_2 & \mathbf{p}_2 \\ \mathbf{0} & 1 \end{bmatrix} \quad (172)$$

$$= \begin{bmatrix} \mathbf{R}_1\mathbf{R}_2 & \mathbf{R}_1\mathbf{p}_2 + \mathbf{p}_1 \\ \mathbf{0} & 1 \end{bmatrix} \in SE(3), \quad (173)$$

identity $\begin{bmatrix} \mathbf{I}_3 & \mathbf{0}_{3 \times 1} \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix}$, *inverse*

$$\begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{R}^\top & -\mathbf{R}^\top \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \in SE(3), \quad (174)$$

and *associativity* $(\mathbf{T}_1\mathbf{T}_2)\mathbf{T}_3 = \mathbf{T}_1(\mathbf{T}_2\mathbf{T}_3)$.

2.6.1 Matrix Lie groups

A *group* is a set with an operation satisfying closure, associativity, identity, and invertibility. A *Lie group* is a group that is also a smooth (differentiable) manifold whose group operation is smooth. A *matrix Lie group* has matrices as its elements. The maps between the vector space, the Lie algebra, and the Lie group are

$$\mathbb{R}^m \xrightarrow{\wedge \text{ (wedge)}} \mathfrak{so}(n) \xrightarrow{\exp} SO(n), \quad SO(n) \xrightarrow{\log} \mathfrak{so}(n) \xrightarrow{\vee \text{ (vee)}} \mathbb{R}^m, \quad (175)$$

and for short we write $\mathbf{X} = \text{Exp}(\mathbf{x}) = \exp(\mathbf{x}^\wedge)$ and $\mathbf{x} = \text{Log}(\mathbf{X}) = \log(\mathbf{X})^\vee$. For $SO(3)$ specifically, $\boldsymbol{\theta} \in \mathbb{R}^3 \rightarrow [\boldsymbol{\theta}]_\times = \boldsymbol{\theta}^\wedge \rightarrow \mathbf{R} = \exp(\boldsymbol{\theta}^\wedge)$, and $\mathbf{R} \in SO(3) \rightarrow [\boldsymbol{\theta}]_\times = \log(\mathbf{R}) \rightarrow \boldsymbol{\theta} = \log(\mathbf{R})^\vee$.

2.6.2 Why this (manifold) machinery?

It *simplifies* estimation: the core idea is to use a smooth manifold to model parameters that are awkward in plain vector space, which makes optimization, interpolation, integration, and differentiation easier or even possible (Fig. 3). For example, a raw angle $\theta \in [-\pi, \pi]$ has a wrap-around cut; the circle/sphere is smooth, and the tangent line at a point models the small change / error state used in optimization.

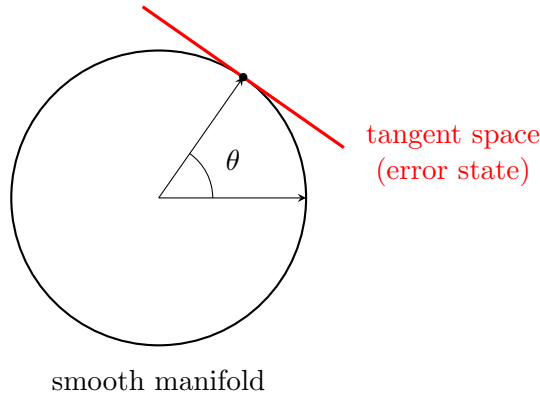


Figure 3: A smooth manifold (here a circle/sphere) avoids the wrap-around of a raw angle; the red tangent line is the local vector space used to model derivatives and error states for optimization.

For $SO(3)$ the left Jacobian also appears directly in the exponential expansion:

$$\mathbf{R}(\boldsymbol{\theta}) = \text{Exp}(\boldsymbol{\theta}) \quad (176)$$

$$= \mathbf{I}_3 + [\boldsymbol{\theta}]_\times \mathbf{J}_l(\boldsymbol{\theta}) = \mathbf{I}_3 + \boldsymbol{\theta}^\wedge \mathbf{J}_l(\boldsymbol{\theta}), \quad (177)$$

$$\mathbf{J}_l(\boldsymbol{\theta}) = \sum_{n=0}^{\infty} \frac{1}{(n+1)!} (\boldsymbol{\theta}^\wedge)^n. \quad (178)$$

2.7 SE(3)

Exp / Log. The $SE(3)$ tangent vector uses $\boldsymbol{\rho}$ for its translation component, $\boldsymbol{\xi} = \begin{bmatrix} \boldsymbol{\theta} \\ \boldsymbol{\rho} \end{bmatrix} \in \mathbb{R}^6$, which is *distinct* from the group translation $\mathbf{p}$ in $\mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix}$; the two are related by $\mathbf{p} = \mathbf{J}_l(\boldsymbol{\theta})\boldsymbol{\rho}$. With

the 4×4 algebra element $\xi^\wedge = \begin{bmatrix} [\boldsymbol{\theta}]_\times & \boldsymbol{\rho} \\ \mathbf{0}_{1 \times 3} & 0 \end{bmatrix}$,

$$\mathbf{T} = \text{Exp}(\xi) = \exp \begin{bmatrix} [\boldsymbol{\theta}]_\times & \boldsymbol{\rho} \\ \mathbf{0} & 0 \end{bmatrix} \quad (179)$$

$$= \begin{bmatrix} \exp([\boldsymbol{\theta}]_\times) & \mathbf{J}_l(\boldsymbol{\theta}) \boldsymbol{\rho} \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix} \quad (180)$$

$$= \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix}, \quad \mathbf{p} = \mathbf{J}_l(\boldsymbol{\theta}) \boldsymbol{\rho}, \quad (181)$$

and inverting,

$$\xi = \text{Log}(\mathbf{T})^\vee, \quad \log(\mathbf{T}) = \log \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \quad (182)$$

$$= \begin{bmatrix} \log(\mathbf{R}) & \mathbf{J}_l^{-1}(\boldsymbol{\theta}) \mathbf{p} \\ \mathbf{0} & 0 \end{bmatrix} \Rightarrow \xi = \begin{bmatrix} \boldsymbol{\theta} \\ \mathbf{J}_l^{-1}(\boldsymbol{\theta}) \mathbf{p} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\theta} \\ \boldsymbol{\rho} \end{bmatrix}. \quad (183)$$

Discussion. Prove $\exp \left(\begin{bmatrix} [\boldsymbol{\theta}]_\times & \boldsymbol{\rho} \\ \mathbf{0} & 0 \end{bmatrix} \right) = \begin{bmatrix} \mathbf{R} & \mathbf{J}_l(\boldsymbol{\theta}) \boldsymbol{\rho} \\ \mathbf{0} & 1 \end{bmatrix}$.

Left/right Jacobians. With $\text{Exp}(\xi + \delta\xi) = \text{Exp}(\xi) \text{Exp}(\mathbf{J}_R(\xi) \delta\xi) = \text{Exp}(\mathbf{J}_L(\xi) \delta\xi) \text{Exp}(\xi)$,

$$\mathbf{J}_R(\xi) = \begin{bmatrix} \mathbf{J}_R(\boldsymbol{\theta}) & \mathbf{0} \\ \mathbf{Q}_R(\xi) & \mathbf{J}_R(\boldsymbol{\theta}) \end{bmatrix}_{6 \times 6}, \quad \mathbf{J}_L(\xi) = \begin{bmatrix} \mathbf{J}_L(\boldsymbol{\theta}) & \mathbf{0} \\ \mathbf{Q}_L(\xi) & \mathbf{J}_L(\boldsymbol{\theta}) \end{bmatrix}_{6 \times 6}, \quad (184)$$

and to first order, with $\xi^\wedge = \begin{bmatrix} [\boldsymbol{\theta}]_\times & \mathbf{0} \\ [\boldsymbol{\rho}]_\times & [\boldsymbol{\theta}]_\times \end{bmatrix}_{6 \times 6}$,

$$\mathbf{J}_L(\xi) \approx \mathbf{I}_6 + \frac{1}{2} \xi^\wedge, \quad \mathbf{J}_R(\xi) \approx \mathbf{I}_6 - \frac{1}{2} \xi^\wedge. \quad (185)$$

Homework. Find $\mathbf{Q}_L$ (and $\mathbf{Q}_R$).

Error states: SE(3) vs SO(3) + $\mathbb{R}^3$. Perturb $\mathbf{T}$ on the right by the SE(3) tangent $\delta\xi = [\delta\boldsymbol{\theta}^\top, \delta\boldsymbol{\rho}^\top]^\top$. Expanding with $\mathbf{J}_l(\delta\boldsymbol{\theta}) \approx \mathbf{I} + \frac{1}{2} [\delta\boldsymbol{\theta}]_\times$ and $\delta\mathbf{R} = \text{Exp}(\delta\boldsymbol{\theta})$,

$$\mathbf{T} \delta\mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \text{Exp}(\delta\boldsymbol{\theta}) & \mathbf{J}_l(\delta\boldsymbol{\theta}) \delta\boldsymbol{\rho} \\ \mathbf{0} & 1 \end{bmatrix} \quad (186)$$

$$= \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \delta\mathbf{R} & (\mathbf{I} + \frac{1}{2} [\delta\boldsymbol{\theta}]_\times) \delta\boldsymbol{\rho} \\ \mathbf{0} & 1 \end{bmatrix} \quad (187)$$

$$\approx \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \delta\mathbf{R} & \delta\boldsymbol{\rho} \\ \mathbf{0} & 1 \end{bmatrix} \quad (188)$$

$$= \begin{bmatrix} \mathbf{R} \delta\mathbf{R} & \mathbf{R} \delta\boldsymbol{\rho} + \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix}. \quad (189)$$

If instead we treat the pose as $SO(3) + \mathbb{R}^3$ (rotation and translation perturbed separately, $\mathbf{p} \rightarrow \mathbf{p} + \delta\mathbf{p}$),

$$\mathbf{T} = \begin{bmatrix} \mathbf{R}\delta\mathbf{R} & \mathbf{p} + \delta\mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \quad (190)$$

$$= \begin{bmatrix} \mathbf{R}\delta\mathbf{R} & \mathbf{p} + \mathbf{R}\mathbf{R}^\top \delta\mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \quad (191)$$

$$= \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \delta\mathbf{R} & \mathbf{R}^\top \delta\mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix}. \quad (192)$$

Comparing the two conventions gives the relation between the error states ($SE(3)$ tangent translation $\delta\boldsymbol{\rho}$ vs. naive position error $\delta\mathbf{p}$),

$$\begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \delta\mathbf{R} & \delta\mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix}_{SO(3)+\mathbb{R}^3} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \delta\mathbf{R} & \delta\boldsymbol{\rho} \\ \mathbf{0} & 1 \end{bmatrix}_{SE(3)} \Rightarrow \begin{cases} \delta\boldsymbol{\rho} = \mathbf{R}^\top \delta\mathbf{p} \\ \delta\mathbf{R}_{SE(3)} = \delta\mathbf{R} \end{cases}. \quad (193)$$

Adjoint. The adjoint transfers algebra elements across the group:

$$SO(3) : \mathbf{R} \text{Exp}(\boldsymbol{\varphi}) \mathbf{R}^\top = \text{Exp}(\mathbf{R}\boldsymbol{\varphi}) \Leftrightarrow \text{Ad}_{\mathbf{R}} = \mathbf{R}, \quad SE(3) : \mathbf{T} \text{Exp}(\boldsymbol{\xi}) \mathbf{T}^{-1} = \text{Exp}(\text{Ad}_{\mathbf{T}} \boldsymbol{\xi}), \quad (194)$$

with $\text{Ad}_{\mathbf{T}} = \begin{bmatrix} \mathbf{R} & \mathbf{0} \\ [\mathbf{p}]_{\times} \mathbf{R} & \mathbf{R} \end{bmatrix}_{6 \times 6}$. The adjoint is itself a Lie group, with $\text{Ad}_{\mathbf{T}} = \exp(\boldsymbol{\xi}^\wedge)$ and $\log(\text{Ad}_{\mathbf{T}})^\vee = \boldsymbol{\xi}$.

Homework. Prove $\mathbf{T} \text{Exp}(\boldsymbol{\xi}) \mathbf{T}^{-1} = \text{Exp}(\text{Ad}_{\mathbf{T}} \boldsymbol{\xi})$.

2.7.1 More operations

(1) Interpolation. For ${}^G_{t_1}\mathbf{T}, {}^G_{t_2}\mathbf{T}$, $t \in [t_1, t_2]$, $\lambda = \frac{t-t_1}{t_2-t_1}$: in $SE(3)$, with ${}^{t_1}_{t_2}\boldsymbol{\xi} = \text{Log}({}^G_{t_1}\mathbf{T}^{-1} {}^G_{t_2}\mathbf{T})$,

$${}^G_t\mathbf{T} = {}^G_{t_1}\mathbf{T} \text{Exp}(\lambda {}^{t_1}_{t_2}\boldsymbol{\xi}); \quad (195)$$

in $SO(3) + \mathbb{R}^3$, the rotation interpolates on the manifold and the position linearly:

$${}^G_t\mathbf{R} = {}^G_{t_1}\mathbf{R} \text{Exp}(\lambda \text{Log}({}^G_{t_1}\mathbf{R}^\top {}^G_{t_2}\mathbf{R})), \quad {}^G_t\mathbf{p} = {}^G_{t_1}\mathbf{p} + \lambda({}^G_{t_2}\mathbf{p} - {}^G_{t_1}\mathbf{p}). \quad (196)$$

(2) Time derivative. For $SE(3)$, with $\mathbf{T} = \mathbf{T} \text{Exp}([\delta\boldsymbol{\theta}; \delta\boldsymbol{\rho}]^\wedge)$ and body velocity, expand the limit using $\exp([\delta\boldsymbol{\theta}]_\times) - \mathbf{I}_3 \approx [\delta\boldsymbol{\theta}]_\times$ and $\mathbf{J}_l(\delta\boldsymbol{\theta}) \approx \mathbf{I} + \frac{1}{2}[\delta\boldsymbol{\theta}]_\times$:

$$\frac{d\mathbf{T}}{dt} = \lim_{\delta t \rightarrow 0} \frac{\mathbf{T}(t + \delta t) - \mathbf{T}(t)}{\delta t} \quad (197)$$

$$= \lim_{\delta t \rightarrow 0} \frac{1}{\delta t} \left(\begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \exp([\delta\boldsymbol{\theta}]_\times) & \mathbf{J}_l(\delta\boldsymbol{\theta})\delta\boldsymbol{\rho} \\ \mathbf{0} & 1 \end{bmatrix} - \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \right) \quad (198)$$

$$= \lim_{\delta t \rightarrow 0} \frac{1}{\delta t} \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \exp([\delta\boldsymbol{\theta}]_\times) - \mathbf{I}_3 & \mathbf{J}_l(\delta\boldsymbol{\theta})\delta\boldsymbol{\rho} \\ \mathbf{0} & 0 \end{bmatrix} \quad (199)$$

$$= \lim_{\delta t \rightarrow 0} \frac{1}{\delta t} \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} [\delta\boldsymbol{\theta}]_\times & \delta\boldsymbol{\rho} \\ \mathbf{0} & 0 \end{bmatrix} \quad (200)$$

$$= \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} [\boldsymbol{\omega}]_\times & \mathbf{v} \\ \mathbf{0} & 0 \end{bmatrix} \quad (201)$$

$$= \begin{bmatrix} \mathbf{R}[\boldsymbol{\omega}]_\times & \mathbf{R}\mathbf{v} \\ \mathbf{0} & 0 \end{bmatrix}, \quad (202)$$

where $\mathbf{v}$ is the body velocity. For $SO(3) + \mathbb{R}^3$ the same computation with

$$\mathbf{T} = \begin{bmatrix} \mathbf{R} \exp([\delta\boldsymbol{\theta}]_\times) & \mathbf{p} + \delta\mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \quad (203)$$

gives

$$\frac{d\mathbf{T}}{dt} = \begin{bmatrix} \mathbf{R}[\boldsymbol{\omega}]_\times & {}^G\mathbf{v} \\ \mathbf{0} & 0 \end{bmatrix}, \quad (204)$$

with ${}^G\mathbf{v}$ the global velocity.

(3) Calibration. For IMU frame $\{I\}$ and camera frame $\{C\}$,

$$\begin{matrix} C_1 \\ C_2 \end{matrix} \mathbf{T} = \begin{matrix} C \\ I \end{matrix} \mathbf{T} \begin{matrix} I_1 \\ I_2 \end{matrix} \mathbf{T} \begin{matrix} C \\ I \end{matrix} \mathbf{T}^{-1}, \quad \text{Exp}(\begin{matrix} C_1 \\ C_2 \end{matrix} \boldsymbol{\xi}) = \text{Exp}(\text{Ad}_{(\begin{matrix} C \\ I \end{matrix} \mathbf{T})} \begin{matrix} I_1 \\ I_2 \end{matrix} \boldsymbol{\xi}) \Rightarrow \begin{matrix} C_1 \\ C_2 \end{matrix} \boldsymbol{\xi} = \text{Ad}_{(\begin{matrix} C \\ I \end{matrix} \mathbf{T})} \begin{matrix} I_1 \\ I_2 \end{matrix} \boldsymbol{\xi}. \quad (205)$$

Homework. Compute the Jacobians $\partial \begin{matrix} C_1 \\ C_2 \end{matrix} \boldsymbol{\xi} / \partial \begin{matrix} I_1 \\ I_2 \end{matrix} \boldsymbol{\xi}$ and $\partial \begin{matrix} C_1 \\ C_2 \end{matrix} \boldsymbol{\xi} / \partial \begin{matrix} C \\ I \end{matrix} \boldsymbol{\xi}$ (and the corresponding $SO(3) + \mathbb{R}^3$ versions).

2.8 $SE_2(3)$

Augmenting the pose with a velocity column gives the 5×5 group element

$$\boldsymbol{\Upsilon} = \begin{bmatrix} \mathbf{R} & \mathbf{p} & \mathbf{v} \\ \mathbf{0}_{1 \times 3} & 1 & 0 \\ \mathbf{0}_{1 \times 3} & 0 & 1 \end{bmatrix}, \quad \boldsymbol{\xi} = \begin{bmatrix} \boldsymbol{\theta} \\ \boldsymbol{\rho} \\ \boldsymbol{\nu} \end{bmatrix} \rightarrow \boldsymbol{\xi}^\wedge = \begin{bmatrix} [\boldsymbol{\theta}]_\times & \boldsymbol{\rho} & \boldsymbol{\nu} \\ \mathbf{0} & 0 & 0 \\ \mathbf{0} & 0 & 0 \end{bmatrix}_{5 \times 5}, \quad (206)$$

where (as in $SE(3)$) the tangent components $\boldsymbol{\rho}, \boldsymbol{\nu}$ are distinct from the group position/velocity $\mathbf{p} = \mathbf{J}_l(\boldsymbol{\theta})\boldsymbol{\rho}$, $\mathbf{v} = \mathbf{J}_l(\boldsymbol{\theta})\boldsymbol{\nu}$. The exponential, logarithm, and adjoint follow the same pattern as $SE(3)$:

$$\text{Exp}(\boldsymbol{\xi}) = \begin{bmatrix} \mathbf{R} & \mathbf{J}_l(\boldsymbol{\theta})\boldsymbol{\rho} & \mathbf{J}_l(\boldsymbol{\theta})\boldsymbol{\nu} \\ \mathbf{0} & 1 & 0 \\ \mathbf{0} & 0 & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R} & \mathbf{p} & \mathbf{v} \\ \mathbf{0} & 1 & 0 \\ \mathbf{0} & 0 & 1 \end{bmatrix}, \quad (207)$$

$$\text{Log}(\boldsymbol{\Upsilon})^\vee = \begin{bmatrix} \boldsymbol{\theta} \\ \mathbf{J}_l^{-1}(\boldsymbol{\theta})\mathbf{p} \\ \mathbf{J}_l^{-1}(\boldsymbol{\theta})\mathbf{v} \end{bmatrix}, \quad \text{Ad}_{\boldsymbol{\Upsilon}} = \begin{bmatrix} \mathbf{R} & \mathbf{0} & \mathbf{0} \\ [\mathbf{p}]_\times \mathbf{R} & \mathbf{R} & \mathbf{0} \\ [\mathbf{v}]_\times \mathbf{R} & \mathbf{0} & \mathbf{R} \end{bmatrix}. \quad (208)$$

2.9 Quaternions

Quaternions are presented last because $SO(3)$, $SE(3)$, and the Lie-group view already cover most needs. There are two main conventions — *Hamilton* (right-handed) and *JPL* (left-handed); we use the **Hamilton** convention. For a thorough treatment see Solà [6] (Hamilton) and Trawny & Roumeliotis [5] (JPL).

A quaternion and its imaginary units satisfy

$$Q = q_w + q_x i + q_y j + q_z k, \quad i^2 = j^2 = k^2 = ijk = -1, \quad ij = k, \quad jk = i, \quad ki = j, \quad (209)$$

written as the 4×1 vector $\bar{q} = [q_w, q_x, q_y, q_z]^\top = [q_w, \mathbf{q}_v^\top]^\top$.

Product. The quaternion product is bilinear and can be written with left/right matrices:

$$\bar{p} \otimes \bar{q} = \mathcal{L}(\bar{p}) \bar{q} = \mathcal{R}(\bar{q}) \bar{p}, \quad (210)$$

$$\mathcal{L}(\bar{p}) = p_w \mathbf{I}_4 + \begin{bmatrix} 0 & -\mathbf{p}_v^\top \\ \mathbf{p}_v & [\mathbf{p}_v]_\times \end{bmatrix}, \quad (211)$$

$$\mathcal{R}(\bar{q}) = q_w \mathbf{I}_4 + \begin{bmatrix} 0 & -\mathbf{q}_v^\top \\ \mathbf{q}_v & -[\mathbf{q}_v]_\times \end{bmatrix}. \quad (212)$$

Identity, unit quaternion, inverse. The identity is $\bar{q} = [1, \mathbf{0}_{3 \times 1}^\top]^\top$. A rotation by angle θ about unit axis $\mathbf{k}$ is the unit quaternion

$$\bar{q} = \begin{bmatrix} \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} \mathbf{k} \end{bmatrix}, \quad \bar{q}^{-1} = \begin{bmatrix} \cos \frac{\theta}{2} \\ -\sin \frac{\theta}{2} \mathbf{k} \end{bmatrix}, \quad \mathbf{R}_{12} = \mathbf{R}_1 \mathbf{R}_2 \leftrightarrow \bar{q}_{12} = \bar{q}_1 \otimes \bar{q}_2. \quad (213)$$

Why $\theta/2$? — rotating a vector. With global frame $\{G\}$ and IMU frame $\{I\}$, rotate ${}^I \mathbf{p}$ via the sandwich product and convert to matrices with $\bar{p} \otimes \bar{q} = \mathcal{R}(\bar{q}) \bar{p}$ and $\bar{p} \otimes \bar{q} = \mathcal{L}(\bar{p}) \bar{q}$:

$$\begin{bmatrix} 0 \\ {}^G \mathbf{p} \end{bmatrix} = {}^G \bar{q} \otimes \begin{bmatrix} 0 \\ {}^I \mathbf{p} \end{bmatrix} \otimes {}^I \bar{q}^{-1} \quad (214)$$

$$= \mathcal{R}({}^G \bar{q}^{-1}) \mathcal{L}({}^G \bar{q}) \begin{bmatrix} 0 \\ {}^I \mathbf{p} \end{bmatrix}. \quad (215)$$

Writing $\bar{q} = [q_w, \mathbf{q}_v^\top]^\top$ and multiplying the two matrices,

$$\begin{aligned} \mathcal{R}(\bar{q}^{-1}) \mathcal{L}(\bar{q}) &= \left(q_w \mathbf{I}_4 + \begin{bmatrix} 0 & \mathbf{q}_v^\top \\ -\mathbf{q}_v & [\mathbf{q}_v]_\times \end{bmatrix} \right) \left(q_w \mathbf{I}_4 + \begin{bmatrix} 0 & -\mathbf{q}_v^\top \\ \mathbf{q}_v & [\mathbf{q}_v]_\times \end{bmatrix} \right) \\ &= q_w^2 \mathbf{I}_4 + 2q_w \begin{bmatrix} 0 & \mathbf{0} \\ \mathbf{0} & [\mathbf{q}_v]_\times \end{bmatrix} + \begin{bmatrix} \mathbf{q}_v^\top \mathbf{q}_v & \mathbf{0} \\ \mathbf{0} & -\mathbf{q}_v \mathbf{q}_v^\top + [\mathbf{q}_v]_\times^2 \end{bmatrix}, \end{aligned} \quad (216)$$

whose lower-right 3×3 block (acting on ${}^I \mathbf{p}$) gives

$${}^G \mathbf{p} = (q_w^2 \mathbf{I}_3 + 2q_w [\mathbf{q}_v]_\times + \mathbf{q}_v \mathbf{q}_v^\top + [\mathbf{q}_v]_\times^2) {}^I \mathbf{p}. \quad (217)$$

Using $[\mathbf{q}_v]_\times^2 = \mathbf{q}_v \mathbf{q}_v^\top - (\mathbf{q}_v^\top \mathbf{q}_v) \mathbf{I}_3$ and the unit-norm condition $q_w^2 + \mathbf{q}_v^\top \mathbf{q}_v = 1$,

$$q_w^2 \mathbf{I}_3 + \mathbf{q}_v \mathbf{q}_v^\top + [\mathbf{q}_v]_\times^2 = q_w^2 \mathbf{I}_3 + (\mathbf{q}_v^\top \mathbf{q}_v) \mathbf{I}_3 + 2 [\mathbf{q}_v]_\times^2 \quad (218)$$

$$= \mathbf{I}_3 + 2 [\mathbf{q}_v]_\times^2, \quad (219)$$

so

$${}^G\mathbf{p} = (\mathbf{I}_3 + 2q_w [\mathbf{q}_v]_{\times} + 2 [\mathbf{q}_v]_{\times}^2)^I \mathbf{p}. \quad (220)$$

Finally substitute $q_w = \cos \frac{\theta}{2}$, $\mathbf{q}_v = \sin \frac{\theta}{2} \mathbf{k}$, so $2q_w [\mathbf{q}_v]_{\times} = 2 \cos \frac{\theta}{2} \sin \frac{\theta}{2} [\mathbf{k}]_{\times} = \sin \theta [\mathbf{k}]_{\times}$ and $2 [\mathbf{q}_v]_{\times}^2 = 2 \sin^2 \frac{\theta}{2} [\mathbf{k}]_{\times}^2 = (1 - \cos \theta) [\mathbf{k}]_{\times}^2$:

$${}^G\mathbf{p} = (\mathbf{I}_3 + \sin \theta [\mathbf{k}]_{\times} + (1 - \cos \theta) [\mathbf{k}]_{\times}^2)^I \mathbf{p} \quad (221)$$

$$= {}^G_I \mathbf{R}^I \mathbf{p} \quad (222)$$

$$= \text{Exp}(\theta \mathbf{k})^I \mathbf{p}, \quad (223)$$

recovering Rodrigues' formula — which is why the half-angle $\theta/2$ appears.

Derivative. With $\delta \boldsymbol{\theta} = \delta \theta \mathbf{k}$ and small-angle $\delta \bar{q} = [\cos \frac{\delta \theta}{2}, \sin \frac{\delta \theta}{2} \mathbf{k}^{\top}]^{\top} \approx [1, \frac{1}{2} \delta \boldsymbol{\theta}^{\top}]^{\top}$,

$${}_t^G \dot{\bar{q}} = \lim_{\Delta t \rightarrow 0} \frac{{}_t^G {}_{t+\Delta t} \bar{q} - {}_t^G \bar{q}}{\Delta t} \quad (224)$$

$$= \lim_{\Delta t \rightarrow 0} \frac{{}_t^G \bar{q} \otimes \begin{bmatrix} \cos \frac{\delta \theta}{2} \\ \sin \frac{\delta \theta}{2} \mathbf{k} \end{bmatrix} - {}_t^G \bar{q} \otimes \begin{bmatrix} 1 \\ \mathbf{0} \end{bmatrix}}{\Delta t} \quad (225)$$

$$= \lim_{\Delta t \rightarrow 0} \frac{{}_t^G \bar{q} \otimes \begin{bmatrix} 0 \\ \frac{1}{2} \delta \boldsymbol{\theta} \end{bmatrix}}{\Delta t} \quad (226)$$

$$= {}_t^G \bar{q} \otimes \begin{bmatrix} 0 \\ \frac{1}{2} \boldsymbol{\omega} \end{bmatrix} \quad (227)$$

$$= \frac{1}{2} \mathcal{R} \left(\begin{bmatrix} 0 \\ \boldsymbol{\omega} \end{bmatrix} \right) {}_t^G \bar{q} \quad (228)$$

$$= \frac{1}{2} \begin{bmatrix} 0 & -\boldsymbol{\omega}^{\top} \\ \boldsymbol{\omega} & -[\boldsymbol{\omega}]_{\times} \end{bmatrix} {}_t^G \bar{q} \quad (229)$$

$$= \frac{1}{2} \Omega(\boldsymbol{\omega}) {}_t^G \bar{q}, \quad (230)$$

where $\Omega(\boldsymbol{\omega}) \triangleq \begin{bmatrix} 0 & -\boldsymbol{\omega}^{\top} \\ \boldsymbol{\omega} & -[\boldsymbol{\omega}]_{\times} \end{bmatrix}$.

Integration. Given ${}_t^G \bar{q}$, $\boldsymbol{\omega}$, δt , with $\delta \boldsymbol{\theta} = \boldsymbol{\omega} \delta t = \delta \theta \mathbf{k}$ and $\delta \bar{q} = [\cos \frac{\delta \theta}{2}, \sin \frac{\delta \theta}{2} \mathbf{k}^{\top}]^{\top}$,

$${}_{t+\delta t}^G \bar{q} = {}_t^G \bar{q} \otimes \delta \bar{q}. \quad (231)$$

3 Lecture 3: Camera Model, Triangulation, Point/Line/Plane

Topics. (1) camera measurements; (2) essential matrix / triangulation; (3) point / line / plane representations.

3.1 The camera measurement model

A camera produces pixel measurements that depend on the camera pose and the observed scene:

$$\mathbf{z}_c = \mathbf{h}_c(\mathbf{x}) + \mathbf{n}_c, \quad (232)$$

where $\mathbf{z}_c$ are the pixel measurements, $\mathbf{x}$ is the state vector (camera pose and point feature), $\mathbf{h}_c(\cdot)$ is the camera model, and $\mathbf{n}_c$ is the pixel noise. The geometry is shown in Fig. 4.

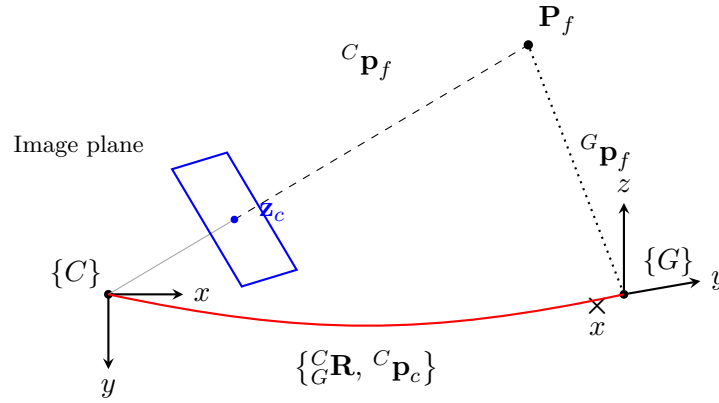


Figure 4: Camera measurement geometry: the feature $\mathbf{P}_f$ projects through the image plane to the pixel $\mathbf{z}_c$ in the camera frame $\{C\}$; the global frame $\{G\}$ and the extrinsic $\{^C_G \mathbf{R}, ^C_G \mathbf{p}_c\}$ relate the two views of the same point ($^C \mathbf{p}_f$ vs. $^G \mathbf{p}_f$).

Simplest pinhole model. With the feature in the camera frame $^C \mathbf{p}_f = [x, y, z]^\top$, the normalized projection is

$$\mathbf{z}_n = \begin{bmatrix} u_n \\ v_n \end{bmatrix} = \begin{bmatrix} x/z \\ y/z \end{bmatrix}, \quad \mathbf{z}_n = \mathbf{h}_p(^C \mathbf{p}_f) + \mathbf{n}_c, \quad ^C \mathbf{p}_f = ^C_G \mathbf{R}^\top (^G \mathbf{p}_f - ^G \mathbf{p}_c). \quad (233)$$

Linearizing the measurement, $\tilde{\mathbf{z}}_n = \mathbf{H}_c \tilde{\mathbf{x}} + \mathbf{n}_c$, with the residual

$$\tilde{\mathbf{z}}_n = \mathbf{z}_n - \mathbf{h}_p(^C \hat{\mathbf{p}}_f) = \begin{bmatrix} u_n \\ v_n \end{bmatrix} - \begin{bmatrix} \hat{x}/\hat{z} \\ \hat{y}/\hat{z} \end{bmatrix}. \quad (234)$$

The state and its error state are

$$\mathbf{x} : \{^C_G \mathbf{R}, ^G \mathbf{p}_c, ^G \mathbf{p}_f\}, \quad ^C_G \mathbf{R} = ^C_G \hat{\mathbf{R}} \text{Exp}(\delta \boldsymbol{\theta}), \quad ^G \mathbf{p}_c = ^G \hat{\mathbf{p}}_c + ^G \tilde{\mathbf{p}}_c, \quad ^G \mathbf{p}_f = ^G \hat{\mathbf{p}}_f + ^G \tilde{\mathbf{p}}_f, \quad (235)$$

with error state $\tilde{\mathbf{x}} = [\delta \boldsymbol{\theta}^\top, ^G \tilde{\mathbf{p}}_c^\top, ^G \tilde{\mathbf{p}}_f^\top]^\top$.

3.2 The measurement Jacobian

By the chain rule, factoring every block through ${}^C\tilde{\mathbf{p}}_f$,

$$\tilde{\mathbf{z}}_n = \mathbf{H}_c \tilde{\mathbf{x}} + \mathbf{n}_c \quad (236)$$

$$= \left[\frac{\partial \tilde{\mathbf{z}}_n}{\partial \delta \boldsymbol{\theta}} \quad \frac{\partial \tilde{\mathbf{z}}_n}{\partial {}^G\tilde{\mathbf{p}}_c} \quad \frac{\partial \tilde{\mathbf{z}}_n}{\partial {}^G\tilde{\mathbf{p}}_f} \right] \tilde{\mathbf{x}} + \mathbf{n}_c \quad (237)$$

$$= \frac{\partial \tilde{\mathbf{z}}_n}{\partial {}^C\tilde{\mathbf{p}}_f} \left[\frac{\partial {}^C\tilde{\mathbf{p}}_f}{\partial \delta \boldsymbol{\theta}} \quad \frac{\partial {}^C\tilde{\mathbf{p}}_f}{\partial {}^G\tilde{\mathbf{p}}_c} \quad \frac{\partial {}^C\tilde{\mathbf{p}}_f}{\partial {}^G\tilde{\mathbf{p}}_f} \right] \tilde{\mathbf{x}} + \mathbf{n}_c. \quad (238)$$

The projection Jacobian, writing ${}^C\mathbf{p}_f = [c_x, c_y, c_z]^\top$, is

$$\mathbf{z}_n = \begin{bmatrix} c_x/c_z \\ c_y/c_z \end{bmatrix}, \quad \frac{\partial \mathbf{z}_n}{\partial {}^C\mathbf{p}_f} = \begin{bmatrix} \frac{1}{c_z} & 0 & -\frac{c_x}{c_z^2} \\ 0 & \frac{1}{c_z} & -\frac{c_y}{c_z^2} \end{bmatrix} = \frac{1}{c_z^2} \begin{bmatrix} c_z & 0 & -c_x \\ 0 & c_z & -c_y \end{bmatrix}. \quad (239)$$

Perturbing ${}^C\mathbf{p}_f$. From ${}^C\mathbf{p}_f = {}^C\mathbf{R}^\top ({}^G\mathbf{p}_f - {}^G\mathbf{p}_c)$ with ${}^C\mathbf{R} = {}^C\hat{\mathbf{R}} \text{Exp}(\delta \boldsymbol{\theta})$ and $\text{Exp}(\delta \boldsymbol{\theta})^\top \approx \mathbf{I} - [\delta \boldsymbol{\theta}]_\times$,

$${}^C\hat{\mathbf{p}}_f + {}^C\tilde{\mathbf{p}}_f = ({}^C\hat{\mathbf{R}} \text{Exp}(\delta \boldsymbol{\theta}))^\top ({}^G\hat{\mathbf{p}}_f + {}^G\tilde{\mathbf{p}}_f - {}^G\hat{\mathbf{p}}_c - {}^G\tilde{\mathbf{p}}_c) \quad (240)$$

$$= (\mathbf{I} - [\delta \boldsymbol{\theta}]_\times) {}^C\hat{\mathbf{R}}^\top ({}^G\hat{\mathbf{p}}_f + {}^G\tilde{\mathbf{p}}_f - {}^G\hat{\mathbf{p}}_c - {}^G\tilde{\mathbf{p}}_c) \quad (241)$$

$$\approx {}^C\hat{\mathbf{R}}^\top ({}^G\hat{\mathbf{p}}_f - {}^G\hat{\mathbf{p}}_c) - [\delta \boldsymbol{\theta}]_\times {}^C\hat{\mathbf{R}}^\top ({}^G\hat{\mathbf{p}}_f - {}^G\hat{\mathbf{p}}_c) + {}^C\hat{\mathbf{R}}^\top ({}^G\tilde{\mathbf{p}}_f - {}^G\tilde{\mathbf{p}}_c). \quad (242)$$

Using $-[\delta \boldsymbol{\theta}]_\times \mathbf{a} = [\mathbf{a}]_\times \delta \boldsymbol{\theta}$ with $\mathbf{a} = {}^C\hat{\mathbf{p}}_f = {}^C\hat{\mathbf{R}}^\top ({}^G\hat{\mathbf{p}}_f - {}^G\hat{\mathbf{p}}_c)$ and matching hat/tilde terms,

$$\begin{aligned} {}^C\hat{\mathbf{p}}_f + {}^C\tilde{\mathbf{p}}_f &= \underbrace{{}^C\hat{\mathbf{R}}^\top ({}^G\hat{\mathbf{p}}_f - {}^G\hat{\mathbf{p}}_c)}_{{}^C\hat{\mathbf{p}}_f} + \left[{}^C\hat{\mathbf{R}}^\top ({}^G\hat{\mathbf{p}}_f - {}^G\hat{\mathbf{p}}_c) \right]_\times \delta \boldsymbol{\theta} \\ &\quad - {}^C\hat{\mathbf{R}}^\top {}^G\tilde{\mathbf{p}}_c + {}^C\hat{\mathbf{R}}^\top {}^G\tilde{\mathbf{p}}_f, \end{aligned} \quad (243)$$

so that, with ${}^C\hat{\mathbf{p}}_f = {}^C\hat{\mathbf{R}}^\top ({}^G\hat{\mathbf{p}}_f - {}^G\hat{\mathbf{p}}_c)$, the three blocks are

$$\frac{\partial {}^C\tilde{\mathbf{p}}_f}{\partial \delta \boldsymbol{\theta}} = [{}^C\hat{\mathbf{p}}_f]_\times, \quad \frac{\partial {}^C\tilde{\mathbf{p}}_f}{\partial {}^G\tilde{\mathbf{p}}_c} = -{}^C\hat{\mathbf{R}}^\top, \quad \frac{\partial {}^C\tilde{\mathbf{p}}_f}{\partial {}^G\tilde{\mathbf{p}}_f} = {}^C\hat{\mathbf{R}}^\top. \quad (244)$$

Therefore $\tilde{\mathbf{z}}_n = \mathbf{H}_c \tilde{\mathbf{x}} + \mathbf{n}_c$ with

$$\mathbf{H}_c = \frac{1}{c_z^2} \begin{bmatrix} c_z & 0 & -c_x \\ 0 & c_z & -c_y \end{bmatrix} \left[[{}^C\hat{\mathbf{p}}_f]_\times \quad -{}^C\hat{\mathbf{R}}^\top \quad {}^C\hat{\mathbf{R}}^\top \right]. \quad (245)$$

3.3 A few important discussions

(1) SLAM. The pose $\{{}^C\mathbf{R}, {}^G\mathbf{p}_c\}$ is the localization of the camera, and ${}^G\mathbf{p}_f$ is a 3D feature of the environment. Solving them together means we solve localization and mapping together $\Rightarrow$ *Simultaneous Localization And Mapping* (SLAM).

(2) **PnP / P3P — known feature, solve pose.** If ${}^G\mathbf{p}_f$ is known, given (u, v) , how do we solve $\{{}_G^C\mathbf{R}, {}^C\mathbf{p}_c\}$? Here $\mathbf{x} : \{{}_G^C\mathbf{R}, {}^C\mathbf{p}_c\}$ is 6-dof, while each measurement $\mathbf{z}_n = \mathbf{h}({}_G^C\mathbf{R}, {}^C\mathbf{p}_c) + \mathbf{n}_c$ contributes 2 dof, so we need at least 3 points ${}^C\mathbf{p}_{f_i} \leftrightarrow (u_i, v_i)$ to solve $\{{}_G^C\mathbf{R}, {}^C\mathbf{p}_c\}$.

- 3 pairs ${}^C\mathbf{p}_{f_i} \leftrightarrow (u_i, v_i)$, **P3P** (perspective-3-point), i.e. 2+1 points: use 3D $\leftrightarrow$ 3D correspondences to determine the same 3D pose.
- n pairs of 3D $\leftrightarrow$ 3D correspondences, **PnP** (perspective- n -point).
- **RANSAC** (random sample consensus) for outlier rejection.

This is the *structure* $\rightarrow$ *motion* direction (known structure, solve motion).

(3) **Triangulation — known poses, solve feature.** If $\{{}_G^C\mathbf{R}, {}^C\mathbf{p}_c\}$ is known, given (u_1, v_1) , (u_2, v_2) , solve $\mathbf{P}_f$ (Fig. 5). Each pixel gives a unit bearing and an unknown range r_i :

$$(u_1, v_1) \rightarrow {}^{C_1}\mathbf{b}_1 = \frac{[u_1, v_1, 1]^\top}{\|[u_1, v_1, 1]^\top\|}, \quad (u_2, v_2) \rightarrow {}^{C_2}\mathbf{b}_2 = \frac{[u_2, v_2, 1]^\top}{\|[u_2, v_2, 1]^\top\|}, \quad (246)$$

$${}^{C_1}\mathbf{p}_f = {}^{C_1}\mathbf{b}_1 r_1, \quad {}^{C_2}\mathbf{p}_f = {}^{C_2}\mathbf{b}_2 r_2, \quad (247)$$

with r_1, r_2 unknown. From ${}^{C_1}\mathbf{p}_f = {}_{C_2}^{C_1}\mathbf{R} {}^{C_2}\mathbf{p}_f + {}^{C_1}\mathbf{p}_{C_2}$ we obtain ${}^{C_1}\mathbf{b}_1 r_1 = {}_{C_2}^{C_1}\mathbf{R} {}^{C_2}\mathbf{b}_2 r_2 + {}^{C_1}\mathbf{p}_{C_2}$.

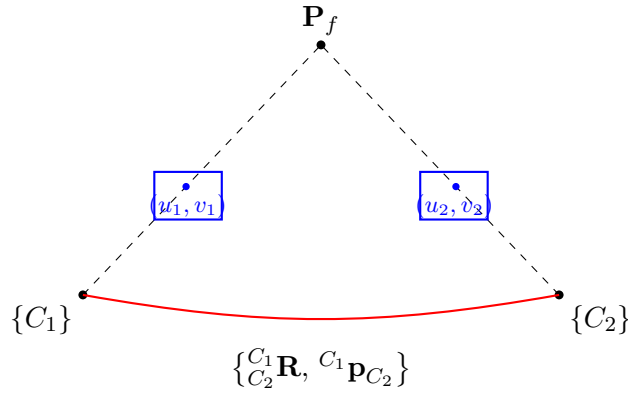


Figure 5: Two-view triangulation: bearings ${}^{C_1}\mathbf{b}_1, {}^{C_2}\mathbf{b}_2$ from pixels $(u_1, v_1), (u_2, v_2)$ and the known relative pose $\{{}_{C_2}^{C_1}\mathbf{R}, {}^{C_1}\mathbf{p}_{C_2}\}$ fix the feature $\mathbf{P}_f$ up to the two ranges r_1, r_2 .

Way 1 (stacked linear system). Collect the two ranges:

$${}^{C_1}\mathbf{b}_1 r_1 - {}_{C_2}^{C_1}\mathbf{R} {}^{C_2}\mathbf{b}_2 r_2 = {}^{C_1}\mathbf{p}_{C_2} \Rightarrow \underbrace{\begin{bmatrix} {}^{C_1}\mathbf{b}_1 & -{}_{C_2}^{C_1}\mathbf{R} {}^{C_2}\mathbf{b}_2 \end{bmatrix}}_{3 \times 2} \begin{bmatrix} r_1 \\ r_2 \end{bmatrix} = {}^{C_1}\mathbf{p}_{C_2}, \quad (248)$$

a 3×2 system; solving r_1, r_2 gives ${}^{C_1}\mathbf{p}_f, {}^{C_2}\mathbf{p}_f$.

Degenerate case. If the motion is pure rotation (no translation), then ${}^{C_1}\mathbf{b}_1 = {}_{C_2}^{C_1}\mathbf{R} {}^{C_2}\mathbf{b}_2$ and ${}^{C_1}\mathbf{p}_{C_2} = \mathbf{0}_{3 \times 1}$, so $r_1 = r_2$ and there is no way to solve r . For a monocular camera, pure rotation is dangerous.

Way 2 (left null space). With ${}^{C_1}\mathbf{b}_1 r_1 = {}_{C_2}^{C_1}\mathbf{R} {}^{C_2}\mathbf{b}_2 r_2 + {}^{C_1}\mathbf{p}_{C_2}$, collect into $\mathbf{B}^\perp = [\mathbf{b}^{\perp_1} \quad \mathbf{b}^{\perp_2}]$ two directions orthogonal to ${}^{C_1}\mathbf{b}_1$ (${}^{C_1}\mathbf{b}_1^\top \mathbf{b}^{\perp_1} = 0$, ${}^{C_1}\mathbf{b}_1^\top \mathbf{b}^{\perp_2} = 0$); projecting removes r_1 :

$$\mathbf{B}^{\perp \top} {}^{C_1}\mathbf{b}_1 r_1 = \mathbf{B}^{\perp \top} ({}_{C_2}^{C_1}\mathbf{R} {}^{C_2}\mathbf{b}_2 r_2 + {}^{C_1}\mathbf{p}_{C_2}) = \mathbf{0}_{2 \times 1} \Rightarrow \underbrace{\mathbf{B}^{\perp \top} {}_{C_2}^{C_1}\mathbf{R} {}^{C_2}\mathbf{b}_2}_{2 \times 1} r_2 = - \underbrace{\mathbf{B}^{\perp \top} {}^{C_1}\mathbf{p}_{C_2}}_{2 \times 1}. \quad (249)$$

(4) **Monocular triangulation is up to scale.** If $\{^C_G \mathbf{R}\}$ is known but for $^{C_1} \mathbf{p}_{C_2}$ we only know the direction $\overrightarrow{^{C_1} \mathbf{p}_{C_2}}$ (with $\|^{C_1} \mathbf{p}_{C_2}\|$ unknown), can we triangulate $^C \mathbf{p}_f$? Let $\|^{C_1} \mathbf{p}_{C_2}\| = \rho$, then $^{C_1} \mathbf{p}_f = \frac{C_1}{C_2} \mathbf{R} \frac{C_2}{C_1} \mathbf{p}_f + ^{C_1} \mathbf{p}_{C_2}$ gives

$$^{C_1} \mathbf{b}_1 r_1 = \frac{C_1}{C_2} \mathbf{R} \frac{C_2}{C_1} \mathbf{b}_2 r_2 + \rho \overrightarrow{^{C_1} \mathbf{p}_{C_2}}. \quad (250)$$

With $\mathbf{A} = [\mathbf{v}^{\perp_1} \ \mathbf{v}^{\perp_2}]^\top$ built from $\overrightarrow{^{C_1} \mathbf{p}_{C_2}}^\top \mathbf{v}^{\perp_1} = 0$, $\overrightarrow{^{C_1} \mathbf{p}_{C_2}}^\top \mathbf{v}^{\perp_2} = 0$, projecting kills the translation term:

$$\mathbf{A} \frac{C_1}{C_2} \mathbf{b}_1 r_1 = \mathbf{A} \frac{C_1}{C_2} \mathbf{R} \frac{C_2}{C_1} \mathbf{b}_2 r_2 + \rho \underbrace{\mathbf{A} \overrightarrow{^{C_1} \mathbf{p}_{C_2}}}_0 \quad (251)$$

$$\Rightarrow \begin{bmatrix} \mathbf{A} \frac{C_1}{C_2} \mathbf{b}_1 & -\mathbf{A} \frac{C_1}{C_2} \mathbf{R} \frac{C_2}{C_1} \mathbf{b}_2 \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \end{bmatrix} = 0. \quad (252)$$

Hence r_1, r_2 are up to scale: a monocular camera does not have scale. The above uses *motion to infer structure*.

(5) **Essential matrix — solve the motion.** Given $\{u_1, v_1\}, \{u_2, v_2\}$, can we solve $\{^C_G \mathbf{R}, ^C \mathbf{p}_c\}$? The three vectors $^{C_1} \mathbf{p}_f, ^{C_2} \mathbf{p}_f, ^{C_1} \mathbf{p}_{C_2}$ are coplanar (Fig. 6), so $^{C_1} \mathbf{p}_{C_2} \times ^{C_1} \mathbf{p}_f \perp \frac{C_1}{C_2} \mathbf{R} \frac{C_2}{C_1} \mathbf{p}_f$:

$$(\lfloor ^{C_1} \mathbf{p}_{C_2} \rfloor_\times \frac{C_1}{C_2} \mathbf{p}_f)^\top \frac{C_1}{C_2} \mathbf{R} \frac{C_2}{C_1} \mathbf{p}_f = 0 \Rightarrow ^{C_1} \mathbf{p}_f^\top \lfloor ^{C_1} \mathbf{p}_{C_2} \rfloor_\times \frac{C_1}{C_2} \mathbf{R} \frac{C_2}{C_1} \mathbf{p}_f = 0. \quad (253)$$

With $^{C_1} \mathbf{p}_f = ^{C_1} \mathbf{b}_1 r_1, ^{C_2} \mathbf{p}_f = ^{C_2} \mathbf{b}_2 r_2$,

$$\boxed{^{C_1} \mathbf{b}_1^\top \lfloor ^{C_1} \mathbf{p}_{C_2} \rfloor_\times \frac{C_1}{C_2} \mathbf{R} \frac{C_2}{C_1} \mathbf{b}_2 = 0, \quad \mathbf{E} = \lfloor ^{C_1} \mathbf{p}_{C_2} \rfloor_\times \frac{C_1}{C_2} \mathbf{R} \quad (\text{essential matrix}).} \quad (254)$$

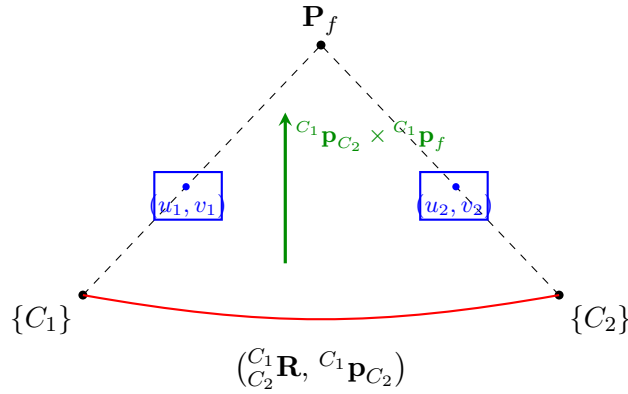


Figure 6: Epipolar geometry: $^{C_1} \mathbf{p}_f, ^{C_2} \mathbf{p}_f$, and the baseline $^{C_1} \mathbf{p}_{C_2}$ are coplanar, giving the essential-matrix constraint $^{C_1} \mathbf{b}_1^\top \mathbf{E} ^{C_2} \mathbf{b}_2 = 0$.

Why is $\mathbf{E}$ 5-dof? Each pair $(u_1, v_1), (u_2, v_2)$ provides 1 constraint, so we need at least **5 pairs of points** to solve $\mathbf{E} \Rightarrow \frac{C_1}{C_2} \mathbf{R}$ (3 dof) and $\frac{C_1}{C_2} \mathbf{p}_{C_2}$ (2 dof, direction only). When selecting pairs of points, RANSAC (random sample consensus) comes in. Finally:

$$\text{normalized coordinates: } ^{C_1} \mathbf{b}_1^\top \lfloor ^{C_1} \mathbf{p}_{C_2} \rfloor_\times \frac{C_1}{C_2} \mathbf{R} \frac{C_2}{C_1} \mathbf{b}_2 = 0 \quad (\text{essential matrix}), \quad (255)$$

$$\text{pixel coordinates: fundamental matrix.} \quad (256)$$

This is the *motion from structure* direction.

3.4 The full intrinsic calibration model

The calibration geometry is the same as Fig. 4 (repeated here as Fig. 7). The camera model is a composition of four functions.

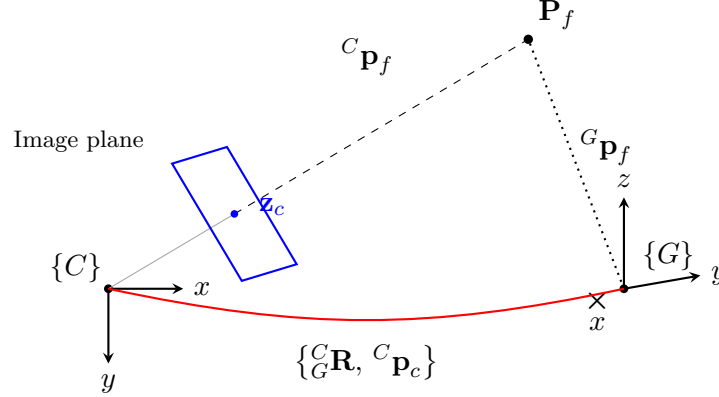


Figure 7: The intrinsic calibration sits inside the same projection geometry: the 3D feature is transformed into $\{C\}$, projected, distorted, and mapped to pixels.

Projection and transform.

$$\mathbf{z}_n = \begin{bmatrix} c_x/c_z \\ c_y/c_z \end{bmatrix} \triangleq \mathbf{h}_p({}^C\mathbf{p}_f), \quad {}^C\mathbf{p}_f = {}^C\mathbf{R}^\top ({}^G\mathbf{p}_f - {}^G\mathbf{p}_c) \triangleq \mathbf{h}_t({}^C\mathbf{R}, {}^G\mathbf{p}_c, {}^G\mathbf{p}_f), \quad (257)$$

$$\mathbf{z}_n = \mathbf{h}_p(\mathbf{h}_t({}^C\mathbf{R}, {}^G\mathbf{p}_c, {}^G\mathbf{p}_f)) + \mathbf{n}. \quad (258)$$

Distortion. Radial-tangential (for example); fisheye uses the equidistant model. With $r^2 = u_n^2 + v_n^2$ and distortion parameters $\mathbf{x}_d = [k_1, k_2, p_1, p_2]^\top$,

$$\begin{bmatrix} u_d \\ v_d \end{bmatrix} = \begin{bmatrix} u_n \\ v_n \end{bmatrix} (1 + k_1 r^2 + k_2 r^4) + \begin{bmatrix} 2p_1 u_n v_n + p_2 (r^2 + 2u_n^2) \\ p_1 (r^2 + 2v_n^2) + 2p_2 u_n v_n \end{bmatrix}, \quad \mathbf{z}_d = \begin{bmatrix} u_d \\ v_d \end{bmatrix} = \mathbf{h}_d(\mathbf{z}_n). \quad (259)$$

Intrinsics / homography. With $\mathbf{x}_{in} = [f_u, f_v, c_u, c_v]$,

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} f_u & 0 & c_u \\ 0 & f_v & c_v \\ 0 & 0 & 1 \end{bmatrix}}_{\mathbf{K}} \begin{bmatrix} u_d \\ v_d \\ 1 \end{bmatrix}, \quad \mathbf{z} = \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} f_u & 0 \\ 0 & f_v \end{bmatrix} \begin{bmatrix} u_d \\ v_d \end{bmatrix} + \begin{bmatrix} c_u \\ c_v \end{bmatrix} \triangleq \mathbf{h}_k(\mathbf{z}_d, \mathbf{x}_{in}). \quad (260)$$

Full composition.

$$\mathbf{z}_c = \mathbf{h}_c(\mathbf{x}) + \mathbf{n}_c = \mathbf{h}_k(\mathbf{z}_d, \mathbf{x}_{in}) + \mathbf{n}_c \quad (261)$$

$$= \mathbf{h}_k(\mathbf{h}_d(\mathbf{z}_n, \mathbf{x}_d), \mathbf{x}_{in}) + \mathbf{n}_c \quad (262)$$

$$= \mathbf{h}_k(\mathbf{h}_d(\mathbf{h}_p({}^C\mathbf{p}_f), \mathbf{x}_d), \mathbf{x}_{in}) + \mathbf{n}_c \quad (263)$$

$$= \mathbf{h}_k\left(\mathbf{h}_d\left(\mathbf{h}_p\left(\mathbf{h}_t({}^C\mathbf{R}, {}^G\mathbf{p}_c, {}^G\mathbf{p}_f)\right), \mathbf{x}_d\right), \mathbf{x}_{in}\right) + \mathbf{n}_c, \quad (264)$$

where $\mathbf{h}_k(\cdot)$ is the camera-intrinsics homography, $\mathbf{h}_d(\cdot)$ the distortion function, $\mathbf{h}_p(\cdot)$ the projection function, $\mathbf{h}_t(\cdot)$ the transform function, $\mathbf{x}_d$ the distortion parameters, and $\mathbf{x}_{in} = \{f_u, f_v, c_u, c_v\}$ the intrinsics. The Jacobians for calibration follow by the chain rule.

3.5 The extrinsic (camera–IMU) calibration

The camera and IMU sensor frames are rigidly connected (Fig. 8).

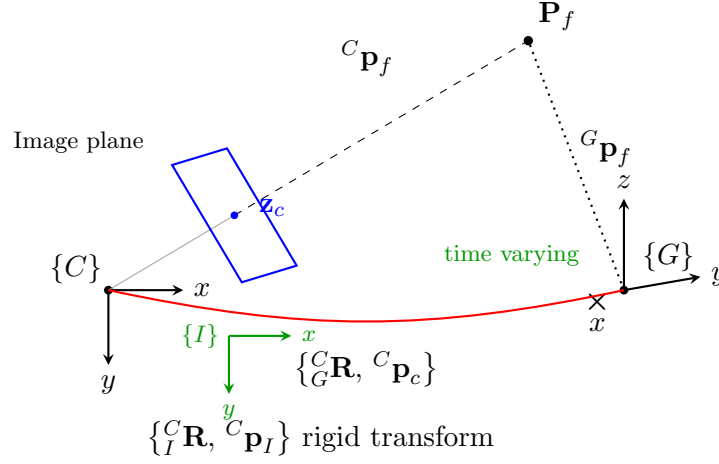


Figure 8: Extrinsic calibration: the IMU frame $\{I\}$ is rigidly fixed to the camera $\{C\}$ via $\{^C_R, ^C_p_I\}$, while the camera-to-global transform is time varying.

$$\begin{bmatrix} ^C_p_f \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} ^C_R & ^C_p_I \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix}}_{^C_T} \underbrace{\begin{bmatrix} ^G_R & ^G_p_I \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix}^{-1}}_{^G_T^{-1}} \begin{bmatrix} ^G_p_f \\ 1 \end{bmatrix}, \quad (265)$$

$$^C_p_f = ^C_R ^I_p_f + ^C_p_I = ^C_R (^I_R (^G_p_f - ^G_p_I)) + ^C_p_I. \quad (266)$$

Time offset. How do we estimate the time offset between camera and IMU? With $t_I = t_c + t_d$,

$$\begin{bmatrix} ^{C_{t_c}}p_f \\ 1 \end{bmatrix} = ^C_T \cdot (^{G_{(t+t_d)}}T)^{-1} \begin{bmatrix} ^Gp_f \\ 1 \end{bmatrix}, \quad (267)$$

$$^{G_{(t+t_d)}}\dot{T} = \begin{bmatrix} ^G_R & ^G_p_I \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix} \begin{bmatrix} [^I\omega]_{\times} & ^I\mathbf{v} \\ \mathbf{0}_{1 \times 3} & 0 \end{bmatrix}. \quad (268)$$

which is the Jacobian for t_d . This is then the full calibration for the vision system.

3.6 Point / line / plane representation

3.6.1 Point representation

In $\mathbb{R}^3$ (closest point),

$$\mathbf{P}_f = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = z \begin{bmatrix} x/z \\ y/z \\ 1 \end{bmatrix} = z \begin{bmatrix} u_n \\ v_n \\ 1 \end{bmatrix}. \quad (269)$$

Inverse depth. Store the inverse of the depth, $\lambda \triangleq 1/z$:

$$\mathbf{x}_f = \begin{bmatrix} x \\ y \\ 1/z \end{bmatrix} = \begin{bmatrix} x \\ y \\ \lambda \end{bmatrix}. \quad (270)$$

What is the motivation to use inverse depth? Transform the feature into a second view and divide by z :

$${}^{C_2}\mathbf{p}_f = {}^{C_2}_{C_1}\mathbf{R}({}^{C_1}\mathbf{p}_f - {}^{C_1}\mathbf{p}_{C_2}) = {}^{C_2}_{C_1}\mathbf{R}\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} - {}^{C_1}\mathbf{p}_{C_2}\right) = {}^{C_2}_{C_1}\mathbf{R}\left(z \begin{bmatrix} x/z \\ y/z \\ 1 \end{bmatrix} - {}^{C_1}\mathbf{p}_{C_2}\right), \quad (271)$$

$$\frac{{}^{C_2}\mathbf{p}_f}{z} = {}^{C_2}_{C_1}\mathbf{R}\left(\begin{bmatrix} x/z \\ y/z \\ 1 \end{bmatrix} - \frac{1}{z}{}^{C_1}\mathbf{p}_{C_2}\right) \Rightarrow \begin{bmatrix} {}^{C_2}x/z \\ {}^{C_2}y/z \\ {}^{C_2}z/z \end{bmatrix} = {}^{C_2}_{C_1}\mathbf{R}\left(\begin{bmatrix} x\lambda \\ y\lambda \\ 1 \end{bmatrix} - \lambda {}^{C_1}\mathbf{p}_{C_2}\right). \quad (272)$$

If $z \rightarrow \infty$, the feature is far away, $z \approx {}^{C_2}z \gg {}^{C_2}x, {}^{C_2}y$, then the left side tends to the bearing measurement in $\{C_2\}$:

$$\begin{bmatrix} {}^{C_2}x/z \\ {}^{C_2}y/z \\ {}^{C_2}z/z \end{bmatrix} \approx \underbrace{\begin{bmatrix} u_{n_2} \\ v_{n_2} \\ 1 \end{bmatrix}}_{\text{measurements in } C_2} \propto {}^{C_2}_{C_1}\mathbf{R}\left(\begin{bmatrix} x\lambda \\ y\lambda \\ 1 \end{bmatrix} - \lambda {}^{C_1}\mathbf{p}_{C_2}\right). \quad (273)$$

The Jacobian with respect to the inverse-depth state $\tilde{\mathbf{x}} = [\tilde{x}, \tilde{y}, \tilde{\lambda}]$ is

$$\frac{\partial[u_{n_2}; v_{n_2}]}{\partial \tilde{\mathbf{x}}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} [\mathbf{H}_\theta \quad \mathbf{H}_p \quad \mathbf{H}_{p_f}], \quad (274)$$

with the three blocks

$$\mathbf{H}_\theta = \left[{}^{C_2}_{C_1}\mathbf{R}([x\lambda; y\lambda; 1] - \lambda {}^{C_1}\mathbf{p}_{C_2}) \right]_{\times}, \quad \mathbf{H}_p = \underbrace{-\lambda {}^{C_2}_{C_1}\mathbf{R}}_{\lambda \rightarrow 0}, \quad (275)$$

$$\mathbf{H}_{p_f} = {}^{C_2}_{C_1}\mathbf{R} \left[\underbrace{\lambda \mathbf{e}_1}_0 \quad \underbrace{\lambda \mathbf{e}_2}_0 \quad [x; y; 0] - {}^{C_1}\mathbf{p}_{C_2} \right]. \quad (276)$$

If the feature is far away, then the rotation is improved, but the Jacobian for translation will not be improved (the $-\lambda {}^{C_2}_{C_1}\mathbf{R}$ and $\lambda \mathbf{e}_1, \lambda \mathbf{e}_2$ columns vanish as $\lambda \rightarrow 0$).

3.6.2 Plane representation

A plane is parameterized by its normal direction and distance to the origin (Fig. 9a):

$$\boldsymbol{\pi} = \begin{bmatrix} \mathbf{n}_\pi \\ d_\pi \end{bmatrix}_{4 \times 1}, \quad \mathbf{n}_\pi : \text{normal direction}, \quad d_\pi : \text{distance to origin}. \quad (277)$$

A plane is 3-dof but $\boldsymbol{\pi}$ has 4 parameters. The *closest-point representation* packs both into one vector,

$$\mathbf{P}_\pi = d_\pi \mathbf{n}_\pi = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad (278)$$

the closest point from the plane to the origin.

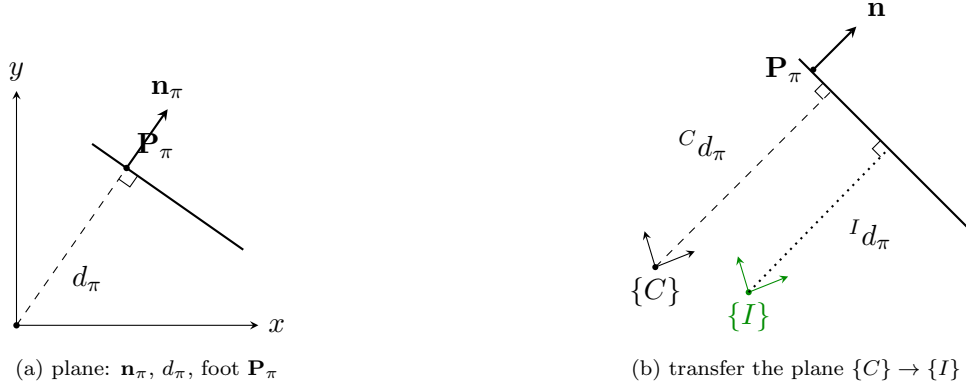


Figure 9: Plane representation (a) and its transfer between the camera and IMU frames (b).

Transfer $\{C\} \rightarrow \{I\}$. For ${}^C\mathbf{p}_\pi \leftrightarrow {}^I\mathbf{p}_\pi$ (Fig. 9b),

$$\begin{cases} {}^I\mathbf{n}_\pi = {}^I_C\mathbf{R} {}^C\mathbf{n}_\pi & \Rightarrow \text{normal direction} \\ {}^I d_\pi = {}^C d_\pi - {}^C\mathbf{n}_\pi^\top \cdot {}^C\mathbf{p}_I & \Rightarrow \text{origin to plane} \end{cases} \quad (279)$$

Hence, using ${}^I d_\pi = {}^C d_\pi - {}^C\mathbf{p}_I^\top {}^C\mathbf{n}_\pi$ and ${}^C\mathbf{p}_I = -{}^C_I\mathbf{R} {}^I\mathbf{p}_C$ (so $-{}^C\mathbf{p}_I^\top = {}^I\mathbf{p}_C^\top {}^I_C\mathbf{R}$),

$$\begin{bmatrix} {}^I\mathbf{n}_\pi \\ {}^I d_\pi \end{bmatrix} = \begin{bmatrix} {}^I_C\mathbf{R} & \mathbf{0}_{3 \times 1} \\ -{}^C\mathbf{p}_I^\top & 1 \end{bmatrix} \begin{bmatrix} {}^C\mathbf{n}_\pi \\ {}^C d_\pi \end{bmatrix} = \begin{bmatrix} {}^I_C\mathbf{R} & \mathbf{0}_{3 \times 1} \\ {}^I\mathbf{p}_C^\top {}^I_C\mathbf{R} & 1 \end{bmatrix} \begin{bmatrix} {}^C\mathbf{n}_\pi \\ {}^C d_\pi \end{bmatrix}. \quad (280)$$

The transformation chain is therefore

$${}^G\mathbf{p}_\pi \rightarrow ({}^C\mathbf{n}_\pi, {}^C d_\pi) \rightarrow ({}^I_C\mathbf{R}, {}^I\mathbf{p}_C) \rightarrow ({}^I\mathbf{n}_\pi, {}^I d_\pi) \rightarrow {}^I\mathbf{p}_\pi. \quad (281)$$

3.6.3 Line representation

How do we represent a 3D line (Fig. 10)? For a line L : $\mathbf{v}_e$ is the 3D direction, d_L is the distance from the origin to the line, and $\mathbf{n}_e$ is the normal direction of the plane (through the line and the origin).

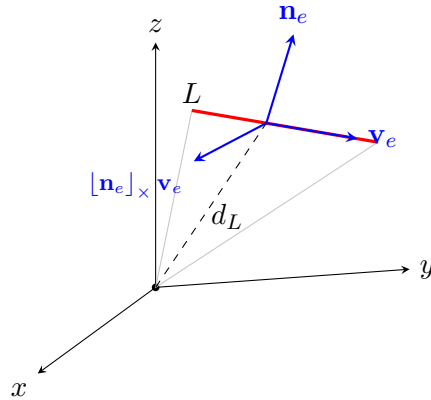


Figure 10: 3D line L : direction $\mathbf{v}_e$, plane normal $\mathbf{n}_e$, third axis $[\mathbf{n}_e]_\times \mathbf{v}_e$, and perpendicular distance d_L from the origin.

We can represent a line with $\{\mathbf{n}_e, \mathbf{v}_e, d_L\}$, where

$$\mathbf{R}_L = [\mathbf{n}_e \ \mathbf{v}_e \ [\mathbf{n}_e]_{\times} \ \mathbf{v}_e] \text{ (a rotation matrix),} \quad d_L \text{ (a distance scalar).} \quad (282)$$

So $L : \{\mathbf{R}_L, d_L\} : \{\bar{\mathbf{q}}_L, d_L\}$, packed as the 4D vector $\mathbf{P}_L = \bar{\mathbf{q}}_L \cdot d_L$, with the error states in Euclidean space. Then

$$\mathbf{P}_L = \hat{\mathbf{P}}_L + \tilde{\mathbf{P}}_L = \bar{\mathbf{q}}_L \cdot d_L = \hat{\mathbf{q}}_L \otimes \delta \mathbf{q}_L \cdot (\hat{d}_L + \tilde{d}_L) \quad (283)$$

$$\Leftrightarrow \hat{\mathbf{P}}_L + \tilde{\mathbf{P}}_L = \hat{\mathbf{q}}_L \otimes \begin{bmatrix} 1 \\ \frac{1}{2} \delta \boldsymbol{\theta}_L \end{bmatrix} (\hat{d}_L + \tilde{d}_L) \quad (284)$$

$$= \hat{\mathbf{q}}_L \otimes \begin{bmatrix} \hat{d}_L + \tilde{d}_L \\ \frac{1}{2} \hat{d}_L \delta \boldsymbol{\theta}_L \end{bmatrix} \quad (285)$$

$$= \hat{\mathbf{q}}_L \otimes \left(\begin{bmatrix} \hat{d}_L \\ \mathbf{0}_{3 \times 1} \end{bmatrix} + \begin{bmatrix} 1 & \mathbf{0} \\ \mathbf{0} & \frac{1}{2} \hat{d}_L \mathbf{I}_3 \end{bmatrix} \begin{bmatrix} \tilde{d}_L \\ \delta \boldsymbol{\theta}_L \end{bmatrix} \right), \quad (286)$$

so that, with $[\hat{\mathbf{q}}_L]_L$ the left quaternion-product matrix,

$$\tilde{\mathbf{P}}_L = [\hat{\mathbf{q}}_L]_L \begin{bmatrix} 1 & \mathbf{0} \\ \mathbf{0} & \frac{1}{2} \hat{d}_L \mathbf{I}_3 \end{bmatrix} \begin{bmatrix} \tilde{d}_L \\ \delta \boldsymbol{\theta}_L \end{bmatrix}. \quad (287)$$

A few discussions on lines with cameras. (1) How to project a 3D line to a 2D line in the image? Given a line $\{\mathbf{n}_e, \mathbf{v}_e, d_L\}$ and camera parameters f_u, f_v, c_u, c_v (Fig. 11a):

$$\begin{bmatrix} u_n \\ v_n \\ 1 \end{bmatrix} \sim \underbrace{\begin{bmatrix} f_u & 0 & c_u \\ 0 & f_v & c_v \\ 0 & 0 & 1 \end{bmatrix}}_{\mathbf{K}} \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{\mathbf{p}_f}, \quad \underbrace{\mathbf{p}}_{\text{pixel on image}} \sim \mathbf{K} \mathbf{p}_f. \quad (288)$$

Every $\mathbf{p}_f$ on the line lies in the plane through the line and the origin, so $\mathbf{p}_f^\top \mathbf{n}_e = 0$. With $\mathbf{p}_f \sim \mathbf{K}^{-1} \mathbf{p}$,

$$(\mathbf{K}^{-1} \mathbf{p})^\top \mathbf{n}_e = 0 \Rightarrow \mathbf{p}^\top \mathbf{K}^{-\top} \mathbf{n}_e = 0 \Rightarrow \mathbf{n}_e^\top \mathbf{K}^{-1} \mathbf{p} = 0 \Rightarrow (\mathbf{K}^{-\top} \mathbf{n}_e)^\top \mathbf{p} = 0, \quad (289)$$

so the 2D image line is

$$\mathbf{l} = \mathbf{K}^{-\top} \mathbf{n}_e, \quad \mathbf{p} \text{ is every pixel on the line.} \quad (290)$$

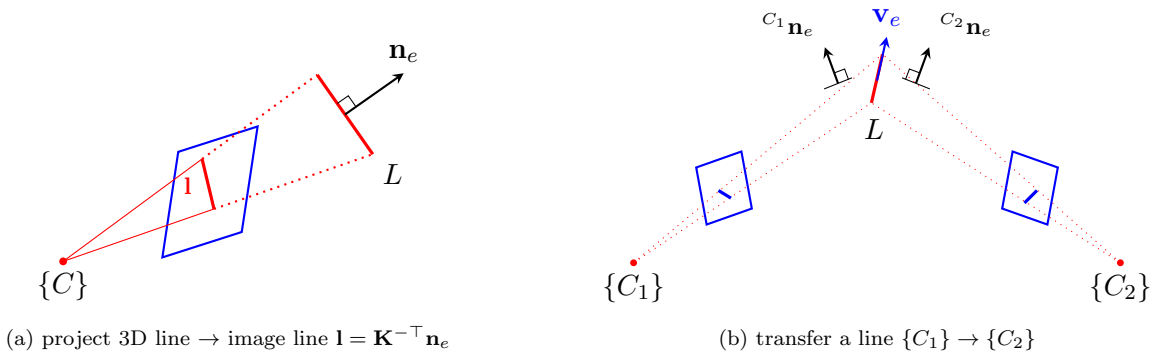


Figure 11: Lines with cameras: projecting a 3D line to a 2D image line (a) and transferring a line between two camera frames (b).

(2) How to transfer a line from frame $\{C_1\}$ to $\{C_2\}$? (Fig. 11b) The line L is given by ${}^{C_1}\mathbf{n}_e$, ${}^{C_1}\mathbf{v}_e$, ${}^{C_1}d_L$ in $\{C_1\}$ and by ${}^{C_2}\mathbf{n}_e$, ${}^{C_2}\mathbf{v}_e$, ${}^{C_2}d_L$ in $\{C_2\}$. Assume ${}^{C_1}\mathbf{p}_f$ is a point on L ; then its moment about the origin is

$$\left[{}^{C_1}\mathbf{p}_f \right]_{\times} {}^{C_1}\mathbf{v}_e = {}^{C_1}\mathbf{n}_e {}^{C_1}d_L. \quad (291)$$

The direction transforms by rotation only, and the point by the full rigid transform:

$${}^{C_2}\mathbf{v}_e = {}_{C_1}^{C_2}\mathbf{R} {}^{C_1}\mathbf{v}_e, \quad {}^{C_2}\mathbf{p}_f = {}_{C_1}^{C_2}\mathbf{R} {}^{C_1}\mathbf{p}_f + {}^{C_2}\mathbf{p}_{C_1}. \quad (292)$$

Hence the moment in $\{C_2\}$ is

$$\left[{}^{C_2}\mathbf{p}_f \right]_{\times} {}^{C_2}\mathbf{v}_e = \left[{}_{C_1}^{C_2}\mathbf{R} {}^{C_1}\mathbf{p}_f + {}^{C_2}\mathbf{p}_{C_1} \right]_{\times} {}_{C_1}^{C_2}\mathbf{R} {}^{C_1}\mathbf{v}_e = {}^{C_2}\mathbf{n}_e {}^{C_2}d_L. \quad (293)$$

Using $[\mathbf{R}\mathbf{a}]_{\times} \mathbf{R} = \mathbf{R} [\mathbf{a}]_{\times}$ on the first term,

$${}_{C_1}^{C_2}\mathbf{R} \left[{}^{C_1}\mathbf{p}_f \right]_{\times} {}^{C_1}\mathbf{v}_e + \left[{}^{C_2}\mathbf{p}_{C_1} \right]_{\times} {}_{C_1}^{C_2}\mathbf{R} {}^{C_1}\mathbf{v}_e = {}^{C_2}\mathbf{n}_e {}^{C_2}d_L, \quad (294)$$

and substituting the moment identity ${}^{C_1}\mathbf{n}_e {}^{C_1}d_L = \left[{}^{C_1}\mathbf{p}_f \right]_{\times} {}^{C_1}\mathbf{v}_e$,

$$\begin{bmatrix} {}_{C_1}^{C_2}\mathbf{R} & \left[{}^{C_2}\mathbf{p}_{C_1} \right]_{\times} {}_{C_1}^{C_2}\mathbf{R} \end{bmatrix} \begin{bmatrix} {}^{C_1}\mathbf{n}_e {}^{C_1}d_L \\ {}^{C_1}\mathbf{v}_e \end{bmatrix} = {}^{C_2}\mathbf{n}_e {}^{C_2}d_L. \quad (295)$$

Stacking the moment and the direction (${}^{C_2}\mathbf{v}_e = {}_{C_1}^{C_2}\mathbf{R} {}^{C_1}\mathbf{v}_e$) gives the full line transfer:

$$\boxed{\begin{bmatrix} {}^{C_2}\mathbf{n}_e {}^{C_2}d_L \\ {}^{C_2}\mathbf{v}_e \end{bmatrix} = \begin{bmatrix} {}_{C_1}^{C_2}\mathbf{R} & \left[{}^{C_2}\mathbf{p}_{C_1} \right]_{\times} {}_{C_1}^{C_2}\mathbf{R} \\ \mathbf{0}_3 & {}_{C_1}^{C_2}\mathbf{R} \end{bmatrix} \begin{bmatrix} {}^{C_1}\mathbf{n}_e {}^{C_1}d_L \\ {}^{C_1}\mathbf{v}_e \end{bmatrix}.} \quad (296)$$

4 Lecture 4: Information, Marginalization, and Estimator Design

Topics. (1) the matrix inversion lemma; (2) covariance vs. information; (3) marginalization and the Schur complement; (4) why the information matrix reflects problem structure; (5) a typical visual-odometry / SLAM pipeline; (6) many ways to handle $\mathbf{z} = \mathbf{h}(\mathbf{x}) + \mathbf{n}$; (7) designing a common estimator — deriving the EKF from MAP by marginalization.

4.1 The matrix inversion lemma

For a 2×2 block matrix, the inverse can be written block by block. In the general form,

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{E} & \mathbf{F} \\ \mathbf{G} & \mathbf{H} \end{bmatrix}, \quad \begin{cases} \mathbf{E} = (\mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C})^{-1}, \\ \mathbf{F} = -\mathbf{E}\mathbf{B}\mathbf{D}^{-1}, \\ \mathbf{G} = -\mathbf{D}^{-1}\mathbf{C}\mathbf{E}, \\ \mathbf{H} = \mathbf{D}^{-1} + \mathbf{D}^{-1}\mathbf{C}\mathbf{E}\mathbf{B}\mathbf{D}^{-1}. \end{cases} \quad (297)$$

For the symmetric case ($\mathbf{C} = \mathbf{B}^\top$, so the inverse is symmetric too),

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^\top & \mathbf{D} \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{E} & \mathbf{F} \\ \mathbf{F}^\top & \mathbf{H} \end{bmatrix}, \quad \begin{cases} \mathbf{E} = (\mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{B}^\top)^{-1}, \\ \mathbf{F} = -\mathbf{E}\mathbf{B}\mathbf{D}^{-1}, \\ \mathbf{H} = \mathbf{D}^{-1} + \mathbf{D}^{-1}\mathbf{B}^\top\mathbf{E}\mathbf{B}\mathbf{D}^{-1} = (\mathbf{D} - \mathbf{B}^\top\mathbf{A}^{-1}\mathbf{B})^{-1}. \end{cases} \quad (298)$$

The term $\mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{B}^\top$ (resp. $\mathbf{D} - \mathbf{B}^\top\mathbf{A}^{-1}\mathbf{B}$) is the *Schur complement*. This is one of the most important matrix operations for the derivations that follow.

4.2 Covariance and information

Recall the measurement model from Lecture 1, $\mathbf{z} = \mathbf{h}(\mathbf{x}) + \mathbf{n}$ with $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$ and $\mathbf{R}$ the noise covariance. The MAP / weighted-least-squares estimate is

$$\max_{\mathbf{x}} p(\mathbf{z} | \mathbf{x}) \Leftrightarrow \min_{\mathbf{x}} \|\mathbf{z} - \mathbf{h}(\mathbf{x})\|_{\mathbf{R}}^2. \quad (299)$$

Given a linearization point $\hat{\mathbf{x}}$, $\mathbf{z} = \mathbf{h}(\hat{\mathbf{x}}) + \mathbf{H}\tilde{\mathbf{x}} + \mathbf{n}$, so with the residual $\tilde{\mathbf{z}} = \mathbf{z} - \mathbf{h}(\hat{\mathbf{x}})$,

$$\tilde{\mathbf{z}} = \mathbf{H}\tilde{\mathbf{x}} + \mathbf{n}. \quad (300)$$

The normal equations and their solution are

$$\boxed{\underbrace{\mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H}}_{\text{information}} \tilde{\mathbf{x}} = \mathbf{H}^\top \mathbf{R}^{-1} \tilde{\mathbf{z}}, \quad \tilde{\mathbf{x}} = (\mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H})^{-1} \mathbf{H}^\top \mathbf{R}^{-1} \tilde{\mathbf{z}}.} \quad (301)$$

To find the covariance of the corrected estimate $\mathbf{x}^\oplus = \hat{\mathbf{x}} + \tilde{\mathbf{x}}$, substitute $\tilde{\mathbf{z}} = \mathbf{H}(\mathbf{x} - \hat{\mathbf{x}}) + \mathbf{n}$:

$$\mathbf{x}^\oplus = \hat{\mathbf{x}} + (\mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H})^{-1} \mathbf{H}^\top \mathbf{R}^{-1} \tilde{\mathbf{z}} \quad (302)$$

$$= \hat{\mathbf{x}} + (\mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H})^{-1} \mathbf{H}^\top \mathbf{R}^{-1} (\mathbf{H}(\mathbf{x} - \hat{\mathbf{x}}) + \mathbf{n}) \quad (303)$$

$$= \mathbf{x} + (\mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H})^{-1} \mathbf{H}^\top \mathbf{R}^{-1} \mathbf{n}. \quad (304)$$

Hence $\mathbf{x} - \mathbf{x}^\oplus = -(\mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H})^{-1} \mathbf{H}^\top \mathbf{R}^{-1} \mathbf{n}$, and

$$\begin{aligned} \mathbf{P}_{\mathbf{x}^\oplus \mathbf{x}^\oplus} &= \mathbb{E}[(\mathbf{x} - \mathbf{x}^\oplus)(\mathbf{x} - \mathbf{x}^\oplus)^\top] \\ &= (\mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H})^{-1} \mathbf{H}^\top \mathbf{R}^{-1} \mathbb{E}(\mathbf{n} \mathbf{n}^\top) \mathbf{R}^{-1} \mathbf{H} (\mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H})^{-1} \end{aligned} \quad (305)$$

$$= (\mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H})^{-1}, \quad (306)$$

using $\mathbb{E}(\mathbf{n} \mathbf{n}^\top) = \mathbf{R}$. Therefore information and covariance are inverses of each other:

information $\Sigma = \mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H}$, covariance $\mathbf{P} = (\mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H})^{-1}$, $\Sigma = \mathbf{P}^{-1}$, $\mathbf{P} = \Sigma^{-1}$.

(307)

This is fundamental for what follows:

- for **optimization / least squares**, we keep track of the information matrix Σ ;
- for **filtering (EKF)**, we keep track of the covariance $\mathbf{P}$.

4.3 Marginalization

Let $\mathbf{x} \sim \mathcal{N}(\hat{\mathbf{x}}, \mathbf{P}_{\mathbf{x}\mathbf{x}})$, where $\hat{\mathbf{x}}$ is the best estimate and $\mathbf{P}_{\mathbf{x}\mathbf{x}}$ the covariance. Partition the state as $\mathbf{x} = [\mathbf{x}_A^\top, \mathbf{x}_B^\top]^\top$, with $\mathbf{x}_A$ and $\mathbf{x}_B$ both part of $\mathbf{x}$:

$$\mathbf{x} \sim \mathcal{N}\left(\begin{bmatrix} \hat{\mathbf{x}}_A \\ \hat{\mathbf{x}}_B \end{bmatrix}, \begin{bmatrix} \mathbf{P}_{AA} & \mathbf{P}_{AB} \\ \mathbf{P}_{BA} & \mathbf{P}_{BB} \end{bmatrix}\right), \quad \mathbf{P}^\top = \mathbf{P} \text{ (SPD)}, \quad \mathbf{P}_{AB} = \mathbf{P}_{BA}^\top. \quad (308)$$

Covariance form (easy). The marginal of either block is read off directly:

$$\mathbf{x}_A \sim \mathcal{N}(\hat{\mathbf{x}}_A, \mathbf{P}_{AA}), \quad \mathbf{x}_B \sim \mathcal{N}(\hat{\mathbf{x}}_B, \mathbf{P}_{BB}). \quad (309)$$

Marginalization in covariance form is straightforward: just keep the corresponding diagonal block.

Information form (Schur complement). For optimization we work with the information system $(\mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H}) \tilde{\mathbf{x}} = \mathbf{H}^\top \mathbf{R}^{-1} \tilde{\mathbf{z}}$, i.e.

$$\Sigma \tilde{\mathbf{x}} = \mathbf{b}, \quad \begin{bmatrix} \Sigma_{AA} & \Sigma_{AB} \\ \Sigma_{AB}^\top & \Sigma_{BB} \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{x}}_A \\ \tilde{\mathbf{x}}_B \end{bmatrix} = \begin{bmatrix} \mathbf{b}_A \\ \mathbf{b}_B \end{bmatrix}, \quad (310)$$

where Σ is the information matrix and $\mathbf{b}$ the residual vector. We seek $\Sigma'_{AA}, \mathbf{b}'_A$ (and $\Sigma'_{BB}, \mathbf{b}'_B$) such that $\Sigma'_{AA} \tilde{\mathbf{x}}_A = \mathbf{b}'_A$ and $\Sigma'_{BB} \tilde{\mathbf{x}}_B = \mathbf{b}'_B$ hold after eliminating the other block.

Solving $\Sigma \tilde{\mathbf{x}} = \mathbf{b}$ via the block inverse,

$$\begin{bmatrix} \tilde{\mathbf{x}}_A \\ \tilde{\mathbf{x}}_B \end{bmatrix} = \begin{bmatrix} \Sigma_{AA} & \Sigma_{AB} \\ \Sigma_{AB}^\top & \Sigma_{BB} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{b}_A \\ \mathbf{b}_B \end{bmatrix} = \begin{bmatrix} \mathbf{P}_{AA} & \mathbf{P}_{AB} \\ \mathbf{P}_{AB}^\top & \mathbf{P}_{BB} \end{bmatrix} \begin{bmatrix} \mathbf{b}_A \\ \mathbf{b}_B \end{bmatrix} = \begin{bmatrix} \mathbf{P}_{AA} \mathbf{b}_A + \mathbf{P}_{AB} \mathbf{b}_B \\ \mathbf{P}_{AB}^\top \mathbf{b}_A + \mathbf{P}_{BB} \mathbf{b}_B \end{bmatrix}, \quad (311)$$

with the blocks of the inverse given by the matrix inversion lemma,

$$\mathbf{P}_{AA} = (\Sigma_{AA} - \Sigma_{AB} \Sigma_{BB}^{-1} \Sigma_{AB}^\top)^{-1}, \quad \mathbf{P}_{AB} = -\mathbf{P}_{AA} \Sigma_{AB} \Sigma_{BB}^{-1}, \quad (312)$$

$$\mathbf{P}_{BB} = \Sigma_{BB}^{-1} + \Sigma_{BB}^{-1} \Sigma_{AB}^\top \mathbf{P}_{AA} \Sigma_{AB} \Sigma_{BB}^{-1} = (\Sigma_{BB} - \Sigma_{AB}^\top \Sigma_{AA}^{-1} \Sigma_{AB})^{-1}. \quad (313)$$

For the A block,

$$\tilde{\mathbf{x}}_A = \mathbf{P}_{AA} \mathbf{b}_A + \mathbf{P}_{AB} \mathbf{b}_B = \mathbf{P}_{AA} \mathbf{b}_A - \mathbf{P}_{AA} \Sigma_{AB} \Sigma_{BB}^{-1} \mathbf{b}_B = \mathbf{P}_{AA} (\mathbf{b}_A - \Sigma_{AB} \Sigma_{BB}^{-1} \mathbf{b}_B), \quad (314)$$

so $\mathbf{P}_{AA}^{-1}\tilde{\mathbf{x}}_A = \mathbf{b}_A - \Sigma_{AB}\Sigma_{BB}^{-1}\mathbf{b}_B$. Using the identity $\mathbf{P}_{AB} = -\mathbf{P}_{AA}\Sigma_{AB}\Sigma_{BB}^{-1} = -\Sigma_{AA}^{-1}\Sigma_{AB}\mathbf{P}_{BB}$, the B block gives symmetrically

$$\tilde{\mathbf{x}}_B = \mathbf{P}_{AB}^\top \mathbf{b}_A + \mathbf{P}_{BB}\mathbf{b}_B = \mathbf{P}_{BB}(\mathbf{b}_B - \Sigma_{AB}^\top \Sigma_{AA}^{-1} \mathbf{b}_A), \quad (315)$$

so $\mathbf{P}_{BB}^{-1}\tilde{\mathbf{x}}_B = \mathbf{b}_B - \Sigma_{AB}^\top \Sigma_{AA}^{-1} \mathbf{b}_A$. Substituting the Schur-complement expressions for $\mathbf{P}_{AA}^{-1}$ and $\mathbf{P}_{BB}^{-1}$:

$$\boxed{\text{marginalize } \mathbf{x}_B, \text{ keep } \mathbf{x}_A : (\Sigma_{AA} - \Sigma_{AB}\Sigma_{BB}^{-1}\Sigma_{AB}^\top)\tilde{\mathbf{x}}_A = \mathbf{b}_A - \Sigma_{AB}\Sigma_{BB}^{-1}\mathbf{b}_B,} \quad (316)$$

$$\boxed{\text{marginalize } \mathbf{x}_A, \text{ keep } \mathbf{x}_B : (\Sigma_{BB} - \Sigma_{AB}^\top \Sigma_{AA}^{-1} \Sigma_{AB})\tilde{\mathbf{x}}_B = \mathbf{b}_B - \Sigma_{AB}^\top \Sigma_{AA}^{-1} \mathbf{b}_A.} \quad (317)$$

Both the new information block and the new residual are Schur complements of the eliminated variable. This is exactly how variables are dropped from a sliding window while preserving their influence on the kept states.

4.4 Why we care about the information matrix

Take visual measurements as an example. A pixel observation depends on a camera pose and a feature,

$$\mathbf{z}_c = \mathbf{h}_c(\mathbf{x}) + \mathbf{n} = \mathbf{h}_c(\mathbf{x}_c, \mathbf{x}_f) + \mathbf{n}, \quad \tilde{\mathbf{z}}_c = \mathbf{H}_c \tilde{\mathbf{x}}_c + \mathbf{H}_f \tilde{\mathbf{x}}_f + \mathbf{n}. \quad (318)$$

Consider three poses $\mathbf{x}_A, \mathbf{x}_B, \mathbf{x}_C$ observing two features f_1, f_2 (Fig. 12). Stacking the four measurements as $\tilde{\mathbf{z}} = \mathbf{H}\tilde{\mathbf{x}} + \mathbf{n}$, with $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and state $\mathbf{x} = [\mathbf{x}_A^\top, \mathbf{x}_B^\top, \mathbf{x}_C^\top, f_1^\top, f_2^\top]^\top$:

$$\begin{bmatrix} \tilde{\mathbf{z}}_{A1} \\ \tilde{\mathbf{z}}_{B1} \\ \tilde{\mathbf{z}}_{B2} \\ \tilde{\mathbf{z}}_{C2} \end{bmatrix} = \underbrace{\begin{bmatrix} \mathbf{H}_A & \mathbf{0} & \mathbf{0} & \mathbf{H}_{A1} & \mathbf{0} \\ \mathbf{0} & \mathbf{H}_B & \mathbf{0} & \mathbf{H}_{B1} & \mathbf{0} \\ \mathbf{0} & \mathbf{H}_B & \mathbf{0} & \mathbf{0} & \mathbf{H}_{B2} \\ \mathbf{0} & \mathbf{0} & \mathbf{H}_C & \mathbf{0} & \mathbf{H}_{C2} \end{bmatrix}}_{\mathbf{H}} \begin{bmatrix} \tilde{\mathbf{x}}_A \\ \tilde{\mathbf{x}}_B \\ \tilde{\mathbf{x}}_C \\ \tilde{f}_1 \\ \tilde{f}_2 \end{bmatrix} + \mathbf{n}. \quad (319)$$

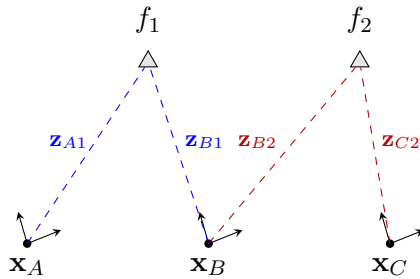


Figure 12: Visual measurement graph: poses $\mathbf{x}_A, \mathbf{x}_B, \mathbf{x}_C$ observe features f_1, f_2 . Each ray $\mathbf{z}_{(\cdot)}$ couples one pose to one feature, which determines the sparsity of $\mathbf{H}$ and of $\Sigma = \mathbf{H}^\top \mathbf{H}$.

The information matrix $\Sigma = \mathbf{H}^\top \mathbf{H}$ has a block-sparse structure (a block (i, j) is nonzero only when states i and j are co-observed in some measurement), shown in Fig. 13. Partitioning into poses $p = \{A, B, C\}$ and features $f = \{f_1, f_2\}$,

$$\Sigma = \mathbf{H}^\top \mathbf{H} \triangleq \begin{bmatrix} \Sigma_{pp} & \Sigma_{pf} \\ \Sigma_{pf}^\top & \Sigma_{ff} \end{bmatrix}. \quad (320)$$

	A	B	C	f_1	f_2	
A	$\otimes$	0	0	$\otimes$	0	Σ_{pp}
B	0	$\otimes$	0	$\otimes$	$\otimes$	
C	0	0	$\otimes$	0	$\otimes$	
f_1	$\otimes$	$\otimes$	0	$\otimes$	0	Σ_{ff}
f_2	0	$\otimes$	$\otimes$	0	$\otimes$	

Figure 13: Sparsity of $\Sigma = \mathbf{H}^\top \mathbf{H}$ ($\otimes$: nonzero block). The pose–pose block Σ_{pp} is block-diagonal, while Σ_{pf} records exactly which pose sees which feature. The information matrix reflects the problem structure.

4.5 A typical visual-odometry / SLAM pipeline

How do the previous two lectures fit together into a working monocular system? A common pipeline incrementally grows poses $C_1, C_2, \dots$ and features $f_1, f_2, \dots$ (Fig. 14):

- (1) **5-point algorithm** on C_1, C_2 : recover ${}^{C_1}\mathbf{R}$ and ${}^{C_1}\mathbf{p}_{C_2}$ (up to scale).
- (2) **Feature triangulation**: with the recovered relative pose, triangulate $f_1, f_2, \dots$.
- (3) **P3P / PnP** for C_3 : with known features, solve ${}^{C_1}\mathbf{R}$, ${}^{C_1}\mathbf{p}_{C_3}$ (then triangulate new features).
- (4) **5-point or P3P/PnP** for C_4 : solve ${}^{C_1}\mathbf{R}$, ${}^{C_1}\mathbf{p}_{C_4}$.
- (5) **Bundle adjustment (BA)** over C_1, C_2, C_3 and $f_1, f_2, f_3, \dots$ (joint refinement of poses and structure).
- (6) **BA** extended to C_1, C_2, C_3, C_4 and $f_1, f_2, f_3, \dots$ as the window grows.
- (7) **Marginalization**: when there are too many poses ($C_1, \dots, C_n$) and features ($f_1, \dots, f_m$), marginalize old states $C_1, f_1, \dots$ to keep the system light-weight.

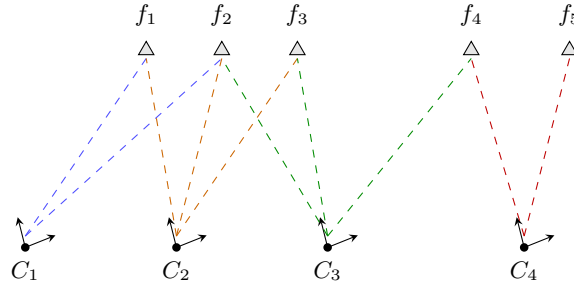


Figure 14: A monocular VO/SLAM pipeline grows poses $C_1, \dots, C_4$ and features $f_1, \dots, f_5$: bootstrap with the 5-point algorithm + triangulation, extend new poses with P3P/PnP, refine with bundle adjustment, and marginalize old states to bound cost. Combined with the previous two lectures, this builds up visual SLAM.

4.6 Designing estimators: many ways to handle $\mathbf{z} = \mathbf{h}(\mathbf{x}) + \mathbf{n}$

Given $\mathbf{z} = \mathbf{h}(\mathbf{x}) + \mathbf{n}$, $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$, there are many ways to linearize and solve, each leading to a different family of algorithms.

- (1) **Iterated re-linearization (GN / LM)**. The typical solution $\min_{\mathbf{x}} \|\mathbf{z} - \mathbf{h}(\mathbf{x})\|_{\mathbf{R}}^2$. Given a linearization $\hat{\mathbf{x}}$, $\tilde{\mathbf{z}} = \mathbf{H}\hat{\mathbf{x}} + \mathbf{n}$; update $\mathbf{x}^\oplus = \hat{\mathbf{x}} + \tilde{\mathbf{x}}$, then *redo* the linearization $\tilde{\mathbf{z}} = \mathbf{H}^\oplus \tilde{\mathbf{x}} + \mathbf{n}$ (Gauss–Newton or Levenberg–Marquardt).

(2) Linearize once + residual correction (FEJ). What if we linearize only once, at a fixed $\hat{\mathbf{x}}^0$? With $\mathbf{H}^0 = \mathbf{H}|_{\hat{\mathbf{x}}^0}$,

$$\mathbf{z} = \mathbf{h}(\hat{\mathbf{x}}^0) + \mathbf{H}^0(\mathbf{x} - \hat{\mathbf{x}}^0) + \mathbf{n} \quad (321)$$

$$\mathbf{z} - \mathbf{h}(\hat{\mathbf{x}}^0) = \mathbf{H}^0(\mathbf{x} - \hat{\mathbf{x}} + \underbrace{\hat{\mathbf{x}} - \hat{\mathbf{x}}^0}_{\Delta \mathbf{x}}) + \mathbf{n} = \mathbf{H}^0 \tilde{\mathbf{x}} + \mathbf{H}^0 \Delta \mathbf{x} + \mathbf{n}, \quad (322)$$

so moving the known offset to the left gives the *residual correction*

$$\underbrace{\mathbf{z} - \mathbf{h}(\hat{\mathbf{x}}^0)}_{\tilde{\mathbf{z}}^0} - \mathbf{H}^0 \Delta \mathbf{x} = \mathbf{H}^0 \tilde{\mathbf{x}} + \mathbf{n}. \quad (323)$$

Keeping the Jacobian fixed at the first estimate is **FEJ** (first-estimate Jacobians) [7], used e.g. in OKVIS [8]; it preserves the correct observability/nullspace structure.

(3) Partial linearization (preintegration / iSAM). For $\mathbf{x} = [\mathbf{x}_A^\top, \mathbf{x}_B^\top]^\top$, $\mathbf{z} = \mathbf{h}(\mathbf{x}_A, \mathbf{x}_B) + \mathbf{n}$, linearize only in $\mathbf{x}_B$ (about $\hat{\mathbf{x}}_B^0$) while keeping $\mathbf{x}_A$ exact:

$$\mathbf{z} = \mathbf{h}(\hat{\mathbf{x}}_A, \hat{\mathbf{x}}_B^0) + \mathbf{H}_A \tilde{\mathbf{x}}_A + \mathbf{H}_B^0 \tilde{\mathbf{x}}_B + \mathbf{H}_B^0 \Delta \mathbf{x}_B + \mathbf{n}, \quad (324)$$

$$\underbrace{\mathbf{z} - \mathbf{h}(\hat{\mathbf{x}}_A, \hat{\mathbf{x}}_B^0)}_{\tilde{\mathbf{z}}^0} - \mathbf{H}_B^0 \Delta \mathbf{x}_B = [\mathbf{H}_A \quad \mathbf{H}_B^0] \begin{bmatrix} \tilde{\mathbf{x}}_A \\ \tilde{\mathbf{x}}_B \end{bmatrix} + \mathbf{n}. \quad (325)$$

This is the idea behind IMU **pre-integration** [9] and **ACI** (analytic combined integration) [10]; **iSAM** [11] uses a similar partial-relinearization idea.

(4) Nullspace projection (MSCKF / structureless). For $\mathbf{x} = [\mathbf{x}_A^\top, \mathbf{x}_B^\top]^\top$, $\tilde{\mathbf{z}} = \mathbf{H}_A \tilde{\mathbf{x}}_A + \mathbf{H}_B \tilde{\mathbf{x}}_B + \mathbf{n}$. If we find $\mathbf{N}$ with $\mathbf{N}^\top \mathbf{H}_B = \mathbf{0}$, then projecting onto the left nullspace removes $\mathbf{x}_B$:

$$\mathbf{N}^\top \tilde{\mathbf{z}} = \mathbf{N}^\top \mathbf{H}_A \tilde{\mathbf{x}}_A + \cancel{\mathbf{N}^\top \mathbf{H}_B \tilde{\mathbf{x}}_B} + \mathbf{N}^\top \mathbf{n}. \quad (326)$$

This is the **MSCKF** [12] (multi-state constraint Kalman filter), a.k.a. the smart / structure-less vision factor: the features are eliminated without ever being added to the state.

(5) Invariant errors. Instead of $\mathbf{x} = \hat{\mathbf{x}} + \tilde{\mathbf{x}}$, put $\mathbf{x}$ on a group $SE(3)$ / $SE_2(3)$ and use a right-invariant error,

$$\mathbf{x} = \hat{\mathbf{x}} \boxplus \tilde{\mathbf{x}} = \text{Exp}(\tilde{\mathbf{x}}) \cdot \hat{\mathbf{x}} \Rightarrow \text{invariant SLAM / VIO}. \quad (327)$$

(6) Decoupled (right-)invariant errors. For $\mathbf{x} = [\mathbf{x}_A^\top, \mathbf{x}_B^\top]^\top$, mix error types per block:

$$\begin{cases} \mathbf{x}_A = \hat{\mathbf{x}}_A \boxplus \tilde{\mathbf{x}}_A = \text{Exp}(\tilde{\mathbf{x}}_A) \cdot \hat{\mathbf{x}}_A & (\text{group / invariant error}) \\ \mathbf{x}_B = \hat{\mathbf{x}}_B + \tilde{\mathbf{x}}_B & (\text{vector error}) \end{cases} \quad (328)$$

which gives decoupled right-invariant SLAM / VIO.

Use your imagination to deal with $\{\mathbf{z} = \mathbf{h}(\mathbf{x}) + \mathbf{n}, \tilde{\mathbf{z}} = \mathbf{H} \tilde{\mathbf{x}} + \mathbf{n}\}$ to design different algorithms.

4.7 Designing a common system: from MAP to the EKF

Consider the canonical prior + motion + measurement system

$$\begin{cases} \mathbf{x}_0 \sim \mathcal{N}(\hat{\mathbf{x}}_0, \mathbf{P}_0) & \rightarrow \text{prior} \\ \mathbf{x}_1 = \mathbf{f}(\mathbf{x}_0) + \mathbf{w}_0, \quad \mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}) & \rightarrow \text{motion model} \\ \mathbf{z}_1 = \mathbf{h}(\mathbf{x}_1) + \mathbf{n}_1, \quad \mathbf{n}_1 \sim \mathcal{N}(\mathbf{0}, \mathbf{R}) & \rightarrow \text{measurement model} \end{cases} \quad (329)$$

The MAP estimate factorizes by Bayes' rule,

$$\max_{\mathbf{x}} p(\mathbf{x} | \mathbf{z}) = \max_{\mathbf{x}} \frac{p(\mathbf{z} | \mathbf{x}) p(\mathbf{x})}{p(\mathbf{z})} = \max_{\mathbf{x}} \frac{p(\mathbf{z}_1 | \mathbf{x}_1) p(\mathbf{x}_1 | \mathbf{x}_0) p(\mathbf{x}_0)}{p(\mathbf{z})}, \quad (330)$$

which is equivalent to the nonlinear least-squares cost

$$\min_{\mathbf{x}} \|\mathbf{z}_1 - \mathbf{h}(\mathbf{x}_1)\|_{\mathbf{R}}^2 + \|\mathbf{x}_1 - \mathbf{f}(\mathbf{x}_0)\|_{\mathbf{Q}}^2 + \|\mathbf{x}_0 - \hat{\mathbf{x}}_0\|_{\mathbf{P}_0}^2. \quad (331)$$

There are two ways to solve it:

1. **Batch:** minimize all three terms jointly $\Rightarrow \hat{\mathbf{x}}_1, \mathbf{P}_1$.
2. **Incremental:**
 - (i) handle the prior + motion terms $\|\mathbf{x}_0 - \hat{\mathbf{x}}_0\|_{\mathbf{P}_0}^2 + \|\mathbf{x}_1 - \mathbf{f}(\mathbf{x}_0)\|_{\mathbf{Q}}^2$;
 - (ii) marginalize $\mathbf{x}_0$, keep $\mathbf{x}_1$, giving $(\hat{\mathbf{x}}_{1|0}, \mathbf{P}_{1|0})$;
 - (iii) add the measurement $\|\mathbf{x}_1 - \hat{\mathbf{x}}_{1|0}\|_{\mathbf{P}_{1|0}}^2 + \|\mathbf{z}_1 - \mathbf{h}(\mathbf{x}_1)\|_{\mathbf{R}}^2 \Rightarrow \hat{\mathbf{x}}_1, \mathbf{P}_1$.

The incremental route is exactly the EKF predict/update, as we now derive.

Propagation (marginalize $\mathbf{x}_0$). Linearize the motion model, $\mathbf{x}_1 = \mathbf{f}(\mathbf{x}_0) + \mathbf{w}_0 \Rightarrow \hat{\mathbf{x}}_1 + \tilde{\mathbf{x}}_1 = \mathbf{f}(\hat{\mathbf{x}}_0) + \mathbf{F}\tilde{\mathbf{x}}_0 + \mathbf{w}_0$, and define the motion residual

$$\mathbf{r}_x \triangleq \hat{\mathbf{x}}_1 - \mathbf{f}(\hat{\mathbf{x}}_0) = \mathbf{F}\tilde{\mathbf{x}}_0 - \tilde{\mathbf{x}}_1 + \mathbf{w}_0, \quad \mathbf{w}_0 = \mathbf{r}_x - [\mathbf{F} \quad -\mathbf{I}] \begin{bmatrix} \tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1 \end{bmatrix}. \quad (332)$$

The prior + motion cost (writing $\mathbf{Q}_0 = \mathbf{Q}$) is

$$\min \|\tilde{\mathbf{x}}_0\|_{\mathbf{P}_0}^2 + \left\| \mathbf{r}_x - [\mathbf{F} \quad -\mathbf{I}] \begin{bmatrix} \tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1 \end{bmatrix} \right\|_{\mathbf{Q}_0}^2. \quad (333)$$

Collecting quadratic and linear terms, the cost C is

$$C = \begin{bmatrix} \tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1 \end{bmatrix}^\top \begin{bmatrix} \mathbf{P}_0^{-1} + \mathbf{F}^\top \mathbf{Q}_0^{-1} \mathbf{F} & -\mathbf{F}^\top \mathbf{Q}_0^{-1} \\ -\mathbf{Q}_0^{-1} \mathbf{F} & \mathbf{Q}_0^{-1} \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1 \end{bmatrix} - 2 \begin{bmatrix} \tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1 \end{bmatrix}^\top \begin{bmatrix} \mathbf{F}^\top \\ -\mathbf{I} \end{bmatrix} \mathbf{Q}_0^{-1} \mathbf{r}_x + \mathbf{r}_x^\top \mathbf{Q}_0^{-1} \mathbf{r}_x. \quad (334)$$

Setting $\partial C / \partial [\tilde{\mathbf{x}}_0; \tilde{\mathbf{x}}_1] = \mathbf{0}$ gives the information system

$$\begin{bmatrix} \mathbf{P}_0^{-1} + \mathbf{F}^\top \mathbf{Q}_0^{-1} \mathbf{F} & -\mathbf{F}^\top \mathbf{Q}_0^{-1} \\ -\mathbf{Q}_0^{-1} \mathbf{F} & \mathbf{Q}_0^{-1} \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1 \end{bmatrix} = \begin{bmatrix} \mathbf{F}^\top \mathbf{Q}_0^{-1} \mathbf{r}_x \\ -\mathbf{Q}_0^{-1} \mathbf{r}_x \end{bmatrix}. \quad (335)$$

Marginalizing $\mathbf{x}_0$ (Schur complement onto the $\mathbf{x}_1$ block) and applying the matrix inversion lemma,

$$\begin{aligned} \left[\mathbf{Q}_0^{-1} - \mathbf{Q}_0^{-1} \mathbf{F} (\mathbf{P}_0^{-1} + \mathbf{F}^\top \mathbf{Q}_0^{-1} \mathbf{F})^{-1} \mathbf{F}^\top \mathbf{Q}_0^{-1} \right] \tilde{\mathbf{x}}_1 &= -\mathbf{Q}_0^{-1} \mathbf{r}_x + \mathbf{Q}_0^{-1} \mathbf{F} (\mathbf{P}_0^{-1} + \mathbf{F}^\top \mathbf{Q}_0^{-1} \mathbf{F})^{-1} \mathbf{F}^\top \mathbf{Q}_0^{-1} \mathbf{r}_x, \\ \underbrace{(\mathbf{Q}_0 + \mathbf{F} \mathbf{P}_0 \mathbf{F}^\top)^{-1}}_{\mathbf{P}_{1|0}^{-1}} \tilde{\mathbf{x}}_1 &= -(\mathbf{Q}_0 + \mathbf{F} \mathbf{P}_0 \mathbf{F}^\top)^{-1} \mathbf{r}_x. \end{aligned} \quad (336)$$

This is the EKF **propagation**: the predicted covariance and mean are

$$\boxed{\mathbf{P}_{1|0} = \mathbf{F} \mathbf{P}_0 \mathbf{F}^\top + \mathbf{Q}_0, \quad \hat{\mathbf{x}}_{1|0} = \mathbf{f}(\hat{\mathbf{x}}_0) \text{ (i.e. } \mathbf{r}_x = \mathbf{0}\text{).}} \quad (337)$$

Update (add the measurement). Now include the measurement $\mathbf{z}_1 = \mathbf{h}(\mathbf{x}_1) + \mathbf{n} \Rightarrow \tilde{\mathbf{z}} = \mathbf{H}\tilde{\mathbf{x}}_1 + \mathbf{n}$. The full cost

$$\|\tilde{\mathbf{z}} - \mathbf{H}\tilde{\mathbf{x}}_1\|_{\mathbf{R}}^2 + \left\| \mathbf{r}_{\mathbf{x}} - [\mathbf{F} \quad -\mathbf{I}] \begin{bmatrix} \tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1 \end{bmatrix} \right\|_{\mathbf{Q}_0}^2 + \|\tilde{\mathbf{x}}_0\|_{\mathbf{P}_0}^2 \quad (338)$$

yields the augmented information system (the measurement adds $\mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H}$ to the $\mathbf{x}_1$ block and $\mathbf{H}^\top \mathbf{R}^{-1} \tilde{\mathbf{z}}$ to its residual),

$$\begin{bmatrix} \mathbf{P}_0^{-1} + \mathbf{F}^\top \mathbf{Q}_0^{-1} \mathbf{F} & -\mathbf{F}^\top \mathbf{Q}_0^{-1} \\ -\mathbf{Q}_0^{-1} \mathbf{F} & \mathbf{Q}_0^{-1} + \mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H} \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1 \end{bmatrix} = \begin{bmatrix} \mathbf{F}^\top \mathbf{Q}_0^{-1} \mathbf{r}_{\mathbf{x}} \\ -\mathbf{Q}_0^{-1} \mathbf{r}_{\mathbf{x}} + \mathbf{H}^\top \mathbf{R}^{-1} \tilde{\mathbf{z}} \end{bmatrix}. \quad (339)$$

Marginalizing $\mathbf{x}_0$ again and recognizing $\mathbf{P}_{1|0}^{-1} = (\mathbf{Q}_0 + \mathbf{F}\mathbf{P}_0\mathbf{F}^\top)^{-1}$,

$$\underbrace{(\mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H} + \mathbf{P}_{1|0}^{-1})}_{\Sigma = \mathbf{P}_{1|1}^{-1}} \tilde{\mathbf{x}}_1 = \mathbf{H}^\top \mathbf{R}^{-1} \tilde{\mathbf{z}} - \underbrace{\mathbf{P}_{1|0}^{-1} \mathbf{r}_{\mathbf{x}}}_{\rightarrow \mathbf{0}}. \quad (340)$$

After propagation $\mathbf{r}_{\mathbf{x}} = \mathbf{0}$, so the second term drops. Applying the matrix inversion lemma to the updated covariance,

$$\mathbf{P}_{1|1} = (\mathbf{P}_{1|0}^{-1} + \mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H})^{-1} = \mathbf{P}_{1|0} - \mathbf{P}_{1|0} \mathbf{H}^\top (\mathbf{H} \mathbf{P}_{1|0} \mathbf{H}^\top + \mathbf{R})^{-1} \mathbf{H} \mathbf{P}_{1|0}, \quad (341)$$

and for the mean correction, using the same lemma to convert information $\rightarrow$ covariance form,

$$\tilde{\mathbf{x}}_1 = (\mathbf{P}_{1|0}^{-1} + \mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H})^{-1} \mathbf{H}^\top \mathbf{R}^{-1} \tilde{\mathbf{z}} \quad (342)$$

$$= (\mathbf{P}_{1|0} - \mathbf{P}_{1|0} \mathbf{H}^\top (\mathbf{H} \mathbf{P}_{1|0} \mathbf{H}^\top + \mathbf{R})^{-1} \mathbf{H} \mathbf{P}_{1|0}) \mathbf{H}^\top \mathbf{R}^{-1} \tilde{\mathbf{z}} \quad (343)$$

$$= \mathbf{P}_{1|0} \mathbf{H}^\top (\mathbf{R}^{-1} - (\mathbf{H} \mathbf{P}_{1|0} \mathbf{H}^\top + \mathbf{R})^{-1} \mathbf{H} \mathbf{P}_{1|0} \mathbf{H}^\top \mathbf{R}^{-1}) \tilde{\mathbf{z}} \quad (344)$$

$$= \mathbf{P}_{1|0} \mathbf{H}^\top (\mathbf{H} \mathbf{P}_{1|0} \mathbf{H}^\top + \mathbf{R})^{-1} ((\mathbf{H} \mathbf{P}_{1|0} \mathbf{H}^\top + \mathbf{R}) \mathbf{R}^{-1} - \mathbf{H} \mathbf{P}_{1|0} \mathbf{H}^\top \mathbf{R}^{-1}) \tilde{\mathbf{z}} \quad (345)$$

$$= \mathbf{P}_{1|0} \mathbf{H}^\top (\mathbf{H} \mathbf{P}_{1|0} \mathbf{H}^\top + \mathbf{R})^{-1} \tilde{\mathbf{z}}. \quad (346)$$

Finally, with the Kalman gain $\mathbf{K} = \mathbf{P}_{1|0} \mathbf{H}^\top (\mathbf{H} \mathbf{P}_{1|0} \mathbf{H}^\top + \mathbf{R})^{-1}$, we recover the EKF **update**:

$$\boxed{\tilde{\mathbf{x}}_1 = \mathbf{P}_{1|0} \mathbf{H}^\top (\mathbf{H} \mathbf{P}_{1|0} \mathbf{H}^\top + \mathbf{R})^{-1} \tilde{\mathbf{z}}, \quad \mathbf{P}_{1|1} = \mathbf{P}_{1|0} - \mathbf{P}_{1|0} \mathbf{H}^\top (\mathbf{H} \mathbf{P}_{1|0} \mathbf{H}^\top + \mathbf{R})^{-1} \mathbf{H} \mathbf{P}_{1|0}.} \quad (347)$$

The extended Kalman filter is thus the incremental (marginalize-then-update) solution of the same MAP problem that batch least squares solves jointly — the two views are connected by the matrix inversion lemma and the Schur complement.

4.8 The relationship between Schur complement and nullspace projection

The Schur-complement marginalization (§4.3) and the nullspace projection (§4.6, item 4, the MSCKF) both eliminate $\mathbf{x}_B$ — it turns out they are the *same* operation. Take $\mathbf{z} = \mathbf{h}(\mathbf{x}_A, \mathbf{x}_B) + \mathbf{n}$, $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$, with $\tilde{\mathbf{z}} = \mathbf{H}_A \tilde{\mathbf{x}}_A + \mathbf{H}_B \tilde{\mathbf{x}}_B + \mathbf{n}$.

(1) **Information form.** The normal equations are

$$\begin{bmatrix} \mathbf{H}_A^\top \mathbf{R}^{-1} \mathbf{H}_A & \mathbf{H}_A^\top \mathbf{R}^{-1} \mathbf{H}_B \\ \mathbf{H}_B^\top \mathbf{R}^{-1} \mathbf{H}_A & \mathbf{H}_B^\top \mathbf{R}^{-1} \mathbf{H}_B \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{x}}_A \\ \tilde{\mathbf{x}}_B \end{bmatrix} = \begin{bmatrix} \mathbf{H}_A^\top \\ \mathbf{H}_B^\top \end{bmatrix} \mathbf{R}^{-1} \tilde{\mathbf{z}}. \quad (348)$$

Marginalizing $\mathbf{x}_B$ and keeping $\mathbf{x}_A$ via the Schur complement gives

$$\boxed{\begin{aligned} & \underbrace{(\mathbf{H}_A^\top \mathbf{R}^{-1} \mathbf{H}_A - \mathbf{H}_A^\top \mathbf{R}^{-1} \mathbf{H}_B (\mathbf{H}_B^\top \mathbf{R}^{-1} \mathbf{H}_B)^{-1} \mathbf{H}_B^\top \mathbf{R}^{-1} \mathbf{H}_A)}_{\text{information}} \tilde{\mathbf{x}}_A \\ &= \underbrace{\mathbf{H}_A^\top \mathbf{R}^{-1} \tilde{\mathbf{z}} - \mathbf{H}_A^\top \mathbf{R}^{-1} \mathbf{H}_B (\mathbf{H}_B^\top \mathbf{R}^{-1} \mathbf{H}_B)^{-1} \mathbf{H}_B^\top \mathbf{R}^{-1} \tilde{\mathbf{z}}}_{\text{residual}}. \end{aligned}} \quad (\text{Eq. 1})$$

(2) **Nullspace form.** Take a (thin) QR-type factorization of $\mathbf{H}_B$,

$$\mathbf{H}_B = [\mathbf{Q}_e \quad \mathbf{Q}_N] \begin{bmatrix} \mathbf{U} \\ \mathbf{0} \end{bmatrix} = \mathbf{Q}_e \mathbf{U} + \mathbf{Q}_N \cdot \mathbf{0} = \mathbf{Q}_e \mathbf{U}, \quad (349)$$

with $\mathbf{U}$ upper triangular and $[\mathbf{Q}_e \quad \mathbf{Q}_N]$ orthogonal, so $\mathbf{Q}_e \mathbf{Q}_e^\top + \mathbf{Q}_N \mathbf{Q}_N^\top = \mathbf{I}$, $\mathbf{Q}_e^\top \mathbf{Q}_e = \mathbf{I}$, $\mathbf{Q}_N^\top \mathbf{Q}_N = \mathbf{I}$, and the left nullspace satisfies $\mathbf{Q}_N^\top \mathbf{H}_B = \mathbf{Q}_N^\top \mathbf{Q}_e \mathbf{U} = \mathbf{0}$. Projecting the residual onto $\mathbf{Q}_N$ removes $\mathbf{x}_B$:

$$\mathbf{Q}_N^\top \tilde{\mathbf{z}} = \mathbf{Q}_N^\top \mathbf{H}_A \tilde{\mathbf{x}}_A + \cancel{\mathbf{Q}_N^\top \mathbf{H}_B \tilde{\mathbf{x}}_B} + \mathbf{Q}_N^\top \mathbf{n} = \mathbf{Q}_N^\top \mathbf{H}_A \tilde{\mathbf{x}}_A + \mathbf{Q}_N^\top \mathbf{n}, \quad (350)$$

with projected noise covariance $\mathbf{Q}_N^\top \mathbf{R} \mathbf{Q}_N$. The resulting normal equations are

$$\boxed{(\mathbf{Q}_N^\top \mathbf{H}_A)^\top (\mathbf{Q}_N^\top \mathbf{R} \mathbf{Q}_N)^{-1} (\mathbf{Q}_N^\top \mathbf{H}_A) \tilde{\mathbf{x}}_A = (\mathbf{Q}_N^\top \mathbf{H}_A)^\top (\mathbf{Q}_N^\top \mathbf{R} \mathbf{Q}_N)^{-1} \mathbf{Q}_N^\top \tilde{\mathbf{z}}.} \quad (\text{Eq. 2})$$

Are Eq. 1 and Eq. 2 equivalent? Consider the isotropic case $\mathbf{R} = \sigma^2 \mathbf{I}$. With $\mathbf{H}_B = \mathbf{Q}_e \mathbf{U}$,

$$\mathbf{H}_B^\top \mathbf{R}^{-1} \mathbf{H}_B = (\mathbf{Q}_e \mathbf{U})^\top \frac{\mathbf{I}}{\sigma^2} \mathbf{Q}_e \mathbf{U} = \frac{\mathbf{U}^\top \mathbf{U}}{\sigma^2}, \quad (351)$$

so the projector that appears in the Schur complement of Eq. 1 is

$$\begin{aligned} \mathbf{H}_A^\top \mathbf{R}^{-1} \mathbf{H}_B (\mathbf{H}_B^\top \mathbf{R}^{-1} \mathbf{H}_B)^{-1} \mathbf{H}_B^\top \mathbf{R}^{-1} \mathbf{H}_A &= \mathbf{H}_A^\top \frac{\mathbf{I}}{\sigma^2} \mathbf{Q}_e \mathbf{U} \sigma^2 (\mathbf{U}^\top \mathbf{U})^{-1} \mathbf{U}^\top \mathbf{Q}_e^\top \frac{\mathbf{I}}{\sigma^2} \mathbf{H}_A \\ &= \frac{1}{\sigma^2} \mathbf{H}_A^\top \mathbf{Q}_e \mathbf{Q}_e^\top \mathbf{H}_A. \end{aligned} \quad (352)$$

Hence the Eq. 1 information becomes, using $\mathbf{Q}_e \mathbf{Q}_e^\top = \mathbf{I} - \mathbf{Q}_N \mathbf{Q}_N^\top$,

$$\frac{1}{\sigma^2} \mathbf{H}_A^\top \mathbf{H}_A - \frac{1}{\sigma^2} \mathbf{H}_A^\top \mathbf{Q}_e \mathbf{Q}_e^\top \mathbf{H}_A = \frac{1}{\sigma^2} \mathbf{H}_A^\top (\mathbf{I} - \mathbf{Q}_e \mathbf{Q}_e^\top) \mathbf{H}_A = \frac{1}{\sigma^2} \mathbf{H}_A^\top \mathbf{Q}_N \mathbf{Q}_N^\top \mathbf{H}_A. \quad (353)$$

The Eq. 2 information, using $\mathbf{Q}_N^\top \mathbf{R} \mathbf{Q}_N = \sigma^2 \mathbf{Q}_N^\top \mathbf{Q}_N = \sigma^2 \mathbf{I}$, is

$$(\mathbf{Q}_N^\top \mathbf{H}_A)^\top (\sigma^2 \mathbf{I})^{-1} (\mathbf{Q}_N^\top \mathbf{H}_A) = \frac{1}{\sigma^2} \mathbf{H}_A^\top \mathbf{Q}_N \mathbf{Q}_N^\top \mathbf{H}_A, \quad (354)$$

which is identical (the residuals match the same way). Therefore, for isotropic noise, Schur-complement marginalization and left-nullspace projection produce exactly the same information and residual — the MSCKF's structureless update is precisely the marginalization of the feature.

5 Lecture 5: The IMU Model and IMU Integration

Topics. (1) the IMU measurement model (gyroscope, accelerometer, biases, noise); (2) continuous vs. discrete noise; (3) the IMU kinematics as a first-order ODE; (4) the big picture: IMU integration in a visual-inertial navigation system; (5) exact integration from t_k to t_{k+1} ; (6) computing $\Delta \mathbf{R}, \Delta \mathbf{p}, \Delta \mathbf{v}$ — preintegration and three methods; (7) mean and covariance propagation with $SO(3) + \mathbb{R}^3$ errors; (8) the linearization using $SE_2(3)$.

5.1 The IMU model

The IMU (inertial measurement unit) is one of the most important sensors for robotics. We use two frames:

- $\{I\}$: the IMU frame (attached to the sensor, see Fig. 15);
- $\{G\}$: the global frame.

The IMU provides two readings: the **gyroscope** reads angular velocity and the **accelerometer** reads acceleration,

$${}^I\boldsymbol{\omega}_m = {}^I\boldsymbol{\omega} + \mathbf{b}_g + \mathbf{n}_g, \quad (355)$$

$${}^I\mathbf{a}_m = {}^I\mathbf{a} + \mathbf{b}_a + \mathbf{n}_a, \quad (356)$$

where

- ${}^I\boldsymbol{\omega}$ is the true angular-rate reading and ${}^I\mathbf{a}$ the true acceleration reading;
- $\mathbf{b}_g$ is the gyro bias, driven by a random walk $\dot{\mathbf{b}}_g = \mathbf{n}_{wg}$; $\mathbf{b}_a$ is the accel bias, driven by a random walk $\dot{\mathbf{b}}_a = \mathbf{n}_{wa}$;
- $\mathbf{n}_g \sim \mathcal{N}(\mathbf{0}, \sigma_g^2 \mathbf{I}_3)$ and $\mathbf{n}_a \sim \mathcal{N}(\mathbf{0}, \sigma_a^2 \mathbf{I}_3)$ are Gaussian noises.

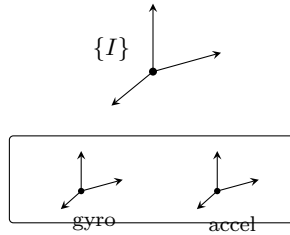


Figure 15: The IMU carries its own frame $\{I\}$; onboard, the gyroscope reads the angular velocity and the accelerometer reads the acceleration.

Another way of modeling the accelerometer makes gravity explicit:

$${}^I\mathbf{a}_m = ({}^I\mathbf{a}_b - {}^I_G\mathbf{R}^G\mathbf{g}) + \mathbf{b}_a + \mathbf{n}_a, \quad {}^G\mathbf{g} = \begin{bmatrix} 0 \\ 0 \\ -9.8 \end{bmatrix}, \quad (357)$$

where ${}^I\mathbf{a}_b$ is the body acceleration and ${}^G\mathbf{g}$ is gravity in $\{G\}$.

We use (355) and (356) as our measurements. Inverting them gives the true readings:

$$\text{true gyro reading: } {}^I\boldsymbol{\omega} = {}^I\boldsymbol{\omega}_m - \mathbf{b}_g - \mathbf{n}_g, \quad (358)$$

$$\text{true accel reading: } {}^I\mathbf{a} = {}^I\mathbf{a}_m - \mathbf{b}_a - \mathbf{n}_a. \quad (359)$$

The IMU state. When we model the IMU, we use the following parameters:

- $\{^G_I \mathbf{R}, ^G_I \mathbf{p}_I\}$: IMU orientation and position — 6 DoF;
- $^G_I \mathbf{v}_I$: IMU velocity — 3 DoF;
- $\mathbf{b}_g$: gyro bias — 3 DoF; $\mathbf{b}_a$: accel bias — 3 DoF.

Then our state vector for the IMU is

$$\mathbf{x}_I = \{^G_I \mathbf{R}, ^G_I \mathbf{p}_I, ^G_I \mathbf{v}_I, \mathbf{b}_g, \mathbf{b}_a\}, \quad 15 \text{ DoF}. \quad (360)$$

Some places use the quaternion $^G_I \mathbf{R} \Leftrightarrow ^G_I \bar{q}$ (Lecture 2); then

$$\mathbf{x}_I = \left[^G_I \bar{q}^\top \quad ^G_I \mathbf{p}_I^\top \quad ^G_I \mathbf{v}_I^\top \quad \mathbf{b}_g^\top \quad \mathbf{b}_a^\top \right]^\top, \quad 16 \times 1 \Rightarrow 15 \text{ DoF}. \quad (361)$$

5.2 Continuous and discrete noise

For the IMU we specify *continuous-time* noise, $\mathbf{n}_c \sim \mathcal{N}(\mathbf{0}, \sigma_c^2 \mathbf{I}_3)$, characterized by

$$\mathbb{E}[\mathbf{n}_c(t)] = \mathbf{0}_{3 \times 1}, \quad \mathbb{E}[\mathbf{n}_c(t + \tau) \mathbf{n}_c^\top(t)] = \sigma_c^2 \mathbf{I}_3 \delta(\tau), \quad (362)$$

where $\delta(\tau)$ is the impulse function: $\delta(\tau) = 1$ if $\tau = 0$ and $\delta(\tau) = 0$ if $\tau \neq 0$.

Consider a random-walk bias driven by this noise, $\dot{\mathbf{b}}(t) = \mathbf{n}_c(t)$. Integrating from t_k to t_{k+1} ,

$$\int_{t_k}^{t_{k+1}} \dot{\mathbf{b}}(t) dt = \int_{t_k}^{t_{k+1}} \mathbf{n}_c(t) dt = \mathbf{b}_{k+1} - \mathbf{b}_k \Rightarrow \Delta \mathbf{b} \triangleq \mathbf{b}_{k+1} - \mathbf{b}_k = \int_{t_k}^{t_{k+1}} \mathbf{n}_c(t) dt. \quad (363)$$

Its mean is $\mathbb{E}(\Delta \mathbf{b}) = \mathbb{E}(\int_{t_k}^{t_{k+1}} \mathbf{n}_c(t) dt) = \mathbf{0}_{3 \times 1}$, and its covariance is

$$\text{Cov}(\Delta \mathbf{b}) \triangleq \mathbf{P}_{\Delta \mathbf{b}} = \mathbb{E}(\Delta \mathbf{b} \Delta \mathbf{b}^\top) = \mathbb{E} \left(\int_{t_k}^{t_{k+1}} \mathbf{n}_c(t) dt \cdot \int_{t_k}^{t_{k+1}} \mathbf{n}_c^\top(t) dt \right) \quad (364)$$

$$= \mathbb{E} \left(\int_{t_k}^{t_{k+1}} \int_{t_k}^{t_{k+1}} \mathbf{n}_c(t) \mathbf{n}_c^\top(s) dt ds \right) \quad (365)$$

$$= \int_{t_k}^{t_{k+1}} \mathbb{E}(\mathbf{n}_c(t) \mathbf{n}_c^\top(t)) dt \quad (366)$$

$$= \int_{t_k}^{t_{k+1}} \sigma_c^2 \mathbf{I}_3 dt = \delta t \sigma_c^2 \mathbf{I}_3, \quad (367)$$

where $\delta t = t_{k+1} - t_k$ (the impulse $\delta(\cdot)$ collapses the double integral). Hence we can **discretize** $\mathbf{n}_c$ in $[t_k, t_{k+1}]$ as

$$\boxed{\mathbf{n}_d \sim \mathcal{N} \left(\mathbf{0}, \frac{\sigma_c^2}{\delta t} \mathbf{I}_3 \right), \quad \delta t = t_{k+1} - t_k.} \quad (368)$$

Then $\Delta \mathbf{b} = \int_{t_k}^{t_{k+1}} \mathbf{n}_c dt = \mathbf{n}_d \cdot \delta t$, and the discrete covariance is consistent with the continuous one:

$$\text{Cov}(\Delta \mathbf{b})_d = \mathbb{E}(\mathbf{n}_d \mathbf{n}_d^\top) \delta t^2 = \delta t^2 \cdot \frac{\sigma_c^2}{\delta t} \mathbf{I}_3 = \sigma_c^2 \delta t \mathbf{I}_3 = \text{Cov}(\Delta \mathbf{b})_c. \checkmark \quad (369)$$

5.3 The IMU kinematics: a first-order ODE

The IMU state evolves by the first-order ordinary differential equation

$$\begin{cases} {}^G\dot{\mathbf{R}} = {}^G\mathbf{R} \left[{}^I\boldsymbol{\omega} \right]_{\times} = {}^G\mathbf{R} \left[{}^I\boldsymbol{\omega}_m - \mathbf{b}_g - \mathbf{n}_g \right]_{\times}, \\ {}^G\dot{\mathbf{p}}_I = {}^G\mathbf{v}_I, \\ {}^G\dot{\mathbf{v}}_I = {}^G\mathbf{a} + {}^G\mathbf{g} = {}^G\mathbf{R} \left({}^I\mathbf{a}_m - \mathbf{b}_a - \mathbf{n}_a \right) + {}^G\mathbf{g}, \\ {}^I\dot{\mathbf{b}}_g = \mathbf{n}_{wg}, \quad {}^I\dot{\mathbf{b}}_a = \mathbf{n}_{wa}. \end{cases} \quad (370)$$

Stacking, we obtain the compact form

$$\boxed{\dot{\mathbf{x}}_I = \mathbf{f}(\mathbf{x}_I, \mathbf{n}_I), \quad \mathbf{n}_I = \begin{bmatrix} \mathbf{n}_g \\ \mathbf{n}_a \\ \mathbf{n}_{wg} \\ \mathbf{n}_{wa} \end{bmatrix}, \quad \begin{array}{ll} \mathbf{n}_g \sim \mathcal{N}(\mathbf{0}, \sigma_g^2 \mathbf{I}_3), & \mathbf{n}_a \sim \mathcal{N}(\mathbf{0}, \sigma_a^2 \mathbf{I}_3), \\ \mathbf{n}_{wg} \sim \mathcal{N}(\mathbf{0}, \sigma_{wg}^2 \mathbf{I}_3), & \mathbf{n}_{wa} \sim \mathcal{N}(\mathbf{0}, \sigma_{wa}^2 \mathbf{I}_3), \end{array}} \quad (371)$$

and, per the previous subsection, the discretized noise over one interval is

$$\mathbf{n}_{Id} = \begin{bmatrix} \mathbf{n}_{gd} \\ \mathbf{n}_{ad} \\ \mathbf{n}_{wgd} \\ \mathbf{n}_{wad} \end{bmatrix}, \quad \sigma_{gd}^2 = \frac{\sigma_g^2}{\delta t}, \quad \sigma_{ad}^2 = \frac{\sigma_a^2}{\delta t}, \quad \sigma_{wgd}^2 = \frac{\sigma_{wg}^2}{\delta t}, \quad \sigma_{wad}^2 = \frac{\sigma_{wa}^2}{\delta t}. \quad (372)$$

The IMU model is done. There are various IMU models, but the basic idea is the same.

5.4 IMU integration: the big picture

Recall the typical linear system from Lecture 1 and put it side by side with our sensor models (Fig. 16):

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{Ax} + \mathbf{Bu} \rightarrow \text{state evolution} \rightarrow \text{IMU model: } \dot{\mathbf{x}}_I = \mathbf{f}(\mathbf{x}_I, \mathbf{n}_I), \\ \mathbf{y} = \mathbf{Cx} + \mathbf{Du} \rightarrow \text{observation / measurements} \rightarrow \text{vision / camera model: } \mathbf{z}_c = \mathbf{h}_c(\mathbf{x}) + \mathbf{n}_c. \end{cases} \quad (373)$$

Together they form a **VINS** (visual-inertial navigation system). These two models are the focus of our tutorials: how to process the camera measurement $\mathbf{z}_c = \mathbf{h}_c(\mathbf{x}) + \mathbf{n}_c$ was covered in Lectures 3 and 4; we will learn how to process the IMU in this Lecture 5.

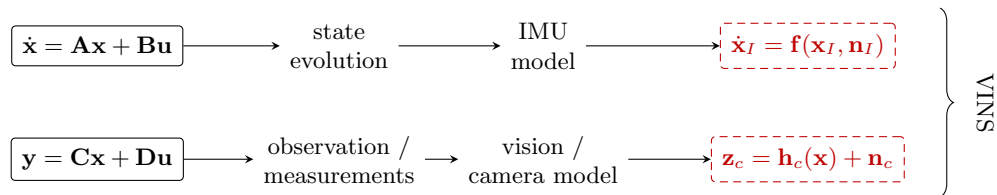


Figure 16: The big picture: the Lecture-1 linear system maps onto the two VINS models — the IMU model plays the state evolution (this lecture) and the camera model plays the observation (Lectures 3–4).

The problem formulation. Given $\mathbf{x}_k \sim \mathcal{N}(\hat{\mathbf{x}}_k, \mathbf{P}_k)$ and the IMU readings $({}^I\boldsymbol{\omega}_{m_k}, {}^I\mathbf{a}_{m_k})$, solve

$$\boxed{\mathbf{x}_{k+1} \sim \mathcal{N}(\hat{\mathbf{x}}_{k+1}, \mathbf{P}_{k+1}) \Rightarrow \text{two targets: } \hat{\mathbf{x}}_{k+1} \text{ and } \mathbf{P}_{k+1}.} \quad (374)$$

That is, the ODE $\dot{\mathbf{x}}_I = \mathbf{f}(\mathbf{x}_I, \mathbf{n}_I)$ carries the *current state distribution* $\mathbf{x}_k \sim \mathcal{N}(\hat{\mathbf{x}}_k, \mathbf{P}_k)$ into the *next state distribution* $\mathbf{x}_{k+1} \sim \mathcal{N}(\hat{\mathbf{x}}_{k+1}, \mathbf{P}_{k+1})$.

How to solve it? From $\dot{\mathbf{x}}_I = \mathbf{f}(\mathbf{x}_I, \mathbf{n}_I)$ we can get the *integrated function*

$$\mathbf{x}_{k+1} = \mathbf{f}_{\text{int}}(\mathbf{x}_k, {}^I\boldsymbol{\omega}_m, {}^I\mathbf{a}_m, \mathbf{n}_k), \quad \begin{cases} \mathbf{x}_{k+1} = \mathbf{x}_{I_{k+1}}, \\ \mathbf{x}_k = \mathbf{x}_{I_k}, \\ \mathbf{n}_k = \mathbf{n}_{I_k} \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_{I_k}). \end{cases} \quad (375)$$

Hence the **mean** is propagated with zero noise,

$$\hat{\mathbf{x}}_{k+1} = \mathbf{f}_{\text{int}}(\hat{\mathbf{x}}_k, {}^I\boldsymbol{\omega}_m, {}^I\mathbf{a}_m, \mathbf{0}) \rightarrow \text{solved } \hat{\mathbf{x}}_{k+1}. \quad (376)$$

Then, linearizing $\mathbf{f}_{\text{int}}$, we get the error-state propagation with the discretized noise $\mathbf{n}_{kd}$,

$$\tilde{\mathbf{x}}_{k+1} = \boldsymbol{\Phi}_{(k+1,k)} \tilde{\mathbf{x}}_k + \mathbf{G}_k \mathbf{n}_{kd}, \quad (377)$$

and therefore the **covariance**

$$\mathbf{P}_{k+1} = \mathbb{E}(\tilde{\mathbf{x}}_{k+1} \tilde{\mathbf{x}}_{k+1}^\top) = \boldsymbol{\Phi}_{(k+1,k)} \mathbf{P}_k \boldsymbol{\Phi}_{(k+1,k)}^\top + \mathbf{G}_k \mathbf{R}_{Id} \mathbf{G}_k^\top \rightarrow \text{solved } \mathbf{P}_{k+1}. \quad (378)$$

Given the big picture, let's explore the detailed derivations.

5.5 Exact integration from t_k to t_{k+1}

The input. Split each true reading into its estimate and error,

$${}^I\boldsymbol{\omega} = {}^I\boldsymbol{\omega}_m - \mathbf{b}_g - \mathbf{n}_g = \underbrace{({}^I\boldsymbol{\omega}_m - \hat{\mathbf{b}}_g)}_{{}^I\tilde{\boldsymbol{\omega}}} + \underbrace{(-\tilde{\mathbf{b}}_g - \mathbf{n}_g)}_{{}^I\tilde{\boldsymbol{\omega}}} = {}^I\hat{\boldsymbol{\omega}} + {}^I\tilde{\boldsymbol{\omega}}, \quad (379)$$

$${}^I\mathbf{a} = {}^I\mathbf{a}_m - \mathbf{b}_a - \mathbf{n}_a = \underbrace{({}^I\mathbf{a}_m - \hat{\mathbf{b}}_a)}_{{}^I\tilde{\mathbf{a}}} + \underbrace{(-\tilde{\mathbf{b}}_a - \mathbf{n}_a)}_{{}^I\tilde{\mathbf{a}}} = {}^I\hat{\mathbf{a}} + {}^I\tilde{\mathbf{a}}. \quad (380)$$

We integrate from k to $k+1$, with $\delta t = t_{k+1} - t_k$.

For rotation.

$${}^G_{I_{k+1}} \mathbf{R} = {}^G_{I_k} \mathbf{R} \cdot {}^{I_k}_{I_{k+1}} \mathbf{R}, \quad \Delta \mathbf{R} \triangleq {}^{I_k}_{I_{k+1}} \mathbf{R}, \quad (381)$$

where $\Delta \mathbf{R}$ is the rotation integration from ${}^I\boldsymbol{\omega}$.

For velocity. From ${}^G\dot{\mathbf{v}}_{I_\tau} = {}^G_{I_\tau}\mathbf{R} {}^{I_\tau}\mathbf{a} + {}^G\mathbf{g}$,

$$\int_{t_k}^{t_{k+1}} {}^G\dot{\mathbf{v}}_{I_\tau} d\tau = \int_{t_k}^{t_{k+1}} ({}^G_{I_\tau}\mathbf{R} {}^{I_\tau}\mathbf{a} + {}^G\mathbf{g}) d\tau. \quad (382)$$

The left-hand side is $\int_{t_k}^{t_{k+1}} {}^G\dot{\mathbf{v}}_{I_\tau} d\tau = {}^G\mathbf{v}_{I_{k+1}} - {}^G\mathbf{v}_{I_k}$, and the right-hand side is

$$\int_{t_k}^{t_{k+1}} ({}^G_{I_\tau}\mathbf{R} {}^{I_\tau}\mathbf{a} + {}^G\mathbf{g}) d\tau = \int_{t_k}^{t_{k+1}} {}^G_{I_\tau}\mathbf{R} {}^{I_\tau}\mathbf{a} d\tau + {}^G\mathbf{g} \delta t \quad (383)$$

$$= \int_{t_k}^{t_{k+1}} {}^G_{I_k}\mathbf{R} {}^{I_k}\mathbf{R} {}^{I_\tau}\mathbf{a} d\tau + {}^G\mathbf{g} \delta t \quad (384)$$

$$= {}^G_{I_k}\mathbf{R} \underbrace{\int_{t_k}^{t_{k+1}} {}^{I_k}\mathbf{R} {}^{I_\tau}\mathbf{a} d\tau}_{\Delta \mathbf{v}} + {}^G\mathbf{g} \delta t. \quad (385)$$

Then

$${}^G\mathbf{v}_{I_{k+1}} = {}^G\mathbf{v}_{I_k} + {}^G_{I_k}\mathbf{R} \Delta \mathbf{v} + {}^G\mathbf{g} \delta t, \quad \Delta \mathbf{v} \triangleq \int_{t_k}^{t_{k+1}} {}^{I_k}\mathbf{R} {}^{I_\tau}\mathbf{a} d\tau, \quad (386)$$

where $\Delta \mathbf{v}$ is the integration of ${}^I\boldsymbol{\omega}$ and ${}^I\mathbf{a}$.

For position. From ${}^G\dot{\mathbf{p}}_{I_\tau} = {}^G\mathbf{v}_{I_\tau}$,

$$\int_{t_k}^{t_{k+1}} {}^G\dot{\mathbf{p}}_{I_\tau} d\tau = \int_{t_k}^{t_{k+1}} {}^G\mathbf{v}_{I_\tau} d\tau. \quad (387)$$

Note that, by the velocity result applied up to time τ (with $\delta\tau = \tau - t_k$),

$${}^G\mathbf{v}_{I_\tau} = {}^G\mathbf{v}_{I_k} + {}^G_{I_k}\mathbf{R} \int_{t_k}^{\tau} {}^{I_k}\mathbf{R} {}^{I_s}\mathbf{a} ds + {}^G\mathbf{g} \delta\tau. \quad (388)$$

The left-hand side is ${}^G\mathbf{p}_{I_{k+1}} - {}^G\mathbf{p}_{I_k}$, and the right-hand side becomes

$$\int_{t_k}^{t_{k+1}} {}^G\mathbf{v}_{I_\tau} d\tau = \int_{t_k}^{t_{k+1}} \left({}^G\mathbf{v}_{I_k} + {}^G_{I_k}\mathbf{R} \int_{t_k}^{\tau} {}^{I_k}\mathbf{R} {}^{I_s}\mathbf{a} ds + {}^G\mathbf{g} \delta\tau \right) d\tau \quad (389)$$

$$= \int_{t_k}^{t_{k+1}} {}^G\mathbf{v}_{I_k} d\tau + \int_{t_k}^{t_{k+1}} {}^G\mathbf{g} \delta\tau d\tau + {}^G_{I_k}\mathbf{R} \int_{t_k}^{t_{k+1}} \int_{t_k}^{\tau} {}^{I_k}\mathbf{R} {}^{I_s}\mathbf{a} ds d\tau \quad (390)$$

$$= {}^G\mathbf{v}_{I_k} \delta t + \frac{1}{2} {}^G\mathbf{g} \delta t^2 + {}^G_{I_k}\mathbf{R} \underbrace{\int_{t_k}^{t_{k+1}} \int_{t_k}^{\tau} {}^{I_k}\mathbf{R} {}^{I_s}\mathbf{a} ds d\tau}_{\Delta \mathbf{p}}. \quad (391)$$

Then

$${}^G\mathbf{p}_{I_{k+1}} = {}^G\mathbf{p}_{I_k} + {}^G\mathbf{v}_{I_k} \delta t + {}^G_{I_k}\mathbf{R} \Delta \mathbf{p} + \frac{1}{2} {}^G\mathbf{g} \delta t^2, \quad \Delta \mathbf{p} \triangleq \int_{t_k}^{t_{k+1}} \int_{t_k}^{\tau} {}^{I_k}\mathbf{R} {}^{I_s}\mathbf{a} ds d\tau, \quad (392)$$

where $\Delta \mathbf{p}$ is the *double* integration of ${}^I\boldsymbol{\omega}$ and ${}^I\mathbf{a}$.

For the biases. From $\dot{\mathbf{b}}_g = \mathbf{n}_{wg}$,

$$\int_{t_k}^{t_{k+1}} \dot{\mathbf{b}}_g d\tau = \int_{t_k}^{t_{k+1}} \mathbf{n}_{wg} d\tau \Rightarrow \mathbf{b}_{g_{k+1}} = \mathbf{b}_{g_k} + \int_{t_k}^{t_{k+1}} \mathbf{n}_{wg} d\tau, \quad (393)$$

and for the accel bias, $\mathbf{b}_{a_{k+1}} = \mathbf{b}_{a_k} + \int_{t_k}^{t_{k+1}} \mathbf{n}_{wa} d\tau$.

Finally. Collecting everything:

$\begin{aligned} {}^G_{I_{k+1}} \mathbf{R} &= {}^G_{I_k} \mathbf{R} \cdot \Delta \mathbf{R}, \\ {}^G \mathbf{p}_{I_{k+1}} &= {}^G \mathbf{p}_{I_k} + {}^G \mathbf{v}_{I_k} \delta t + {}^G_{I_k} \mathbf{R} \Delta \mathbf{p} + \frac{1}{2} {}^G \mathbf{g} \delta t^2, \\ {}^G \mathbf{v}_{I_{k+1}} &= {}^G \mathbf{v}_{I_k} + {}^G_{I_k} \mathbf{R} \Delta \mathbf{v} + {}^G \mathbf{g} \delta t, \\ \mathbf{b}_{g_{k+1}} &= \mathbf{b}_{g_k} + \int_{t_k}^{t_{k+1}} \mathbf{n}_{wg} d\tau, \\ \mathbf{b}_{a_{k+1}} &= \mathbf{b}_{a_k} + \int_{t_k}^{t_{k+1}} \mathbf{n}_{wa} d\tau, \end{aligned}$	$\begin{aligned} \Delta \mathbf{R} &= {}^{I_k}_{I_{k+1}} \mathbf{R}, \\ \Delta \mathbf{p} &= \int_{t_k}^{t_{k+1}} \int_{t_k}^{\tau} {}^{I_k}_{I_s} \mathbf{R} {}^{I_s} \mathbf{a} ds d\tau, \\ \Delta \mathbf{v} &= \int_{t_k}^{t_{k+1}} {}^{I_k}_{I_\tau} \mathbf{R} {}^{I_\tau} \mathbf{a} d\tau, \\ \Delta \mathbf{b}_g &= \int_{t_k}^{t_{k+1}} \mathbf{n}_{wg} d\tau, \\ \Delta \mathbf{b}_a &= \int_{t_k}^{t_{k+1}} \mathbf{n}_{wa} d\tau. \end{aligned}$	(394)
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This is exactly the integrated function $\mathbf{x}_{k+1} = \mathbf{f}_{\text{int}}(\mathbf{x}_k, {}^I \boldsymbol{\omega}_m, {}^I \mathbf{a}_m, \mathbf{n}_k)$.

To this step, no assumption or approximation is made — every step is accurate. The accuracy of the integration is decided by how $\Delta \mathbf{R}$, $\Delta \mathbf{p}$, $\Delta \mathbf{v}$ are computed from ${}^I \boldsymbol{\omega}$ and ${}^I \mathbf{a}$.

5.6 Computing $\Delta \mathbf{R}$, $\Delta \mathbf{p}$, $\Delta \mathbf{v}$: preintegration and three methods

The way to compute $\Delta \mathbf{R}$, $\Delta \mathbf{p}$, $\Delta \mathbf{v}$ given $\boldsymbol{\omega}_m$ and $\mathbf{a}_m$ defines the algorithms. A few discussions follow.

(1) Connection to preintegration. $\Delta \mathbf{R}$, $\Delta \mathbf{p}$, $\Delta \mathbf{v}$ are also called the *pre-integrated measurements*: they are only affected by ${}^I \boldsymbol{\omega}_m$, ${}^I \mathbf{a}_m$ and $\mathbf{b}_g$, $\mathbf{b}_a$ (not by the global state). Inverting (394),

$$\Delta \mathbf{R} \triangleq {}^{I_k}_{I_{k+1}} \mathbf{R} = {}^G_{I_k} \mathbf{R}^\top {}^G_{I_{k+1}} \mathbf{R}, \quad (395)$$

$$\Delta \mathbf{p} \triangleq \int_{t_k}^{t_{k+1}} \int_{t_k}^{\tau} {}^{I_k}_{I_s} \mathbf{R} {}^{I_s} \mathbf{a} ds d\tau = {}^G_{I_k} \mathbf{R}^\top ({}^G \mathbf{p}_{I_{k+1}} - {}^G \mathbf{p}_{I_k} - {}^G \mathbf{v}_{I_k} \delta t - \frac{1}{2} {}^G \mathbf{g} \delta t^2), \quad (396)$$

$$\Delta \mathbf{v} \triangleq \int_{t_k}^{t_{k+1}} {}^{I_k}_{I_\tau} \mathbf{R} {}^{I_\tau} \mathbf{a} d\tau = {}^G_{I_k} \mathbf{R}^\top ({}^G \mathbf{v}_{I_{k+1}} - {}^G \mathbf{v}_{I_k} - {}^G \mathbf{g} \delta t), \quad (397)$$

$$\Delta \mathbf{b}_g \triangleq \int_{t_k}^{t_{k+1}} \mathbf{n}_{wg} d\tau = \mathbf{b}_{g_{k+1}} - \mathbf{b}_{g_k}, \quad (398)$$

$$\Delta \mathbf{b}_a \triangleq \int_{t_k}^{t_{k+1}} \mathbf{n}_{wa} d\tau = \mathbf{b}_{a_{k+1}} - \mathbf{b}_{a_k}. \quad (399)$$

So the IMU integration/propagation and pre-integration are connected: they compute the same quantities.

(2) Discrete preintegration on the manifold. For example, IMU preintegration on manifold (Forster, Carlone, et al.) [9] and VINS-Mono (Tong Qin, Shaojie Shen) [13]. Assume ${}^{I_k}\boldsymbol{\omega}$ and ${}^{I_k}\mathbf{a}$ are constant in $[t_k, t_{k+1}]$ and $\delta t = t_{k+1} - t_k$ is small, so ${}^{I_k}_{I_\tau}\mathbf{R} \approx \mathbf{I}$ as $\delta t \rightarrow 0$:

$$\Delta \mathbf{R} \triangleq {}^{I_k}_{I_{k+1}}\mathbf{R} \approx \text{Exp}({}^{I_k}\boldsymbol{\omega} \cdot \delta t), \quad (400)$$

$$\Delta \mathbf{p} = \int_{t_k}^{t_{k+1}} \int_{t_k}^{\tau} {}^{I_k}_{I_s}\mathbf{R} {}^{I_s}\mathbf{a} ds d\tau \approx \int_{t_k}^{t_{k+1}} \int_{t_k}^{\tau} \mathbf{I}_3 \cdot {}^{I_k}\mathbf{a} ds d\tau = \frac{1}{2} {}^{I_k}\mathbf{a} \delta t^2, \quad (401)$$

$$\Delta \mathbf{v} = \int_{t_k}^{t_{k+1}} {}^{I_k}_{I_\tau}\mathbf{R} {}^{I_\tau}\mathbf{a} d\tau \approx {}^{I_k}\mathbf{a} \delta t. \quad (402)$$

Then the propagation becomes

$$\begin{cases} {}^G_{I_{k+1}}\mathbf{R} = {}^G_{I_k}\mathbf{R} \cdot \text{Exp}({}^{I_k}\boldsymbol{\omega} \delta t), \\ {}^G\mathbf{p}_{I_{k+1}} = {}^G\mathbf{p}_{I_k} + {}^G\mathbf{v}_{I_k} \delta t + \frac{1}{2} {}^G_{I_k}\mathbf{R} {}^{I_k}\mathbf{a} \delta t^2 + \frac{1}{2} {}^G\mathbf{g} \delta t^2, \\ {}^G\mathbf{v}_{I_{k+1}} = {}^G\mathbf{v}_{I_k} + {}^G_{I_k}\mathbf{R} {}^{I_k}\mathbf{a} \delta t + {}^G\mathbf{g} \delta t, \\ \mathbf{b}_{g_{k+1}} = \mathbf{b}_{g_k} + \int_{t_k}^{t_{k+1}} \mathbf{n}_{wg} d\tau, \quad \mathbf{b}_{a_{k+1}} = \mathbf{b}_{a_k} + \int_{t_k}^{t_{k+1}} \mathbf{n}_{wa} d\tau. \end{cases} \quad (403)$$

This is easy and simple, but it might lose some accuracy for fast rotation and low sampling rate (δt larger).

(3) Analytic integration: ACI² and CPI. ACI² — analytic combined IMU integration (Yulin/Chuchu) [14]; CPI — continuous preintegration (Kevin/Patrick) [15, 10]. If we assume only that ${}^{I_k}\boldsymbol{\omega}$ and ${}^{I_k}\mathbf{a}$ are constant in $[t_k, t_{k+1}]$ (without the small- δt approximation), then ${}^{I_k}_{I_\tau}\mathbf{R} = \text{Exp}({}^{I_k}\boldsymbol{\omega} \delta \tau)$ and

$$\Delta \mathbf{R} \triangleq {}^{I_k}_{I_{k+1}}\mathbf{R} = \text{Exp}({}^{I_k}\boldsymbol{\omega} \cdot \delta t), \quad (404)$$

$$\Delta \mathbf{p} = \int_{t_k}^{t_{k+1}} \int_{t_k}^{\tau} {}^{I_k}_{I_s}\mathbf{R} {}^{I_s}\mathbf{a} ds d\tau = \int_{t_k}^{t_{k+1}} \int_{t_k}^{\tau} \text{Exp}({}^{I_k}\boldsymbol{\omega} \delta s) {}^{I_k}\mathbf{a} ds d\tau, \quad (405)$$

$$\Delta \mathbf{v} = \int_{t_k}^{t_{k+1}} {}^{I_k}_{I_\tau}\mathbf{R} {}^{I_\tau}\mathbf{a} d\tau = \int_{t_k}^{t_{k+1}} \text{Exp}({}^{I_k}\boldsymbol{\omega} \delta \tau) {}^{I_k}\mathbf{a} d\tau, \quad (406)$$

which admit closed-form (analytic) solutions. We now use this form for the detailed mean and covariance propagation.

5.7 Mean and covariance propagation with $SO(3) + \mathbb{R}^3$ errors

The input, discretized. As in §5.5, but discretizing the noises over the interval ($\mathbf{n}_g \rightarrow \mathbf{n}_{gd}$, $\mathbf{n}_a \rightarrow \mathbf{n}_{ad}$):

$${}^I\boldsymbol{\omega} = {}^I\hat{\boldsymbol{\omega}} + {}^I\tilde{\boldsymbol{\omega}}, \quad \begin{cases} {}^I\hat{\boldsymbol{\omega}} = {}^I\boldsymbol{\omega}_m - \hat{\mathbf{b}}_g \\ {}^I\tilde{\boldsymbol{\omega}} = -\tilde{\mathbf{b}}_g - \mathbf{n}_{gd} \end{cases} \quad {}^I\mathbf{a} = {}^I\hat{\mathbf{a}} + {}^I\tilde{\mathbf{a}}, \quad \begin{cases} {}^I\hat{\mathbf{a}} = {}^I\mathbf{a}_m - \hat{\mathbf{b}}_a \\ {}^I\tilde{\mathbf{a}} = -\tilde{\mathbf{b}}_a - \mathbf{n}_{ad} \end{cases} \quad (407)$$

Let's integrate $\Delta \mathbf{R}$, $\Delta \mathbf{p}$, $\Delta \mathbf{v}$, $\Delta \mathbf{b}_g$, $\Delta \mathbf{b}_a$ with these perturbed inputs.

Perturbing $\Delta \mathbf{R}$. With $\hat{\boldsymbol{\theta}} \triangleq {}^I\hat{\boldsymbol{\omega}} \delta t$ and the right Jacobian $\mathbf{J}_r$ from Lecture 2,

$$\Delta \mathbf{R} = \text{Exp}({}^I\boldsymbol{\omega} \delta t) = \text{Exp}({}^I\hat{\boldsymbol{\omega}} + {}^I\tilde{\boldsymbol{\omega}} \delta t) \quad (408)$$

$$= \text{Exp}({}^I\hat{\boldsymbol{\omega}} \delta t) \text{Exp}(\mathbf{J}_r(\hat{\boldsymbol{\theta}}) {}^I\tilde{\boldsymbol{\omega}} \delta t) \quad (409)$$

$$\approx \text{Exp}(\hat{\boldsymbol{\theta}}) \left(\mathbf{I} + \left[\mathbf{J}_r(\hat{\boldsymbol{\theta}}) \delta t {}^I\tilde{\boldsymbol{\omega}} \right]_{\times} \right). \quad (410)$$

Perturbing $\Delta \mathbf{p}$. Substituting the split inputs and expanding to first order (using $[\mathbf{x}]_{\times} \mathbf{y} = -[\mathbf{y}]_{\times} \mathbf{x}$ and dropping second-order error terms),

$$\Delta \mathbf{p} = \int_{t_k}^{t_{k+1}} \int_{t_k}^{\tau} \text{Exp}((I_k \hat{\omega} + I_k \tilde{\omega}) \delta s) ds d\tau (I_k \hat{\mathbf{a}} + I_k \tilde{\mathbf{a}}) \quad (411)$$

$$= \int_{t_k}^{t_{k+1}} \int_{t_k}^{\tau} \text{Exp}(I_k \hat{\omega} \delta s) \left(\mathbf{I} + [\mathbf{J}_r(I_k \hat{\omega} \delta s) \delta s I_k \tilde{\omega}]_{\times} \right) ds d\tau (I_k \hat{\mathbf{a}} + I_k \tilde{\mathbf{a}}) \quad (412)$$

$$= \underbrace{\int_{t_k}^{t_{k+1}} \int_{t_k}^{\tau} \text{Exp}(I_k \hat{\omega} \delta s) ds d\tau I_k \hat{\mathbf{a}}}_{\Delta \hat{\mathbf{p}}} + \underbrace{\int_{t_k}^{t_{k+1}} \int_{t_k}^{\tau} \text{Exp}(I_k \hat{\omega} \delta s) ds d\tau I_k \tilde{\mathbf{a}}}_{\Xi_2} - \underbrace{\int_{t_k}^{t_{k+1}} \int_{t_k}^{\tau} \text{Exp}(I_k \hat{\omega} \delta s) [I_k \hat{\mathbf{a}}]_{\times} \mathbf{J}_r(I_k \hat{\omega} \delta s) \delta s ds d\tau I_k \tilde{\omega}}_{\Xi_4} \quad (413)$$

$$= \Delta \hat{\mathbf{p}} + \Xi_2 I_k \tilde{\mathbf{a}} - \Xi_4 I_k \tilde{\omega}. \quad (414)$$

Perturbing $\Delta \mathbf{v}$. By the same expansion,

$$\Delta \mathbf{v} = \int_{t_k}^{t_{k+1}} \text{Exp}((I_k \hat{\omega} + I_k \tilde{\omega}) \delta \tau) d\tau (I_k \hat{\mathbf{a}} + I_k \tilde{\mathbf{a}}) \quad (415)$$

$$= \int_{t_k}^{t_{k+1}} \text{Exp}(I_k \hat{\omega} \delta \tau) \left(\mathbf{I} + [\mathbf{J}_r(I_k \hat{\omega} \delta \tau) \delta \tau I_k \tilde{\omega}]_{\times} \right) d\tau (I_k \hat{\mathbf{a}} + I_k \tilde{\mathbf{a}}) \quad (416)$$

$$= \underbrace{\int_{t_k}^{t_{k+1}} \text{Exp}(I_k \hat{\omega} \delta \tau) d\tau I_k \hat{\mathbf{a}}}_{\Delta \hat{\mathbf{v}}} + \underbrace{\int_{t_k}^{t_{k+1}} \text{Exp}(I_k \hat{\omega} \delta \tau) d\tau I_k \tilde{\mathbf{a}}}_{\Xi_1} - \underbrace{\int_{t_k}^{t_{k+1}} \text{Exp}(I_k \hat{\omega} \delta \tau) [I_k \hat{\mathbf{a}}]_{\times} \mathbf{J}_r(I_k \hat{\omega} \delta \tau) \delta \tau d\tau I_k \tilde{\omega}}_{\Xi_3} \quad (417)$$

$$= \Delta \hat{\mathbf{v}} + \Xi_1 I_k \tilde{\mathbf{a}} - \Xi_3 I_k \tilde{\omega}. \quad (418)$$

The perturbed propagation. Then we have

$$\begin{cases} {}^G_{I_{k+1}} \mathbf{R} = {}^G_{I_k} \mathbf{R} \cdot \text{Exp}(I_k \hat{\omega} \delta t) \cdot \text{Exp}(\mathbf{J}_r(\hat{\theta}) \delta t I_k \tilde{\omega}), \\ {}^G \mathbf{p}_{I_{k+1}} = {}^G \mathbf{p}_{I_k} + {}^G \mathbf{v}_{I_k} \delta t + {}^G_{I_k} \mathbf{R} (\Delta \hat{\mathbf{p}} + \Xi_2 I_k \tilde{\mathbf{a}} - \Xi_4 I_k \tilde{\omega}) + \frac{1}{2} {}^G \mathbf{g} \delta t^2, \\ {}^G \mathbf{v}_{I_{k+1}} = {}^G \mathbf{v}_{I_k} + {}^G_{I_k} \mathbf{R} (\Delta \hat{\mathbf{v}} + \Xi_1 I_k \tilde{\mathbf{a}} - \Xi_3 I_k \tilde{\omega}) + {}^G \mathbf{g} \delta t, \\ \mathbf{b}_{g_{k+1}} = \mathbf{b}_{g_k} + \mathbf{n}_{wgd} \delta t, \\ \mathbf{b}_{a_{k+1}} = \mathbf{b}_{a_k} + \mathbf{n}_{wad} \delta t. \end{cases} \quad (419)$$

First: the mean propagation. Setting all errors and noises to zero,

$$\boxed{\begin{cases} {}^G_{I_{k+1}} \hat{\mathbf{R}} = {}^G_{I_k} \hat{\mathbf{R}} \cdot \text{Exp}(I_k \hat{\omega} \delta t), \\ {}^G \hat{\mathbf{p}}_{I_{k+1}} = {}^G \hat{\mathbf{p}}_{I_k} + {}^G \hat{\mathbf{v}}_{I_k} \delta t + {}^G_{I_k} \hat{\mathbf{R}} \Delta \hat{\mathbf{p}} + \frac{1}{2} {}^G \mathbf{g} \delta t^2, \\ {}^G \hat{\mathbf{v}}_{I_{k+1}} = {}^G \hat{\mathbf{v}}_{I_k} + {}^G_{I_k} \hat{\mathbf{R}} \Delta \hat{\mathbf{v}} + {}^G \mathbf{g} \delta t, \\ \hat{\mathbf{b}}_{g_{k+1}} = \hat{\mathbf{b}}_{g_k}, \quad \hat{\mathbf{b}}_{a_{k+1}} = \hat{\mathbf{b}}_{a_k}. \end{cases}} \quad (420)$$

Hence $\hat{\mathbf{x}}_k \rightarrow \hat{\mathbf{x}}_{k+1}$: solved.

Second: the linearization for covariance propagation. Use $SO(3) + \mathbb{R}^3 + \mathbb{R}^3 + \mathbb{R}^3 + \mathbb{R}^3$ for example, i.e., a right rotation perturbation ${}^G\mathbf{R} = {}^G\hat{\mathbf{R}} \text{Exp}(\delta\boldsymbol{\theta})$ and additive errors elsewhere. For the rotation,

$${}^G_{I_{k+1}}\hat{\mathbf{R}} \text{Exp}(\delta\boldsymbol{\theta}_{k+1}) = {}^G_{I_k}\hat{\mathbf{R}} \text{Exp}(\delta\boldsymbol{\theta}_k) \text{Exp}({}^{I_k}\hat{\boldsymbol{\omega}} \delta t) \text{Exp}(\mathbf{J}_r({}^{I_k}\hat{\boldsymbol{\omega}} \delta t) \delta t {}^{I_k}\tilde{\boldsymbol{\omega}}), \quad (421)$$

so, moving $\text{Exp}(\delta\boldsymbol{\theta}_k)$ across $\text{Exp}({}^{I_k}\hat{\boldsymbol{\omega}} \delta t) = \Delta\hat{\mathbf{R}}$ with the adjoint identity,

$$\delta\boldsymbol{\theta}_{k+1} = {}^{I_k}_{I_{k+1}}\hat{\mathbf{R}}^\top \delta\boldsymbol{\theta}_k + \mathbf{J}_r({}^{I_k}\hat{\boldsymbol{\omega}} \delta t) \delta t {}^{I_k}\tilde{\boldsymbol{\omega}}. \quad (422)$$

For the position and velocity (substituting ${}^G\mathbf{R} = {}^G_{I_k}\hat{\mathbf{R}} \text{Exp}(\delta\boldsymbol{\theta}_k) \approx {}^G_{I_k}\hat{\mathbf{R}}(\mathbf{I} + [\delta\boldsymbol{\theta}_k]_\times)$ and keeping first-order terms),

$${}^G\tilde{\mathbf{p}}_{I_{k+1}} = {}^G\tilde{\mathbf{p}}_{I_k} + {}^G\tilde{\mathbf{v}}_{I_k} \delta t - {}^G_{I_k}\hat{\mathbf{R}} [\Delta\hat{\mathbf{p}}]_\times \delta\boldsymbol{\theta}_k + {}^G_{I_k}\hat{\mathbf{R}} (\Xi_2 {}^{I_k}\tilde{\mathbf{a}} - \Xi_4 {}^{I_k}\tilde{\boldsymbol{\omega}}), \quad (423)$$

$${}^G\tilde{\mathbf{v}}_{I_{k+1}} = {}^G\tilde{\mathbf{v}}_{I_k} - {}^G_{I_k}\hat{\mathbf{R}} [\Delta\hat{\mathbf{v}}]_\times \delta\boldsymbol{\theta}_k + {}^G_{I_k}\hat{\mathbf{R}} (\Xi_1 {}^{I_k}\tilde{\mathbf{a}} - \Xi_3 {}^{I_k}\tilde{\boldsymbol{\omega}}), \quad (424)$$

$$\tilde{\mathbf{b}}_{g_{k+1}} = \tilde{\mathbf{b}}_{g_k} + \mathbf{n}_{wgd} \delta t, \quad \tilde{\mathbf{b}}_{a_{k+1}} = \tilde{\mathbf{b}}_{a_k} + \mathbf{n}_{wad} \delta t, \quad (425)$$

where the input errors are

$${}^{I_k}\tilde{\mathbf{a}} = -\tilde{\mathbf{b}}_a - \mathbf{n}_{ad}, \quad {}^{I_k}\tilde{\boldsymbol{\omega}} = -\tilde{\mathbf{b}}_g - \mathbf{n}_{gd}. \quad (426)$$

Hence, stacking the error state $\tilde{\mathbf{x}} = [\delta\boldsymbol{\theta}^\top, {}^G\tilde{\mathbf{p}}_I^\top, {}^G\tilde{\mathbf{v}}_I^\top, \tilde{\mathbf{b}}_g^\top, \tilde{\mathbf{b}}_a^\top]^\top$,

$$\tilde{\mathbf{x}}_{k+1} = \boldsymbol{\Phi}_{(k+1,k)} \tilde{\mathbf{x}}_k + \mathbf{G}_k \mathbf{n}_{Id}, \quad \mathbf{n}_{Id} = \begin{bmatrix} \mathbf{n}_{gd} \\ \mathbf{n}_{ad} \\ \mathbf{n}_{wgd} \\ \mathbf{n}_{wad} \end{bmatrix}, \quad (427)$$

with the state-transition matrix

$$\boldsymbol{\Phi}_{(k+1,k)} = \begin{bmatrix} {}^{I_k}_{I_{k+1}}\hat{\mathbf{R}}^\top & \mathbf{0}_3 & \mathbf{0}_3 & -\mathbf{J}_r({}^{I_k}\hat{\boldsymbol{\omega}} \delta t) \delta t & \mathbf{0}_3 \\ -{}^G_{I_k}\hat{\mathbf{R}} [\Delta\hat{\mathbf{p}}]_\times & \mathbf{I}_3 & \mathbf{I}_3 \delta t & {}^G_{I_k}\hat{\mathbf{R}} \Xi_4 & -{}^G_{I_k}\hat{\mathbf{R}} \Xi_2 \\ -{}^G_{I_k}\hat{\mathbf{R}} [\Delta\hat{\mathbf{v}}]_\times & \mathbf{0}_3 & \mathbf{I}_3 & {}^G_{I_k}\hat{\mathbf{R}} \Xi_3 & -{}^G_{I_k}\hat{\mathbf{R}} \Xi_1 \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3 \end{bmatrix}, \quad (428)$$

and the noise Jacobian

$$\mathbf{G}_k = \begin{bmatrix} -\mathbf{J}_r({}^{I_k}\hat{\boldsymbol{\omega}} \delta t) \delta t & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 \\ {}^G_{I_k}\hat{\mathbf{R}} \Xi_4 & -{}^G_{I_k}\hat{\mathbf{R}} \Xi_2 & \mathbf{0}_3 & \mathbf{0}_3 \\ {}^G_{I_k}\hat{\mathbf{R}} \Xi_3 & -{}^G_{I_k}\hat{\mathbf{R}} \Xi_1 & \mathbf{0}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3 \delta t & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3 \delta t \end{bmatrix}. \quad (429)$$

Hence the covariance propagation

$$\boxed{\mathbf{P}_{k+1} = \boldsymbol{\Phi}_{(k+1,k)} \mathbf{P}_k \boldsymbol{\Phi}_{(k+1,k)}^\top + \mathbf{G}_k \mathbf{Q}_d \mathbf{G}_k^\top \rightarrow \text{solved.}} \quad (430)$$

5.8 The linearization using $SE_2(3)$

Now the linearization using $SE_2(3)$ (Lecture 2). The states live on $SE_2(3) + \mathbb{R}^3 + \mathbb{R}^3$:

$$\mathbf{x}_I = (\mathbf{X}_n, \mathbf{x}_b), \quad \mathbf{X}_n = ({}_I^G \mathbf{R}, {}^G \mathbf{p}_I, {}^G \mathbf{v}_I) = \begin{bmatrix} {}_I^G \mathbf{R} & {}^G \mathbf{p}_I & {}^G \mathbf{v}_I \\ \mathbf{0}_{1 \times 3} & 1 & 0 \\ \mathbf{0}_{1 \times 3} & 0 & 1 \end{bmatrix}, \quad \mathbf{x}_b = (\mathbf{b}_g, \mathbf{b}_a). \quad (431)$$

The error states. Define a left (group) error on $\mathbf{X}_n$ and additive errors on the biases,

$$\mathbf{x}_I = \hat{\mathbf{x}}_I \boxplus \delta \mathbf{x}_I = (\text{Exp}(\delta \mathbf{x}_n) \hat{\mathbf{X}}_n, \hat{\mathbf{x}}_b + \tilde{\mathbf{x}}_b), \quad (432)$$

which, writing $\delta \mathbf{x}_n = [\delta \boldsymbol{\theta}^\top, \delta \mathbf{p}^\top, \delta \mathbf{v}^\top]^\top$ and using the $SE_2(3)$ exponential (with the left Jacobian $\mathbf{J}_l$),

$$\begin{cases} {}_I^G \mathbf{R} = \text{Exp}(\delta \boldsymbol{\theta}) {}_I^G \hat{\mathbf{R}}, \\ {}^G \mathbf{p}_I = \text{Exp}(\delta \boldsymbol{\theta}) {}^G \hat{\mathbf{p}}_I + \mathbf{J}_l(\delta \boldsymbol{\theta}) \delta \mathbf{p}, \\ {}^G \mathbf{v}_I = \text{Exp}(\delta \boldsymbol{\theta}) {}^G \hat{\mathbf{v}}_I + \mathbf{J}_l(\delta \boldsymbol{\theta}) \delta \mathbf{v}, \\ \mathbf{b}_g = \hat{\mathbf{b}}_g + \delta \mathbf{b}_g, \quad \mathbf{b}_a = \hat{\mathbf{b}}_a + \delta \mathbf{b}_a. \end{cases} \quad (433)$$

The propagation as a group factorization. The propagation equations

$$\begin{cases} {}_{I_{k+1}}^G \mathbf{R} = {}_{I_k}^G \mathbf{R} \cdot \Delta \mathbf{R}, \\ {}_{I_{k+1}}^G \mathbf{p}_{I_{k+1}} = {}_{I_k}^G \mathbf{p}_{I_k} + {}_{I_k}^G \mathbf{v}_{I_k} \delta t + {}_{I_k}^G \mathbf{R} \Delta \mathbf{p} + \frac{1}{2} {}_{I_k}^G \mathbf{g} \delta t^2, \\ {}_{I_{k+1}}^G \mathbf{v}_{I_{k+1}} = {}_{I_k}^G \mathbf{v}_{I_k} + {}_{I_k}^G \mathbf{R} \Delta \mathbf{v} + {}_{I_k}^G \mathbf{g} \delta t, \\ \mathbf{b}_{g_{k+1}} = \mathbf{b}_{g_k} + \mathbf{n}_{wg d} \delta t, \quad \mathbf{b}_{a_{k+1}} = \mathbf{b}_{a_k} + \mathbf{n}_{wad} \delta t \end{cases} \quad (434)$$

factor into a product of three $SE_2(3)$ elements,

$$\boxed{\mathbf{X}_{n_{k+1}} = \boldsymbol{\Gamma}_k \cdot \boldsymbol{\Psi}(\mathbf{X}_{n_k}) \cdot \mathbf{Y}_k}, \quad (435)$$

with

$$\boldsymbol{\Gamma}_k = \begin{bmatrix} \mathbf{I}_3 & \frac{1}{2} {}_{I_k}^G \mathbf{g} \delta t^2 & {}_{I_k}^G \mathbf{g} \delta t \\ \mathbf{0}_{1 \times 3} & 1 & 0 \\ \mathbf{0}_{1 \times 3} & 0 & 1 \end{bmatrix}, \quad \mathbf{Y}_k = \begin{bmatrix} \Delta \mathbf{R} & \Delta \mathbf{p} & \Delta \mathbf{v} \\ \mathbf{0}_{1 \times 3} & 1 & 0 \\ \mathbf{0}_{1 \times 3} & 0 & 1 \end{bmatrix}, \quad (436)$$

$$\boldsymbol{\Psi}(\mathbf{X}_{n_k}) = \begin{bmatrix} {}_{I_k}^G \mathbf{R} & {}_{I_k}^G \mathbf{p}_{I_k} + {}_{I_k}^G \mathbf{v}_{I_k} \delta t & {}_{I_k}^G \mathbf{v}_{I_k} \\ \mathbf{0}_{1 \times 3} & 1 & 0 \\ \mathbf{0}_{1 \times 3} & 0 & 1 \end{bmatrix}. \quad (437)$$

The properties of $\boldsymbol{\Psi}(\cdot)$. The map $\boldsymbol{\Psi}$ satisfies

$$\begin{cases} \boldsymbol{\Psi}(\mathbf{X}_{n_i} \cdot \mathbf{X}_{n_j}) = \boldsymbol{\Psi}(\mathbf{X}_{n_i}) \cdot \boldsymbol{\Psi}(\mathbf{X}_{n_j}), \\ \boldsymbol{\Psi}(\text{Exp}(\delta \mathbf{x}_n)) \approx \text{Exp}(\mathbf{F} \delta \mathbf{x}_n), \\ \boldsymbol{\Psi}(\text{Exp}(\delta \mathbf{x}_n) \hat{\mathbf{X}}_n) = \boldsymbol{\Psi}(\text{Exp}(\delta \mathbf{x}_n)) \cdot \boldsymbol{\Psi}(\hat{\mathbf{X}}_n) = \text{Exp}(\mathbf{F} \delta \mathbf{x}_n) \boldsymbol{\Psi}(\hat{\mathbf{X}}_n), \end{cases} \quad \mathbf{F} \triangleq \begin{bmatrix} \mathbf{I}_3 & \mathbf{0}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{I}_3 & \mathbf{I}_3 \delta t \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3 \end{bmatrix}. \quad (438)$$

The linearization. Hence we can do the linearization as

$$\mathbf{X}_{n_{k+1}} = \text{Exp}(\delta \mathbf{x}_{n_{k+1}}) \hat{\mathbf{X}}_{n_{k+1}} = \mathbf{\Gamma}_k \Psi(\mathbf{X}_{n_k}) \mathbf{Y}_k \quad (439)$$

$$= \mathbf{\Gamma}_k \Psi(\text{Exp}(\delta \mathbf{x}_{n_k}) \hat{\mathbf{X}}_{n_k}) \hat{\mathbf{Y}}_k \cdot \tilde{\mathbf{Y}}_k \quad (440)$$

$$\approx \mathbf{\Gamma}_k \text{Exp}(\mathbf{F} \delta \mathbf{x}_{n_k}) \Psi(\hat{\mathbf{X}}_{n_k}) \hat{\mathbf{Y}}_k \tilde{\mathbf{Y}}_k \quad (441)$$

$$= \text{Exp}(\text{Ad}_{\mathbf{\Gamma}_k} \mathbf{F} \delta \mathbf{x}_{n_k}) \underbrace{\mathbf{\Gamma}_k \Psi(\hat{\mathbf{X}}_{n_k}) \hat{\mathbf{Y}}_k}_{\hat{\mathbf{X}}_{n_{k+1}}} \tilde{\mathbf{Y}}_k \quad (442)$$

$$= \text{Exp}(\text{Ad}_{\mathbf{\Gamma}_k} \mathbf{F} \delta \mathbf{x}_{n_k}) \hat{\mathbf{X}}_{n_{k+1}} \tilde{\mathbf{Y}}_k \quad (443)$$

$$= \text{Exp}(\Phi_{nn} \delta \mathbf{x}_{n_k}) \hat{\mathbf{X}}_{n_{k+1}} \tilde{\mathbf{Y}}_k, \quad \boxed{\Phi_{nn} \triangleq \text{Ad}_{\mathbf{\Gamma}_k} \mathbf{F}} \quad (444)$$

where the preintegration factor splits as $\mathbf{Y}_k = \hat{\mathbf{Y}}_k \tilde{\mathbf{Y}}_k$,

$$\mathbf{Y}_k = \hat{\mathbf{Y}}_k \tilde{\mathbf{Y}}_k = \begin{bmatrix} \Delta \hat{\mathbf{R}} & \Delta \hat{\mathbf{p}} & \Delta \hat{\mathbf{v}} \\ \mathbf{0}_{1 \times 3} & 1 & 0 \\ \mathbf{0}_{1 \times 3} & 0 & 1 \end{bmatrix} \begin{bmatrix} \Delta \tilde{\mathbf{R}} & \Delta \hat{\mathbf{R}}^\top \Delta \tilde{\mathbf{p}} & \Delta \hat{\mathbf{R}}^\top \Delta \tilde{\mathbf{v}} \\ \mathbf{0}_{1 \times 3} & 1 & 0 \\ \mathbf{0}_{1 \times 3} & 0 & 1 \end{bmatrix} = \hat{\mathbf{Y}}_k \cdot \text{Exp} \left(\begin{bmatrix} \delta \Delta \boldsymbol{\theta} \\ \Delta \hat{\mathbf{R}}^\top \Delta \tilde{\mathbf{p}} \\ \Delta \hat{\mathbf{R}}^\top \Delta \tilde{\mathbf{v}} \end{bmatrix} \right), \quad (445)$$

with $\delta \Delta \boldsymbol{\theta} = \text{Log}(\Delta \tilde{\mathbf{R}})$ (to first order). From the perturbations of §5.7,

$$\begin{bmatrix} \delta \Delta \boldsymbol{\theta} \\ \Delta \hat{\mathbf{R}}^\top \Delta \tilde{\mathbf{p}} \\ \Delta \hat{\mathbf{R}}^\top \Delta \tilde{\mathbf{v}} \end{bmatrix} = \begin{bmatrix} \mathbf{I}_3 & \mathbf{0}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \Delta \hat{\mathbf{R}}^\top & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \Delta \hat{\mathbf{R}}^\top \end{bmatrix} \begin{bmatrix} \delta \Delta \boldsymbol{\theta} \\ \Delta \tilde{\mathbf{p}} \\ \Delta \tilde{\mathbf{v}} \end{bmatrix} \quad (446)$$

$$= \begin{bmatrix} \mathbf{I}_3 & \mathbf{0}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \Delta \hat{\mathbf{R}}^\top & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \Delta \hat{\mathbf{R}}^\top \end{bmatrix} \begin{bmatrix} -\mathbf{J}_r(\Delta \hat{\boldsymbol{\theta}}) \delta t & \mathbf{0}_3 \\ \boldsymbol{\Xi}_4 & -\boldsymbol{\Xi}_2 \\ \boldsymbol{\Xi}_3 & -\boldsymbol{\Xi}_1 \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{b}}_g + \mathbf{n}_{gd} \\ \tilde{\mathbf{b}}_a + \mathbf{n}_{ad} \end{bmatrix} \quad (447)$$

$$= \begin{bmatrix} -\mathbf{J}_r(\Delta \hat{\boldsymbol{\theta}}) \delta t & \mathbf{0}_3 \\ \Delta \hat{\mathbf{R}}^\top \boldsymbol{\Xi}_4 & -\Delta \hat{\mathbf{R}}^\top \boldsymbol{\Xi}_2 \\ \Delta \hat{\mathbf{R}}^\top \boldsymbol{\Xi}_3 & -\Delta \hat{\mathbf{R}}^\top \boldsymbol{\Xi}_1 \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{b}}_g + \mathbf{n}_{gd} \\ \tilde{\mathbf{b}}_a + \mathbf{n}_{ad} \end{bmatrix}, \quad (448)$$

with $\Delta \hat{\boldsymbol{\theta}} = {}^{I_k} \hat{\boldsymbol{\omega}} \delta t$. Finally, move $\tilde{\mathbf{Y}}_k$ across $\hat{\mathbf{X}}_{n_{k+1}}$ with the adjoint:

$$\text{Exp}(\delta \mathbf{x}_{n_{k+1}}) = \text{Exp}(\Phi_{nn} \delta \mathbf{x}_{n_k}) \hat{\mathbf{X}}_{n_{k+1}} \tilde{\mathbf{Y}}_k \hat{\mathbf{X}}_{n_{k+1}}^{-1} \quad (449)$$

$$= \text{Exp}(\Phi_{nn} \delta \mathbf{x}_{n_k}) \text{Exp} \left(\text{Ad}_{\hat{\mathbf{X}}_{n_{k+1}}} \begin{bmatrix} -\mathbf{J}_r(\Delta \hat{\boldsymbol{\theta}}) \delta t & \mathbf{0}_3 \\ \Delta \hat{\mathbf{R}}^\top \boldsymbol{\Xi}_4 & -\Delta \hat{\mathbf{R}}^\top \boldsymbol{\Xi}_2 \\ \Delta \hat{\mathbf{R}}^\top \boldsymbol{\Xi}_3 & -\Delta \hat{\mathbf{R}}^\top \boldsymbol{\Xi}_1 \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{b}}_g + \mathbf{n}_{gd} \\ \tilde{\mathbf{b}}_a + \mathbf{n}_{ad} \end{bmatrix} \right), \quad (450)$$

so that

$$\delta \mathbf{x}_{n_{k+1}} = \Phi_{nn} \delta \mathbf{x}_{n_k} + \Phi_{nb} \begin{bmatrix} \tilde{\mathbf{b}}_g + \mathbf{n}_{gd} \\ \tilde{\mathbf{b}}_a + \mathbf{n}_{ad} \end{bmatrix}, \quad \boxed{\Phi_{nb} \triangleq \text{Ad}_{\hat{\mathbf{X}}_{n_{k+1}}} \begin{bmatrix} -\mathbf{J}_r(\Delta \hat{\boldsymbol{\theta}}) \delta t & \mathbf{0}_3 \\ \Delta \hat{\mathbf{R}}^\top \boldsymbol{\Xi}_4 & -\Delta \hat{\mathbf{R}}^\top \boldsymbol{\Xi}_2 \\ \Delta \hat{\mathbf{R}}^\top \boldsymbol{\Xi}_3 & -\Delta \hat{\mathbf{R}}^\top \boldsymbol{\Xi}_1 \end{bmatrix}} \quad (451)$$

The stacked system. Then

$$\begin{cases} \delta \mathbf{x}_{n_{k+1}} = \Phi_{nn} \delta \mathbf{x}_{n_k} + \Phi_{nb} \begin{bmatrix} \tilde{\mathbf{b}}_g \\ \tilde{\mathbf{b}}_a \end{bmatrix} + \Phi_{nb} \begin{bmatrix} \mathbf{n}_{gd} \\ \mathbf{n}_{ad} \end{bmatrix}, \\ \tilde{\mathbf{b}}_{g_{k+1}} = \tilde{\mathbf{b}}_{g_k} + \mathbf{n}_{wgd} \delta t, \quad \tilde{\mathbf{b}}_{a_{k+1}} = \tilde{\mathbf{b}}_{a_k} + \mathbf{n}_{wad} \delta t, \end{cases} \quad (452)$$

which stacks into $\tilde{\mathbf{x}}_{k+1} = \Phi_{(k+1,k)} \tilde{\mathbf{x}}_k + \mathbf{G}_k \mathbf{n}_{Id}$ with

$$\Phi_{(k+1,k)} = \begin{bmatrix} \Phi_{nn} & \Phi_{nb} \\ \mathbf{0}_{6 \times 9} & \mathbf{I}_6 \end{bmatrix} = \begin{bmatrix} \mathbf{I}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \Phi_{14} & \mathbf{0}_3 \\ \frac{1}{2} \begin{bmatrix} G \mathbf{g} \end{bmatrix}_{\times} \delta t^2 & \mathbf{I}_3 & \mathbf{I}_3 \delta t & \Phi_{24} & \Phi_{25} \\ \begin{bmatrix} G \mathbf{g} \end{bmatrix}_{\times} \delta t & \mathbf{0}_3 & \mathbf{I}_3 & \Phi_{34} & \Phi_{35} \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3 \end{bmatrix}, \quad (453)$$

where the first three block-columns are exactly $\Phi_{nn} = \text{Ad}_{\Gamma_k} \mathbf{F}$ and the Φ_{i4}, Φ_{i5} blocks are the columns of Φ_{nb} ($\Phi_{15} = \mathbf{0}_3$ since the accel error does not enter the rotation), and

$$\mathbf{G}_k = \begin{bmatrix} \Phi_{nb} & \mathbf{0}_{9 \times 6} \\ \mathbf{0}_{6 \times 6} & \mathbf{I}_6 \delta t \end{bmatrix} = \begin{bmatrix} \mathbf{G}_{11} & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 \\ \mathbf{G}_{21} & \mathbf{G}_{22} & \mathbf{0}_3 & \mathbf{0}_3 \\ \mathbf{G}_{31} & \mathbf{G}_{32} & \mathbf{0}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3 \delta t & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3 \delta t \end{bmatrix}, \quad (454)$$

with $[\mathbf{G}_{i1}, \mathbf{G}_{i2}]$ the blocks of Φ_{nb} . Hence, as before,

$$\boxed{\tilde{\mathbf{x}}_{k+1} = \Phi_{(k+1,k)} \tilde{\mathbf{x}}_k + \mathbf{G}_k \mathbf{n}_{Id} \quad \Rightarrow \quad \mathbf{P}_{k+1} = \Phi_{(k+1,k)} \mathbf{P}_k \Phi_{(k+1,k)}^{\top} + \mathbf{G}_k \mathbf{Q}_d \mathbf{G}_k^{\top} \quad \rightarrow \quad \text{solved.}} \quad (455)$$

Note that Φ_{nn} in (453) depends only on gravity and δt — not on the state estimate — which is the hallmark of the (right-)invariant error design mentioned in Lecture 4.

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